

Setup

```
In [1]: #Load Libraries
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib as mpl
import matplotlib.colors as colors
import matplotlib.cm as cmx

import urllib.request as url
import glob
import csv
from pathlib import Path
from scipy.stats import spearmanr
import pymannkendall as mk

from capstone_glambie.glambie.const import regions
from capstone_glambie.help_functions.plotting_functions import two_region_comparison_plot_2
from capstone_glambie.help_functions.plotting_functions import three_runs_cumulative_comparison_plot
from capstone_glambie.help_functions.plotting_functions import three_runs_relative_difference_plot
from capstone_glambie.help_functions.plotting_functions import derivative_to_cumulative
from capstone_glambie.help_functions.plotting_functions import four_runs_annual_cumulative_plot
from capstone_glambie.help_functions.plotting_functions import three_runs_annual_cumulative_plot

sns.set_theme(palette='colorblind', style='white', font_scale=1.2)
sns.set_style({'axes.linewidth': 2, 'axes.edgecolor': 'lightgrey'})

colors_list = sns.color_palette(palette='colorblind', n_colors=18, desat=0.9)
```

```
In [2]: #define data directories
reference_run_dir = 'capstone_glambie/input/glambie_runs_clean/reanalysis_reference_rgi6/calendar_years/'
rgi7_dir = 'capstone_glambie/input/glambie_runs_clean/RGI7_2025-11-24_17-01/calendar_years/'
d_dir_2012 = 'capstone_glambie/input/glambie_runs_clean/final_squared_area_2012_2026/calendar_years/'
d_dir_2023 = 'capstone_glambie/input/glambie_runs_clean/final_squared_area_2023_2026-02-10_19-30/calenda

#method runs
methods_regression_proportional = 'capstone_glambie/input/glambie_runs_clean/methods_regression-proportion
methods_regression_seasonal_homogenization = 'capstone_glambie/input/glambie_runs_clean/methods_regression
methods_start_vs_end_proportional = 'capstone_glambie/input/glambie_runs_clean/methods_start_vs_end-prop
methods_start_vs_end_seasonal_homogenization = 'capstone_glambie/input/glambie_runs_clean/methods_start_v

method_run_dirs = [
    Path(methods_regression_proportional),
    Path(methods_regression_seasonal_homogenization),
    Path(methods_start_vs_end_proportional),
    Path(methods_start_vs_end_seasonal_homogenization),
]

#dataset comparison
glambie_runs_base_dir = 'capstone_glambie/input/glambie_runs_clean/calendar_years/'
datasets_including_gravimetry = 'capstone_glambie/input/glambie_runs_clean/datasets_including_gravimetry
datasets_including_all = 'capstone_glambie/input/glambie_runs_clean/datasets_including_all/calendar_year
datasets_including_except_gravimetry = 'capstone_glambie/input/glambie_runs_clean/datasets_including_exc

dataset_run_dirs = [
    Path(datasets_including_all),
    Path(datasets_including_gravimetry),
    Path(datasets_including_except_gravimetry),
]
```

```
In [3]: #Load in global data for reference run
global_reference = pd.read_csv(
    os.path.join(reference_run_dir, '0_global.csv')
).drop('Unnamed: 0', axis=1)
```

```
#get a list of all the regions/folders
list_of_folders = sorted(os.listdir(reference_run_dir))
```

Reanalysis of the GlaMBIE project

1. Introduction

This paper is a reanalysis performed on the paper “Community estimate of global glacier mass changes from 2000 to 2023” published by Zemp et al. (2025). This study presents the results from the Glacier Mass Balance Intercomparison Exercise (GlaMBIE) which is a project collecting measurements of glacier mass balance from many different techniques from the research community. The paper aims to combine datasets from four observation methods (glaciological, DEM differencing, altimetry, gravimetry). To do this, they homogenize the data in space, time and unit, separate the temporal variability from the long term trend and produce combined estimates per region, method and globally.

They find that global glacier mass decreased about -6,542 +/- 387 gigatonnes (Gt) from 2000 to 2023 (about -273 +/- 16 Gt/yr) which corresponds to a mass loss of about 5% globally or up to 39% regionally. They also report an increase in loss rate by 36 +/- 10% from the first half (2000-2011) to the second half (2012-2023) of the examined period.

Short overview of methods

GlaMBIE combines 233 regional estimates of glacier mass changes from 35 research teams, covering 19 predefined glacier regions (based on the Randolph Glacier Inventory, RGI 6) (RGI Consortium, 2017).

1. Homogenization: Datasets are converted to specific mass changes in meter water equivalent (m w.e.) using the glacier area which is derived from inventory data (RGI 6). Elevation-change methods assume an ice density for conversion (850 kg/m^3 + period dependent conversion uncertainty). They are aligned to an annual grid by aggregating monthly or seasonal data to annual sums. Multi-annual data are corrected to hydrological year boundaries assuming linear change.
2. Separation of annual variability and trend: Each time series mean is subtracted. This isolates the annual anomaly from the long-term trend, avoiding artefacts from differing observation periods. The annual anomalies from multiple datasets within each method are averaged. The averaged anomaly is then re-trended by adding back the long-term trend (mean) of each individual input dataset, resulting in one time series per method. This produces one combined estimate: (i) glaciological + DEM differencing, (ii) altimetry, and (iii) gravimetry.
3. The previous step then is repeated for each method to obtain a single combined result per region. Lastly, those regional results are combined into a global estimate.

A benefit of this approach is that glaciological observations, which have a high interannual resolution, can be combined with DEM differencing, which provides precise estimates of overall mass changes.

The following uncertainties are taken into account (see Zemp et al., 2025 and the following functions from the GlaMBIE):

- propagated observational uncertainties from input datasets (`data_catalogue.py: average_timeseries_in_catalogue`),
- variability (spread) among input datasets within a method (`data_catalogue.py: average_timeseries_in_catalogue`),
- homogenization corrections (e.g., density assumption `density_uncertainty.py: get_density_uncertainty_over_survey_period` , `timeseries.py: convert_timeseries_to_unit_mwe` , temporal alignment `timeseries.py: shift_timeseries_to_annual_grid_with_seasonal_homogenization`).

Compared with previous global assessments, the estimates by Zemp et al. (2025) are generally in line with earlier studies, indicating similar rates and patterns of worldwide glacier mass loss (Table 1). The main novelty of Zemp et al. (2025) is that they incorporate multiple observation methods and compare them. With this approach of combining observation methods, they are able to provide estimates at annual resolution for a long observation period.

Study	Period	Global estimate [Gt/yr]
Gardner et al. (2013)	2003–2009	-259 ± 28
Zemp et al. (2019)	2006–2016	-332 ± 144
Wouters et al. (2019)	2002–2016	-280
Hugonnet et al. (2021)	2000–2019	-267 ± 16
IPCC AR6	2000–2019	-267 ± 16
Zemp et al. (2025)	2000–2023	-273 ± 16

2. Reproducing GlaMBIE's results

Access and run the model

The GlaMBIE team has made their code publicly available on Github. It can be accessed here:

<https://github.com/earthwave/glambie>. The GlaMBIE model works as a python package, that can be installed in an environment with python 3.11.

We followed their instructions on Installation for Development (found in their readme file under the header `Installation For Development (or when Google Cloud authentication is not possible)`). This included cloning the Github repository to our local devices, installing their testing requirements and the glambie package in editable mode (`pip install -e .`).

The GlaMBIE team provides both their input and their output data. It can be downloaded through the following link: https://wgms.ch/data_glambie/. However, the model uses the GLaMBIE Submission System (Google bucket with the input data) to access the input data. As we want to be able to run the model, we needed access to this as well. In order to get the access, we contacted support@earthwave.co.uk. Furthermore, we installed `google cloud command line interface` as well as the pip keyring backend as described in the installation instructions provided by the GlaMBIE team.

In order to run the model, we ran the following code which triggers the model to be run using the configuration file "0_parent_config.yaml" with the following command:

```
python -m glambie configuration/0_parent_config.yaml
```

Processing results

Running the model with configuration file '0_parent_config.yaml' allowed us to successfully reproduce Zemp et al.'s result. However, the output of their model is in a different format than the results they published. In order to be able to compare them to see if we successfully reproduced the results, as well as for convenience in further analysis, we first had to put the data in the right format. In order to do this, we created a notebook written specifically so that whenever we ran the model we only had to change one or two variables in this script in order to put the results in the right format. This script can be found in the github and is called `scripts/data_right_format_V4.ipynb`. It was run for all model runs except for model 2012 and 2023 (as described in the heading "Non-Static area change rate") as they required a different glacier area calculation. The code used to process the data for these two models can be found on github as well in the file `scripts/area_data_right_format_V3.ipynb`.

In order to create too large blocks of code, some of the code creating the plots has been put into functions that are defined in the file `help_functions/plotting_functions.py`.

3. Performed tests

After successfully reproducing the results, there were a number of tests we performed, which are all discussed in their own chapter below:

1. We tested the effect of different trend removal and temporal homogenization methods
2. We looked at the relative change, as opposed to the absolute change
3. We tested two things regarding their glacier area calculating
 - 3.1 The effect of RGI 6 vs 7
 - 3.2 The effect a non-static glacier area change rate
4. We tested the effect of including the excluded datasets.

1. Test effect of different trend removal and temporal homogenization methods

The GLaMBIE model incorporates two different methods for trend estimation and temporal homogenization, respectively. Temporal homogenization is performed at two stages in the model. Firstly, individual input datasets that do not fall on the desired annual grid (calendar vs hydrological year) are corrected to it, assuming linear change by default. Secondly, the final regional results are corrected from hydrological to calendar years using seasonal homogenization with a monthly glaciological calibration series. Optionally, this can also be applied at the first stage.

- Proportional: Shifts the time series to the nearest annual grid boundaries and scales the change values by the ratio of the new to old period length, assuming uniform (linear) mass change over time. (`timeseries.py: shift_timeseries_to_annual_grid_proportionally`)
- Seasonal homogenization: Uses a monthly calibration series and a sine-wave model of the seasonal cycle (based on Zemp et al., 2019), so that the temporal shift respects the typical timing of mass change within a year. (`timeseries.py: shift_timeseries_to_annual_grid_with_seasonal_homogenization`, `timeseries_combination_helpers.py: calibrate_timeseries_with_trends`)

Trend extraction determines how the long-term trend is derived from the input timeseries. Two methods are available:

- `start_vs_end`: Computes the long-term trend as the net total change, i.e. the sum of all interval changes (`np.nansum(changes)`). This method is always used for data at annual resolution. (`timeseries_helpers.py: get_total_trend`)
- `regression`: Converts the timeseries to a cumulative series, fits a linear regression, and returns the total change as slope × time span. Only applied to interannual timeseries, where it produces a trend that is less sensitive to anomalies at the start or end of the record. (`timeseries_helpers.py: get_total_trend`, `timeseries_helpers.py: get_slope_of_timeseries_with_linear_regression`)

By default, the study employs seasonal homogenization for the temporal homogenization and regression-based trend extraction when processing interannual datasets. To assess the sensitivity of the results to these methodological choices, we ran the GLaMBIE model for all four possible combinations of the processing approaches and quantified the resulting differences. Specifically, we calculated the absolute difference (Gt), the relative difference (%), and the Spearman rank correlation of the annual changes with respect to the default configuration (regression, proportional).

```
In [4]: method_label_map = {
    "regression_proportional": "Regression + proportional",
    "regression_seasonal_homogenization": "Regression + seasonal homogenization",
    "start_vs_end_proportional": "Start vs end + proportional",
    "start_vs_end_seasonal_homogenization": "Start vs end + seasonal homogenization",
}

def get_method_label(folder_name):
    key = folder_name.replace("methods_", "").replace("-", "_")
    if key in method_label_map:
        return method_label_map[key]
    return folder_name

for region_file in list_of_folders:
    display_name = region_file.split('_', 1)[-1].replace('.csv', '').replace('_', ' ').title()

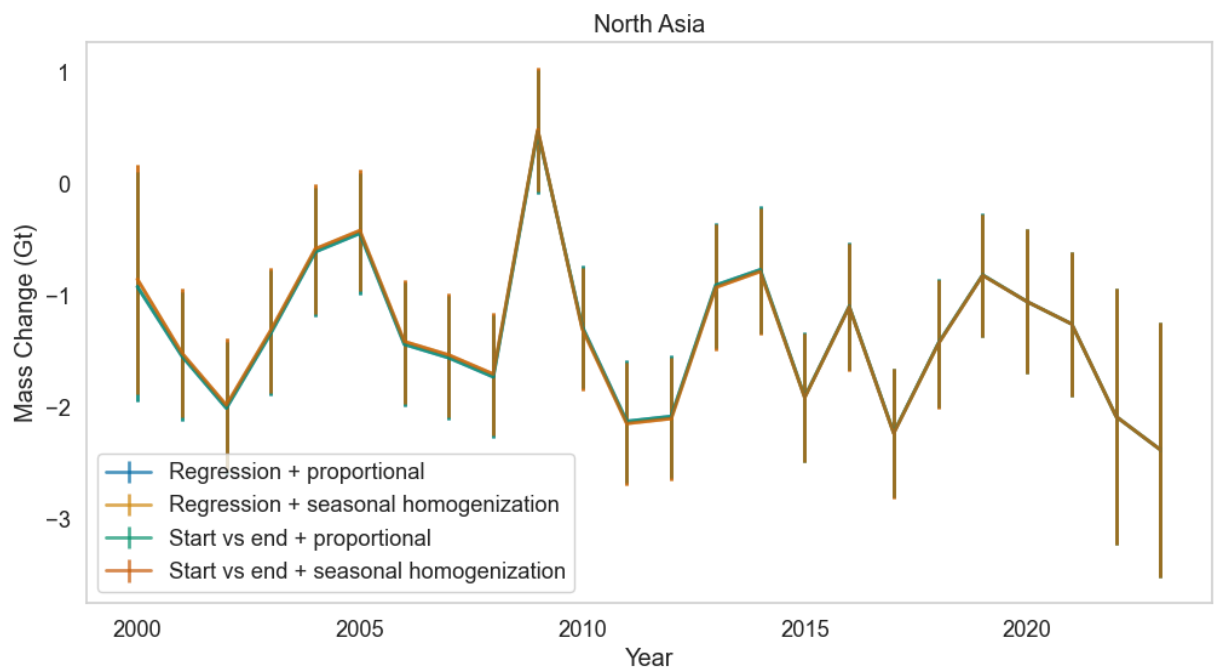
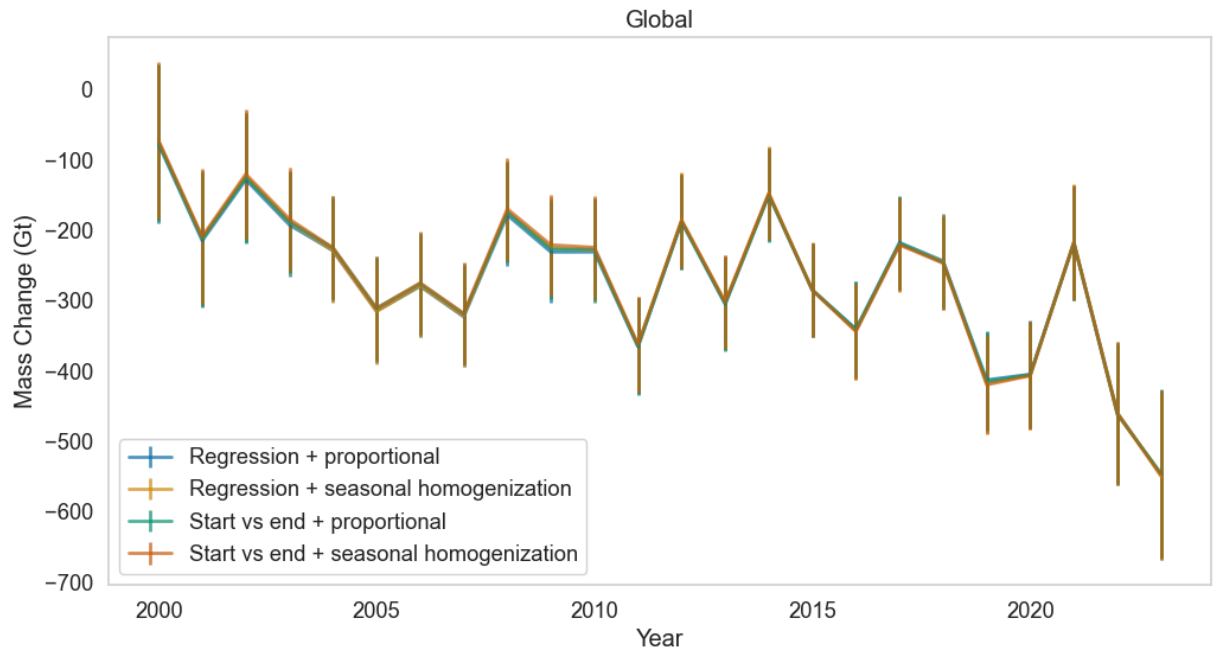
    # plot
    fig, ax = plt.subplots(figsize=(12, 6))
    for i, run_dir in enumerate(method_run_dirs):
        df = pd.read_csv(run_dir / region_file)
```

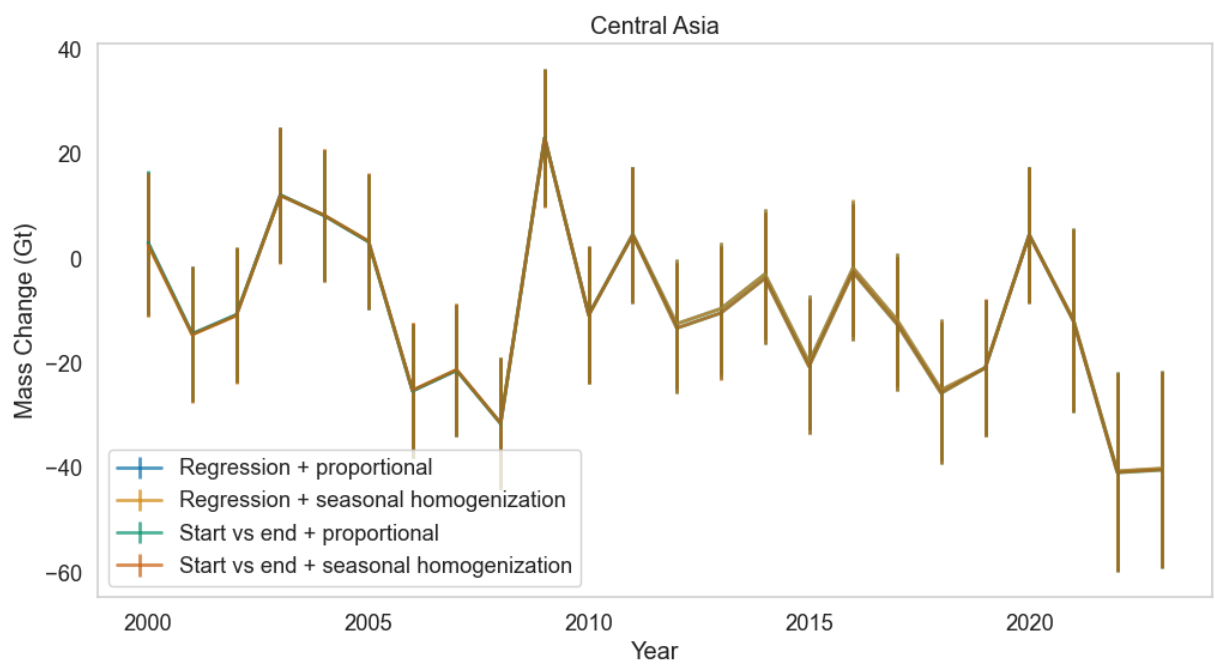
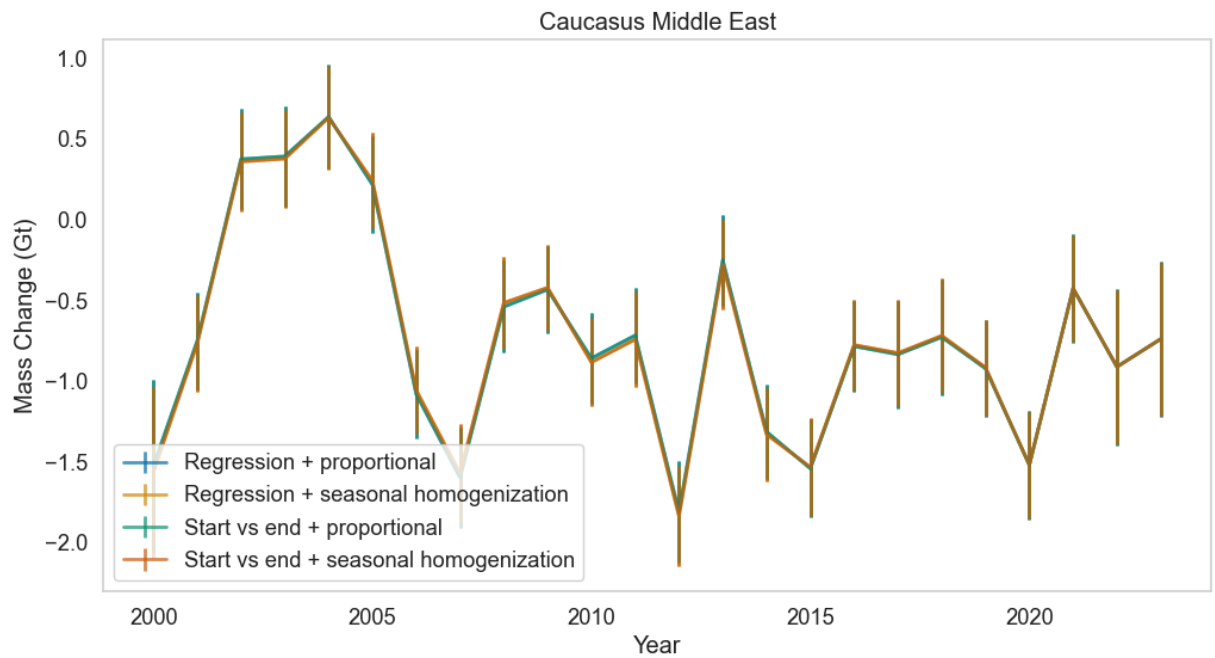
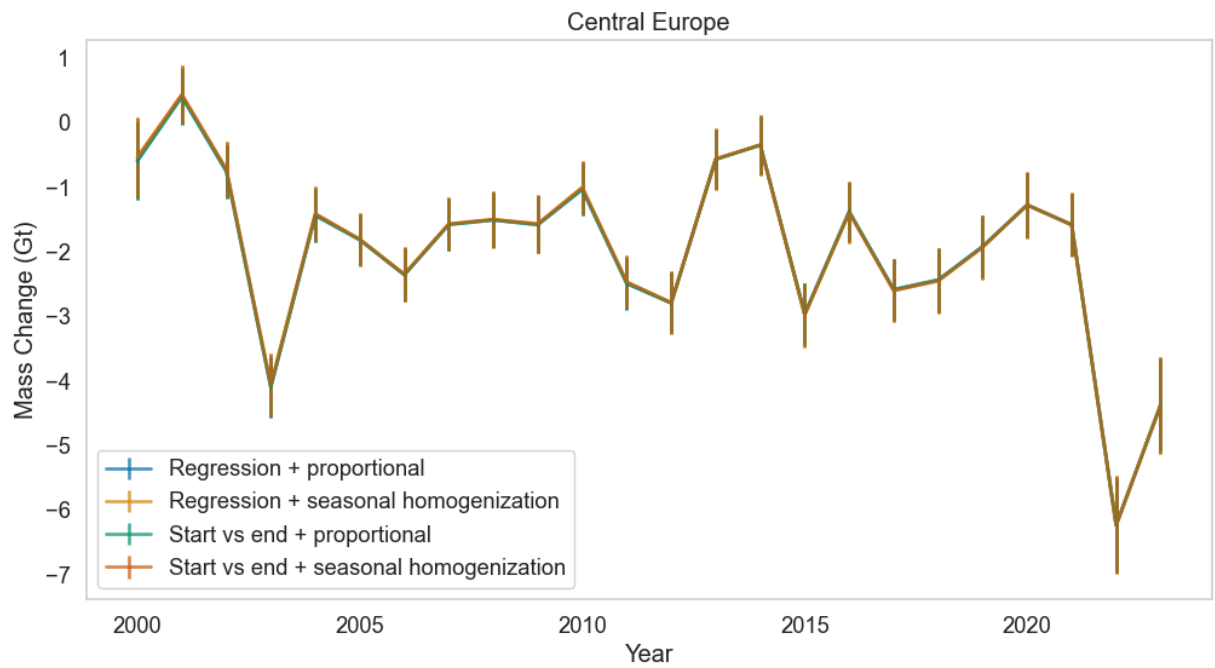


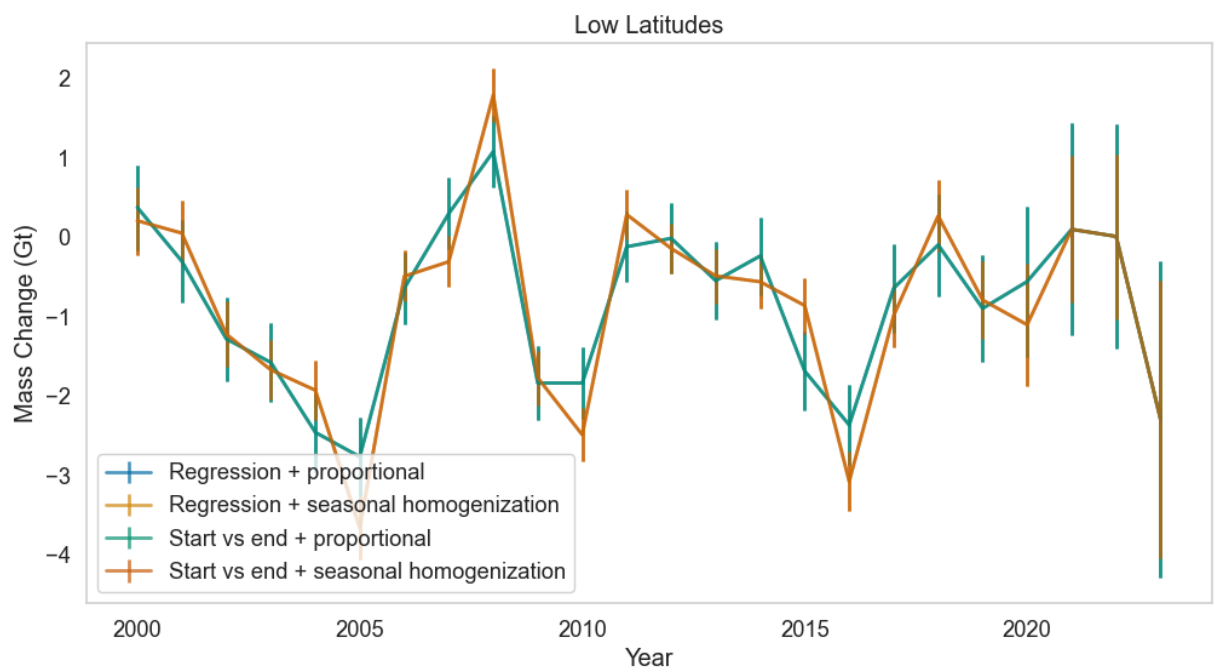
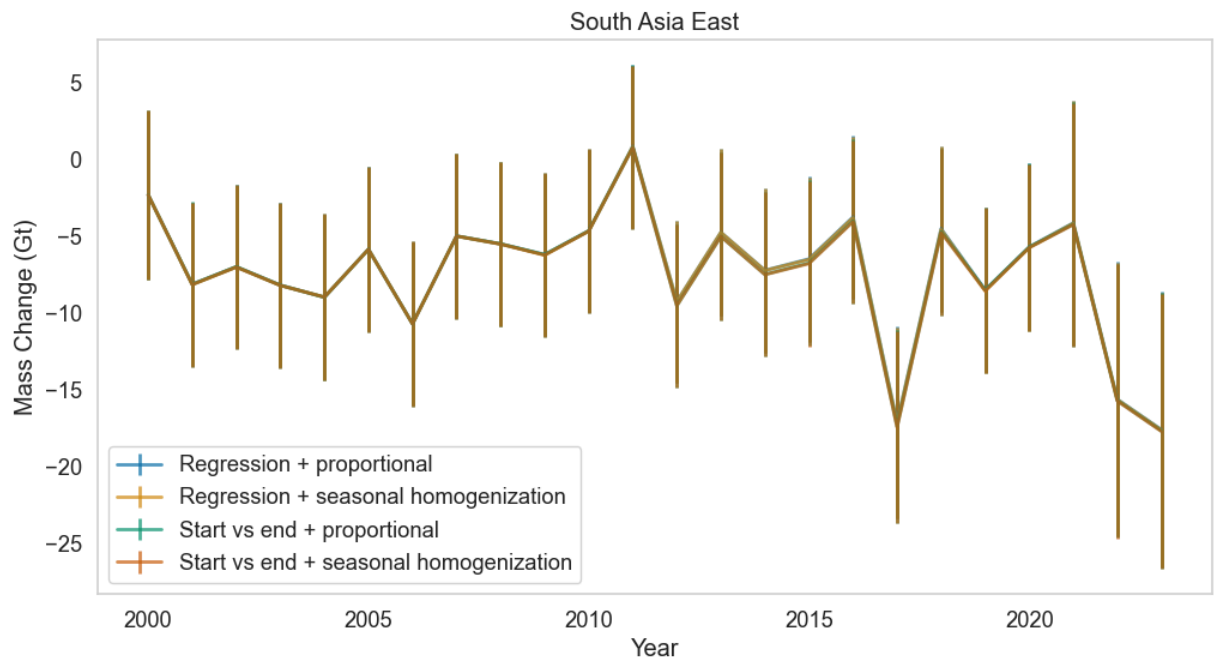
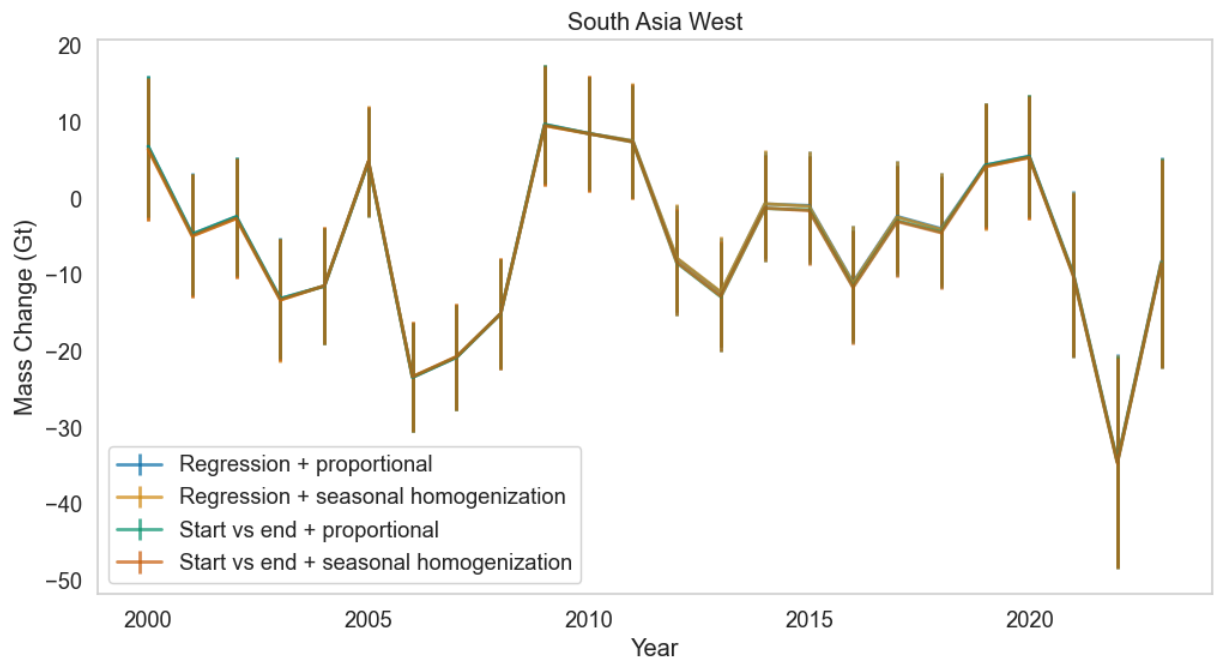
```

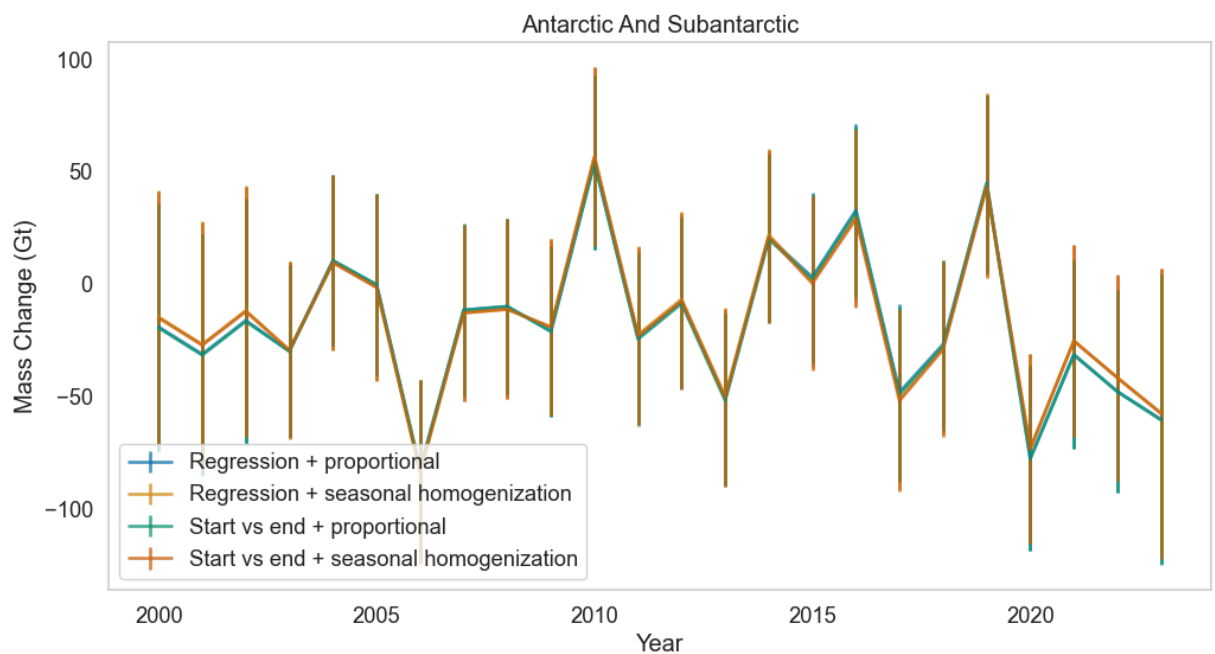
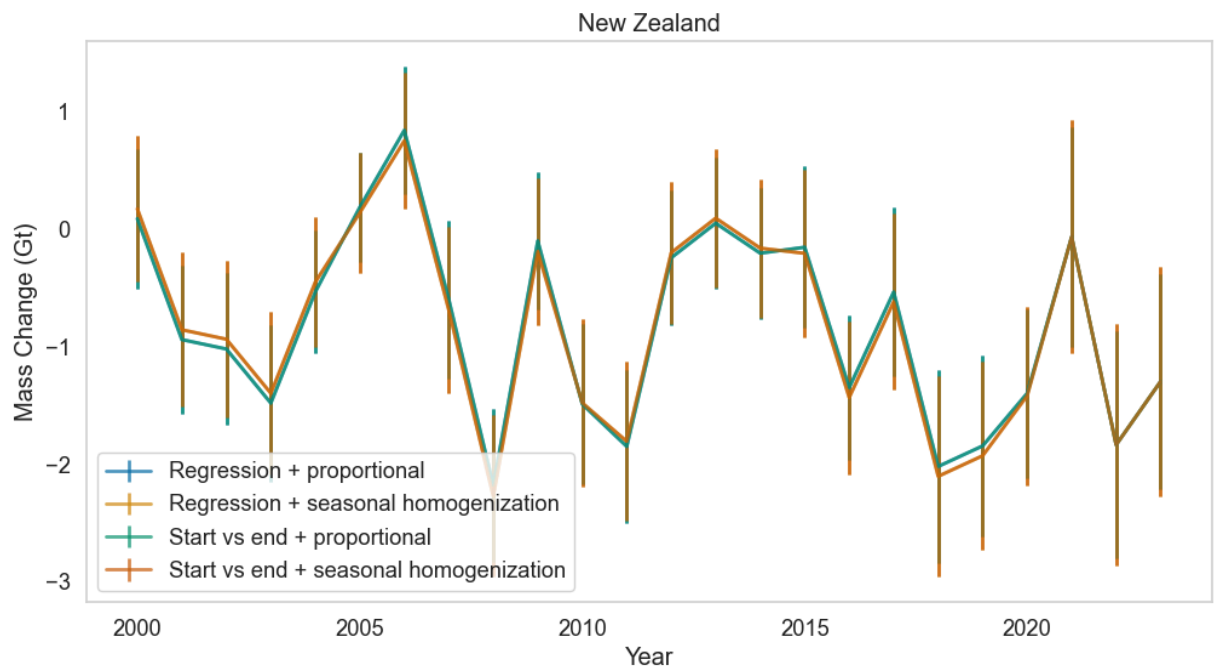
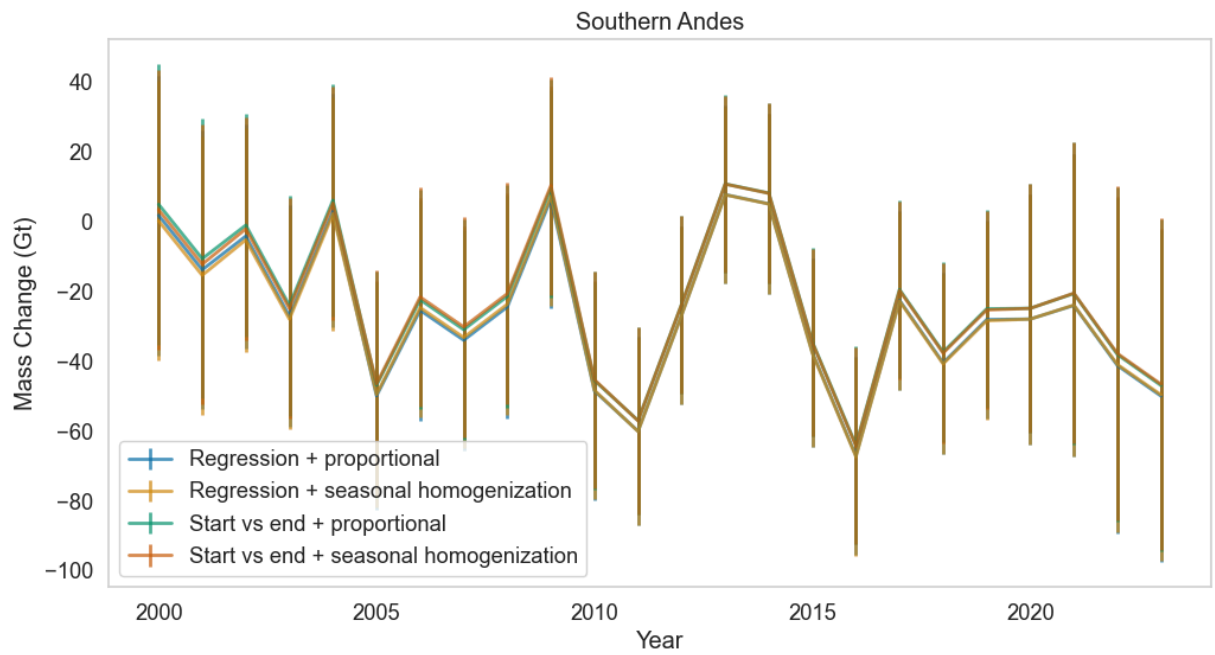
label = get_method_label(run_dir.parent.name)
color = colors_list[i % len(colors_list)]
ax.errorbar(df["start_dates"], df["combined_gt"], yerr=df.get("combined_gt_errors"),
            linewidth=2, color=color, label=label, alpha=0.7)
ax.set_title(display_name)
ax.set_xlabel("Year")
ax.set_ylabel("Mass Change (Gt)")
ax.legend(loc="lower left")
plt.show()

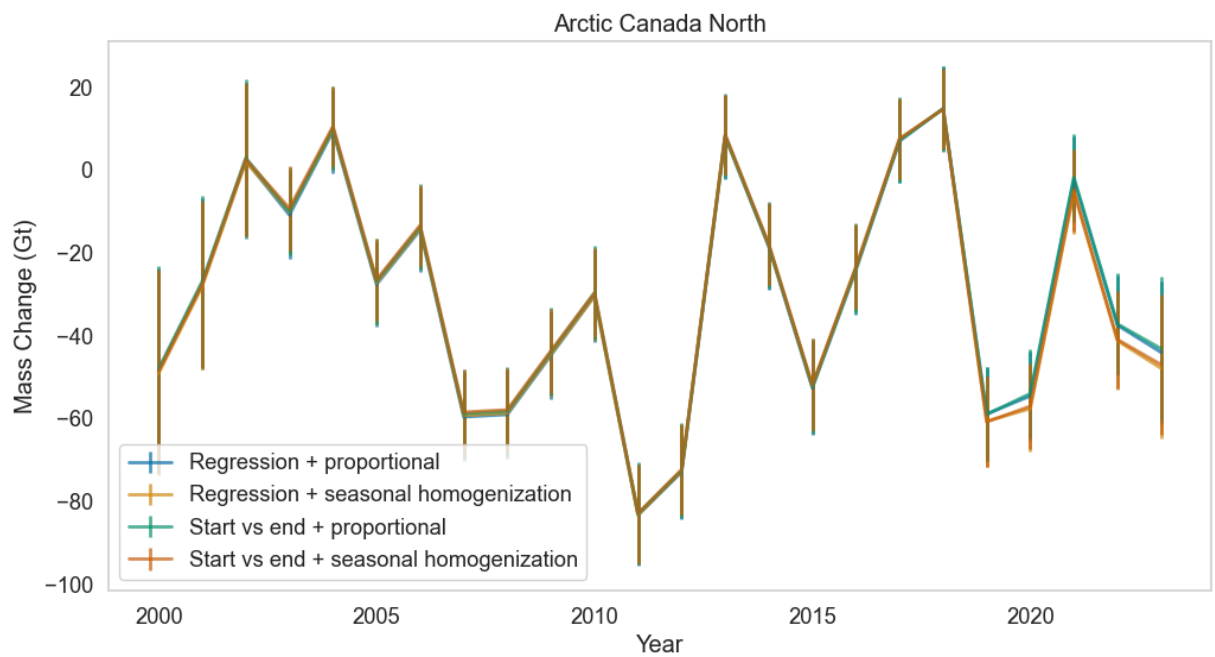
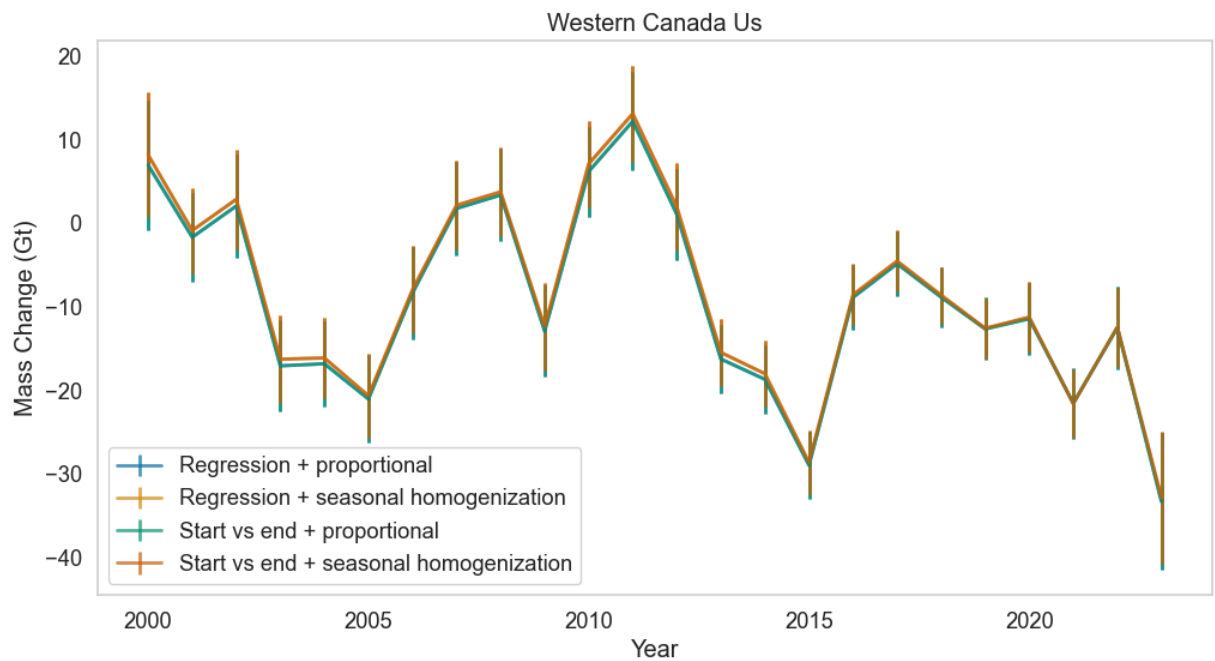
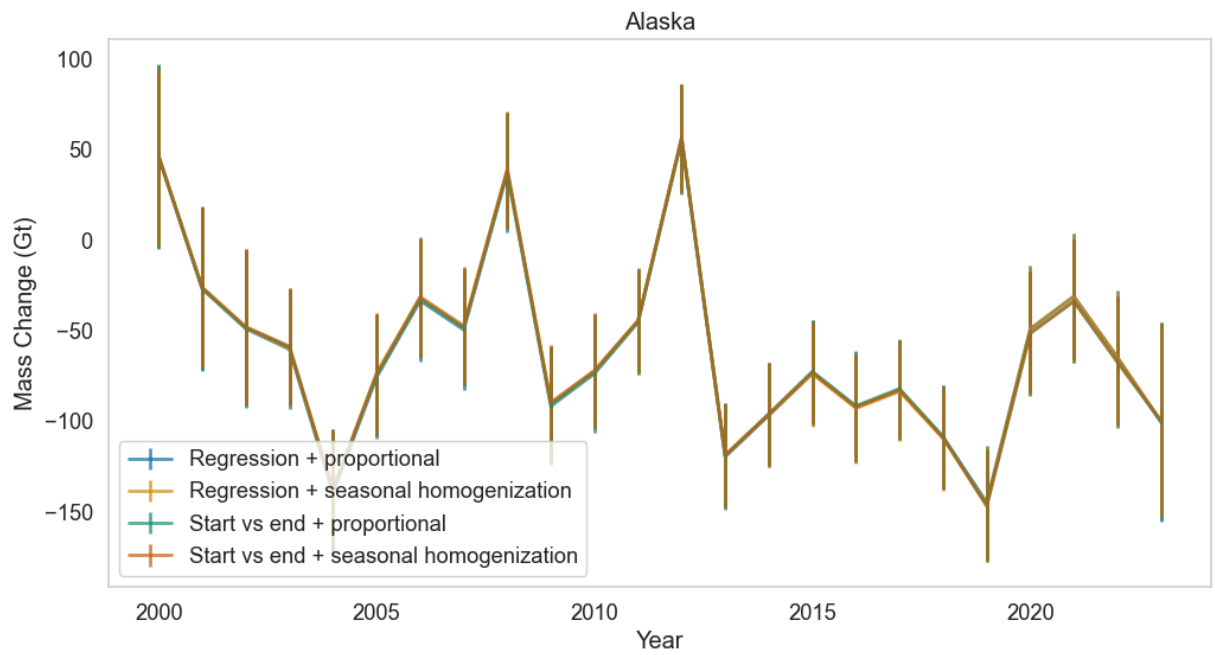
```



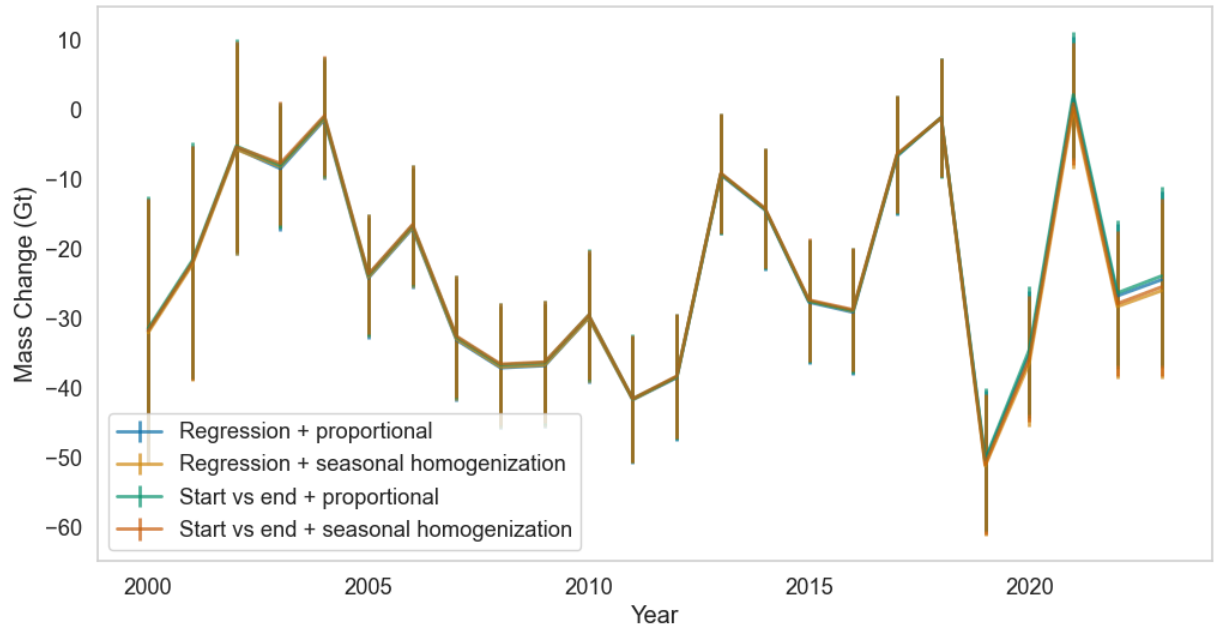




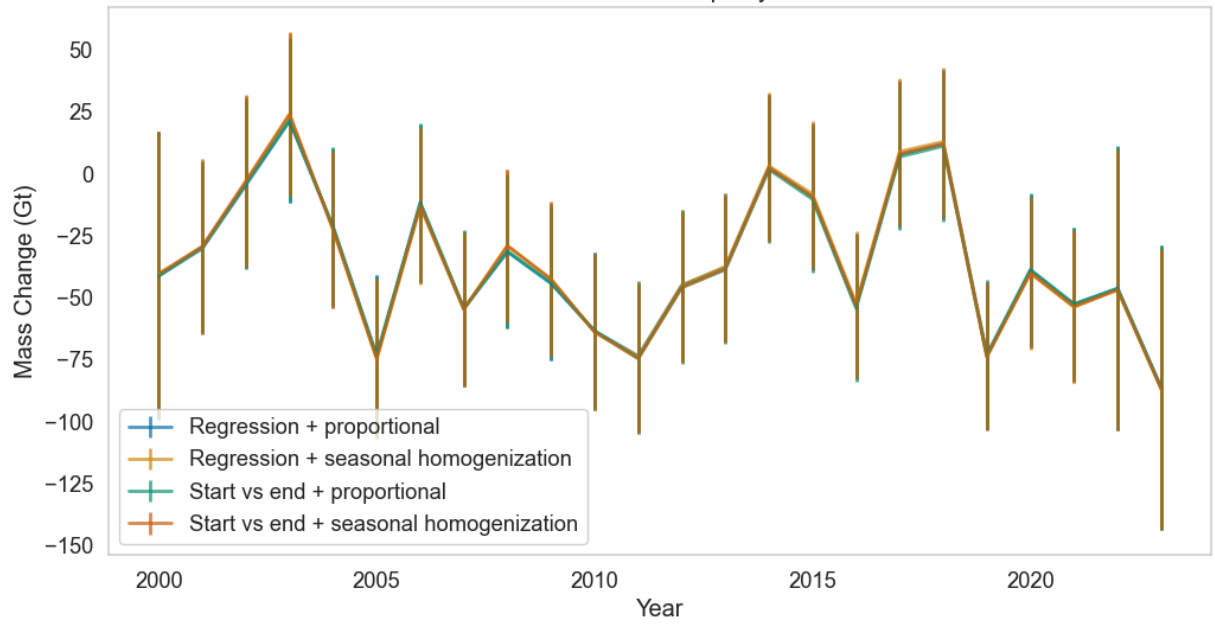




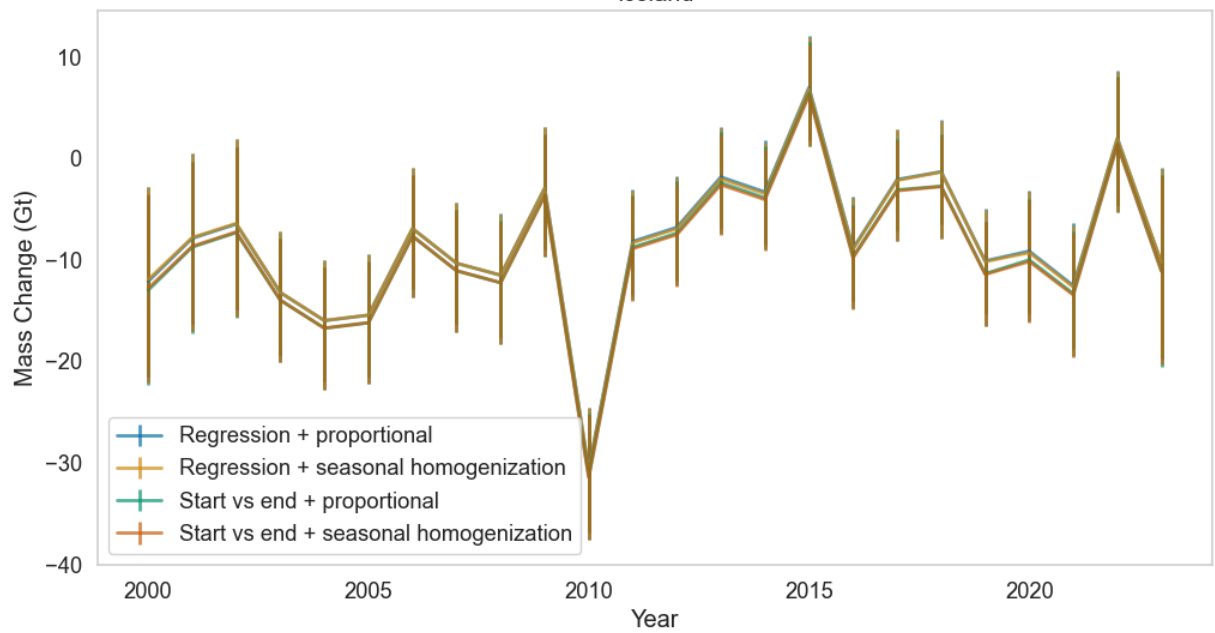
Arctic Canada South

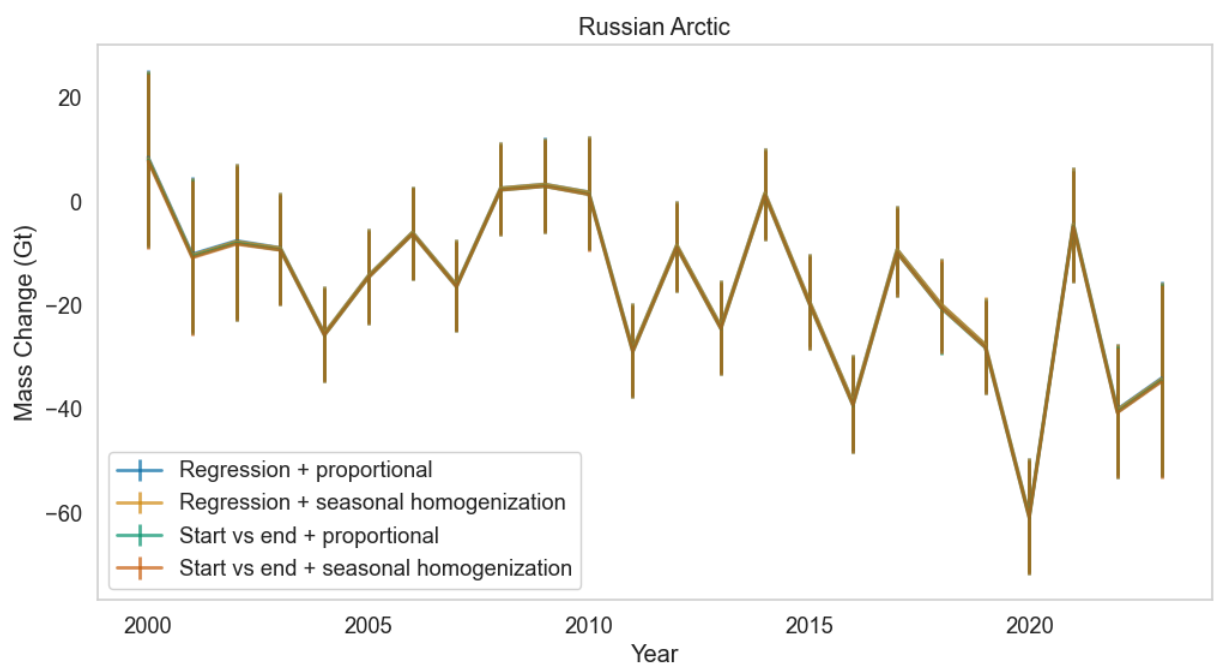
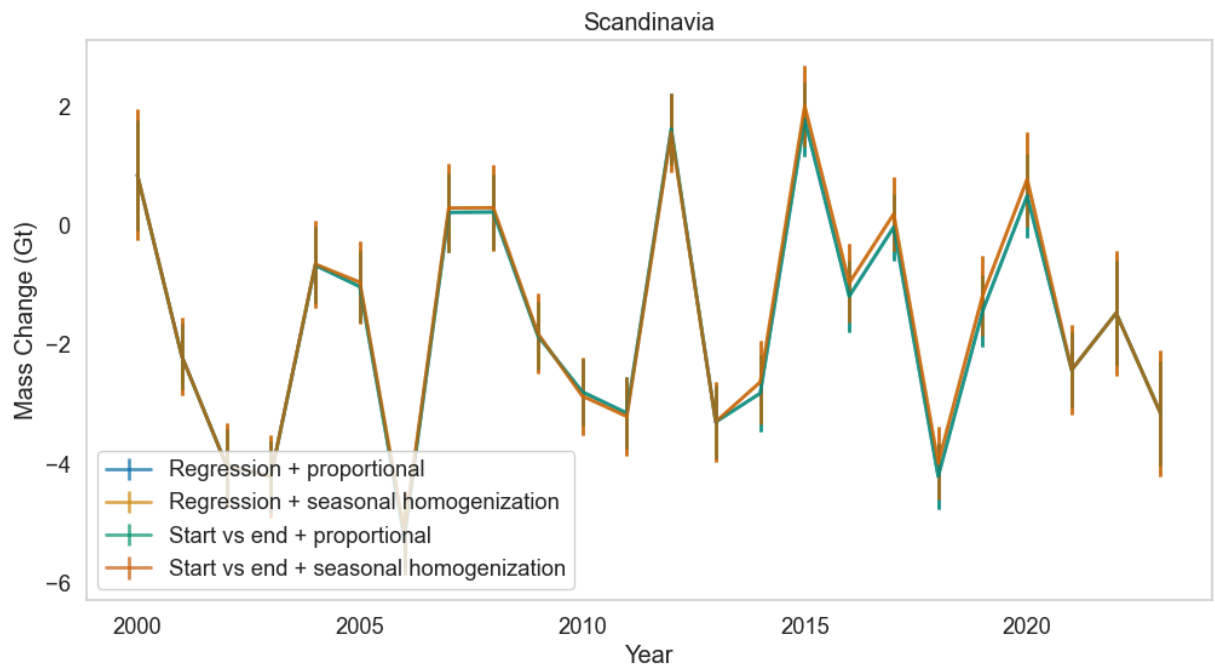
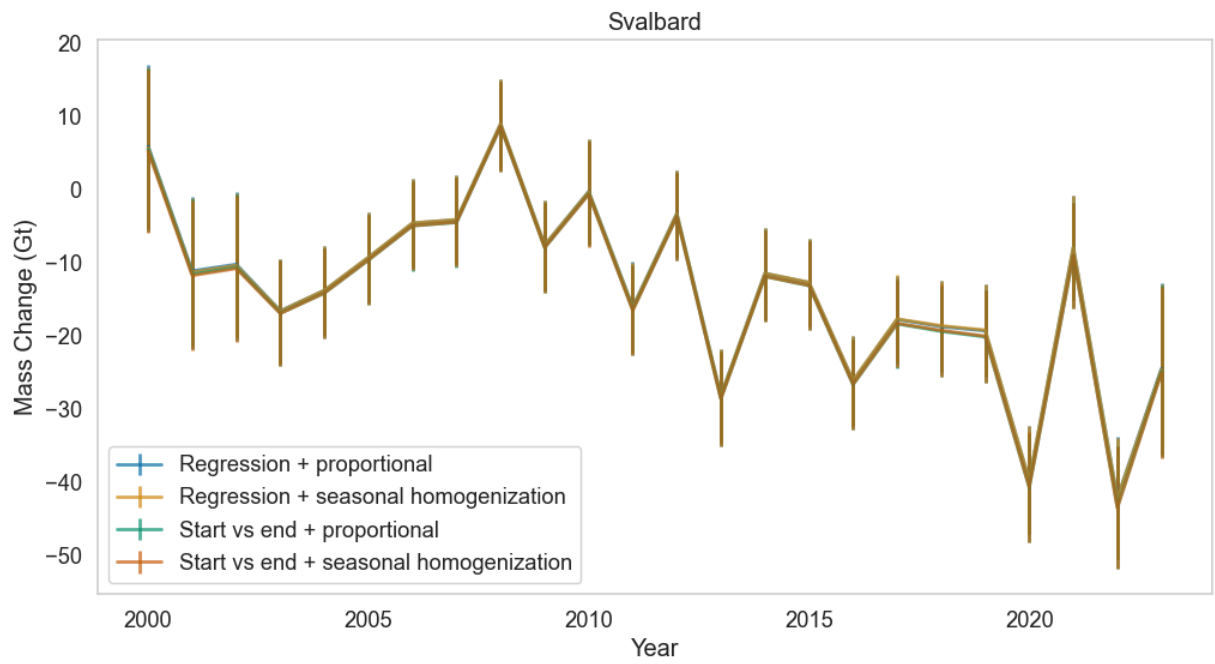


Greenland Periphery



Iceland





Annual glacier mass change (Gt) with uncertainty bands for four combinations of trend extraction and temporal homogenization methods.

```
In [5]: all_rows = []

for region_file in list_of_folders:
    display_name = region_file.split('_', 1)[-1].replace('.csv', '').replace('_', ' ').title()

    method_labels = []
    mass_change_per_method = []
    for run_dir in method_run_dirs:
        df = pd.read_csv(run_dir / region_file).set_index("start_dates")
        method_labels.append(get_method_label(run_dir.parent.name))
        mass_change_per_method.append(df["combined_gt"])

    ref_total = mass_change_per_method[0].sum()
    for i in range(len(mass_change_per_method)):
        total = mass_change_per_method[i].sum()
        corr, _ = spearmanr(mass_change_per_method[0], mass_change_per_method[i], nan_policy="omit")
        all_rows.append({
            "Region": display_name,
            "Run": method_labels[i],
            "Absolute Difference (Gt)": round(total - ref_total, 2),
            "Relative Difference (%)": round(
                ((total - ref_total) / abs(ref_total)) * 100, 2
            ) if ref_total != 0 else 0,
            "Correlation": round(corr, 4),
        })

combined_df = pd.DataFrame(all_rows).set_index(["Region", "Run"])
combined_df = combined_df.drop("Regression + proportional", level="Run")
display(combined_df)
```


		Absolute Difference (Gt)	Relative Difference (%)	Correlation
Region	Run			
Global	Regression + seasonal homogenization	26.12	0.40	0.9957
	Start vs end + proportional	31.98	0.49	0.9974
	Start vs end + seasonal homogenization	58.11	0.89	0.9913
North Asia	Regression + seasonal homogenization	0.17	0.53	0.9974
	Start vs end + proportional	0.00	0.00	1.0000
	Start vs end + seasonal homogenization	0.17	0.53	0.9974
Central Europe	Regression + seasonal homogenization	0.23	0.48	0.9983
	Start vs end + proportional	0.00	0.00	1.0000
	Start vs end + seasonal homogenization	0.23	0.48	0.9983
Caucasus Middle East	Regression + seasonal homogenization	-0.05	-0.31	0.9957
	Start vs end + proportional	0.00	0.00	1.0000
	Start vs end + seasonal homogenization	-0.05	-0.31	0.9957
Central Asia	Regression + seasonal homogenization	-0.02	-0.01	0.9991
	Start vs end + proportional	-6.66	-2.65	0.9983
	Start vs end + seasonal homogenization	-6.68	-2.65	0.9974
South Asia West	Regression + seasonal homogenization	-1.73	-1.28	1.0000
	Start vs end + proportional	-4.53	-3.37	0.9983
	Start vs end + seasonal homogenization	-6.26	-4.65	0.9983
South Asia East	Regression + seasonal homogenization	-0.45	-0.26	1.0000
	Start vs end + proportional	-2.12	-1.20	0.9983
	Start vs end + seasonal homogenization	-2.57	-1.46	0.9983
Low Latitudes	Regression + seasonal homogenization	-0.83	-4.08	0.9165
	Start vs end + proportional	0.00	0.00	1.0000
	Start vs end + seasonal homogenization	-0.83	-4.08	0.9165
Southern Andes	Regression + seasonal homogenization	-3.08	-0.49	0.9983
	Start vs end + proportional	73.79	11.70	1.0000
	Start vs end + seasonal homogenization	70.71	11.21	0.9991
New Zealand	Regression + seasonal homogenization	-0.21	-1.05	0.9887
	Start vs end + proportional	0.00	0.00	1.0000

		Absolute Difference (Gt)	Relative Difference (%)	Correlation
Region	Run			
Antarctic And Subantarctic	Start vs end + seasonal homogenization	-0.21	-1.05	0.9887
	Regression + seasonal homogenization	26.34	6.16	0.9904
	Start vs end + proportional	-9.19	-2.15	0.9991
Alaska	Start vs end + seasonal homogenization	17.15	4.01	0.9904
	Regression + seasonal homogenization	5.75	0.39	0.9965
	Start vs end + proportional	8.38	0.57	0.9983
Western Canada Us	Start vs end + seasonal homogenization	14.14	0.96	0.9957
	Regression + seasonal homogenization	12.56	5.66	0.9965
	Start vs end + proportional	0.00	0.00	1.0000
Arctic Canada North	Start vs end + seasonal homogenization	12.56	5.66	0.9965
	Regression + seasonal homogenization	-12.66	-1.73	0.9957
	Start vs end + proportional	12.99	1.78	0.9965
Arctic Canada South	Start vs end + seasonal homogenization	0.33	0.05	0.9957
	Regression + seasonal homogenization	-5.20	-0.94	0.9991
	Start vs end + proportional	7.87	1.43	0.9991
Greenland Periphery	Start vs end + seasonal homogenization	2.67	0.48	0.9983
	Regression + seasonal homogenization	5.50	0.65	0.9974
	Start vs end + proportional	-5.32	-0.63	0.9974
Iceland	Start vs end + seasonal homogenization	0.18	0.02	0.9983
	Regression + seasonal homogenization	-1.02	-0.51	0.9991
	Start vs end + proportional	-19.03	-9.56	0.9957
Svalbard	Start vs end + seasonal homogenization	-20.05	-10.07	0.9957
	Regression + seasonal homogenization	-0.33	-0.10	1.0000
	Start vs end + proportional	-12.44	-3.76	1.0000
Scandinavia	Start vs end + seasonal homogenization	-12.77	-3.86	1.0000
	Regression + seasonal homogenization	1.79	4.47	0.9974
	Start vs end + proportional	0.00	0.00	1.0000
	Start vs end + seasonal homogenization	1.79	4.47	0.9974

		Absolute Difference (Gt)	Relative Difference (%)	Correlation
Region	Run			
Russian Arctic	Regression + seasonal homogenization	-0.64	-0.17	1.0000
	Start vs end + proportional	-11.75	-3.06	1.0000
	Start vs end + seasonal homogenization	-12.39	-3.22	1.0000

Absolute difference (Gt), relative difference (%), and Spearman correlation of annual mass changes for three alternative processing method combinations relative to the reference run (regression + proportional), for selected regions.

These methodological choices have a limited effect on the global total mass change: the largest difference from the reference (regression + proportional) is about 58 Gt for "start_vs_end + seasonal_homogenization", with a relative difference 0.89% (compared to total change of -6,542 +/- 387 Gt over the modelled time period). The pearman correlation between the annual change time series and the reference remains high (> 0.99) for all runs, confirming that the temporal pattern is very similar across methods. Results are therefore robust to the choice of trend removal and homogenization at the global level.

At the regional level, method choice matters more, with relative differences ranging from about -10% (Iceland) to +12% (Southern Andes). Not all regions are affected by trend extration, likely due to missing interannual datasets. The effect of temporal homogenization is generally smaller but present in all regions.

Overall, we consider the authors' chosen combination - regression for trend extraction and proportional for shifting to annual boundaries - to be a reasonable default. For trend extraction, the regression method is more robust, as it is less sensitive to anomalies at the start or end of a time series.

For temporal homogenization, it is not clear to us which method is superior, and this requires further investigation. The seasonal homogenization method is theoretically preferable where strong seasonal mass balance cycles exist, as it explicitly models intra-annual variability via a sine-wave calibration series rather than assuming uniform change (Zemp and Welty, 2023). However, its advantage over the proportional method could depend on the quality of the monthly calibration data available per region. For regions with sparse seasonal observations, the proportional method may in practice be equally or more reliable. Future work should evaluate both methods to determine where the additional complexity of seasonal homogenization yields a meaningful improvement in accuracy. While these approaches matter on a regional level - and thus so some extent show uncertainty in the estimates - they do not fundamentally change the global results.

2. Relative change

Zemp et al. (2025) report an increase in mass change. Looking only at the absolute mass change, as done in the paper, could understate the severity of ice loss. This could be especially relevant for regions where glaciers have lost a large fraction of their mass: as the glaciers shrink, the same absolute mass loss can correspond to a larger percentage loss per year, i.e. an acceleration in relative terms. We want to see whether glacier mass loss is getting worse than absolute numbers suggest. Thus, we compute the relative change as comparison to absolute change.

We use the glacier mass in 2000, as reported by Zemp et al. (2025) as the baseline and compute the annual relative change as a percentage of the remaining mass each year. Additionally, we propagate uncertainty.

```
In [6]: # Load glacier mass data
mass_2000 = pd.read_csv('capstone_glambie/input/glambie_reference/glacier_mass_2000.csv', sep=';')
mass_2000 = mass_2000.set_index('File')
results = []

for csv_path in sorted(Path(reference_run_dir).glob("*.csv")):
    filename = csv_path.name

    region = mass_2000.loc[filename, 'Region']
    current_mass = mass_2000.loc[filename, 'Mass']
    current_mass_error = mass_2000.loc[filename, 'Error']
```

```

df = pd.read_csv(csv_path).sort_values('start_dates').reset_index(drop=True)

for _, row in df.iterrows():
    annual_change = row['combined_gt']
    annual_change_error = row['combined_gt_errors']

    # Relative change: fraction of annual change compared to current total mass
    relative_change = annual_change / current_mass

    # Error propagation in quadrature (see Taylor, 1997)
    # Assuming independence as done by Zemp et al. (2025)
    relative_change_error = abs(relative_change) * np.sqrt(
        (annual_change_error / annual_change) ** 2 + # uncertainty contributed by the annual change
        (current_mass_error / current_mass) ** 2 # uncertainty contributed by glacier mass
    )

    results.append({
        'region': region,
        'year_start': int(row['start_dates']),
        'year_end': int(row['end_dates']),
        'glacier_mass_at_start': current_mass,
        'glacier_mass_error_at_start': current_mass_error,
        'glacier_mass_at_end': current_mass + annual_change,
        'annual_change_gt': annual_change,
        'annual_change_error': annual_change_error,
        'relative_change_pct': relative_change * 100,
        'relative_change_error_pct': relative_change_error * 100
    })

    current_mass = current_mass + annual_change
    current_mass_error = np.sqrt(current_mass_error**2 + annual_change_error**2)

results_df = pd.DataFrame(results)

```

```

In [7]: # Define colors from the established palette
blue = colors_list[0]
orange = colors_list[1]

# Plotting only a few regions as examples
target_regions = ["Global", "Alaska", "Arctic Canada north", "Southern Andes",
                  "Iceland", "Caucasus and Middle East", "Central Europe",
                  "Russian Arctic", "Antarctic and subantarctic islands"]

for region in target_regions:
    region_data = results_df[results_df['region'] == region].sort_values('year_start')

    # Glacier Mass Data
    x_mass = region_data['year_start'].values
    y_mass = region_data['glacier_mass_at_start'].values
    y_err_mass = region_data['glacier_mass_error_at_start'].values

    # Relative Change Data
    x_rel = region_data['year_start'].values
    y_rel = region_data['relative_change_pct'].values
    y_err_rel = region_data['relative_change_error_pct'].values

    # Create a figure with two subplots
    fig, (ax_top, ax_bottom) = plt.subplots(nrows=2, figsize=(10, 8), sharex=True)

    # Upper subplot: Relative change (%) + twin for Absolute change (Gt)
    ax_top.plot(
        x_rel, y_rel, marker='o',
        label='Relative Mass Change', alpha=0.8, linewidth=2, color=blue)
    ax_top.fill_between(x_rel, y_rel - y_err_rel, y_rel + y_err_rel, alpha=0.1, color=blue)
    ax_top.set_ylabel('Relative Mass Change (%)', fontsize=14)
    ax_top.set_title(f'{region}')
    ax_top.grid(True, alpha=0.3)

    # Twin axis for absolute change (Gt)
    ax_top_twin = ax_top.twinx()
    ax_top_twin.plot(region_data['year_start'], region_data['annual_change_gt'],
                     marker='s', label='Absolute Mass Change', alpha=0.6, linewidth=2, color=orange)
    ax_top_twin.fill_between(region_data['year_start'],
                             region_data['annual_change_gt'] - region_data['annual_change_error'],
                             region_data['annual_change_gt'] + region_data['annual_change_error'],

```

```

        alpha=0.1, color=orange)
ax_top_twin.set_ylabel('Mass Change (Gt)', fontsize=14)

# Symmetric y-limits and zero lines for upper panel
max_rel = max(
    abs(region_data['relative_change_pct'].min()),
    abs(region_data['relative_change_pct'].max()))
max_abs = max(
    abs(region_data['annual_change_gt'].min()),
    abs(region_data['annual_change_gt'].max()))
ax_top.set_ylim(-max_rel * 1.55, max_rel * 1.4)
ax_top_twin.set_ylim(-max_abs * 1.55, max_abs * 1.4)
ax_top.axhline(y=0, color='k', linestyle='--', alpha=0.5)
ax_top_twin.axhline(y=0, color='k', linestyle='--', alpha=0.5)

h1, l1 = ax_top.get_legend_handles_labels()
h2, l2 = ax_top_twin.get_legend_handles_labels()
ax_top.legend(h1 + h2, l1 + l2, loc="upper right", frameon=True, fontsize=13)

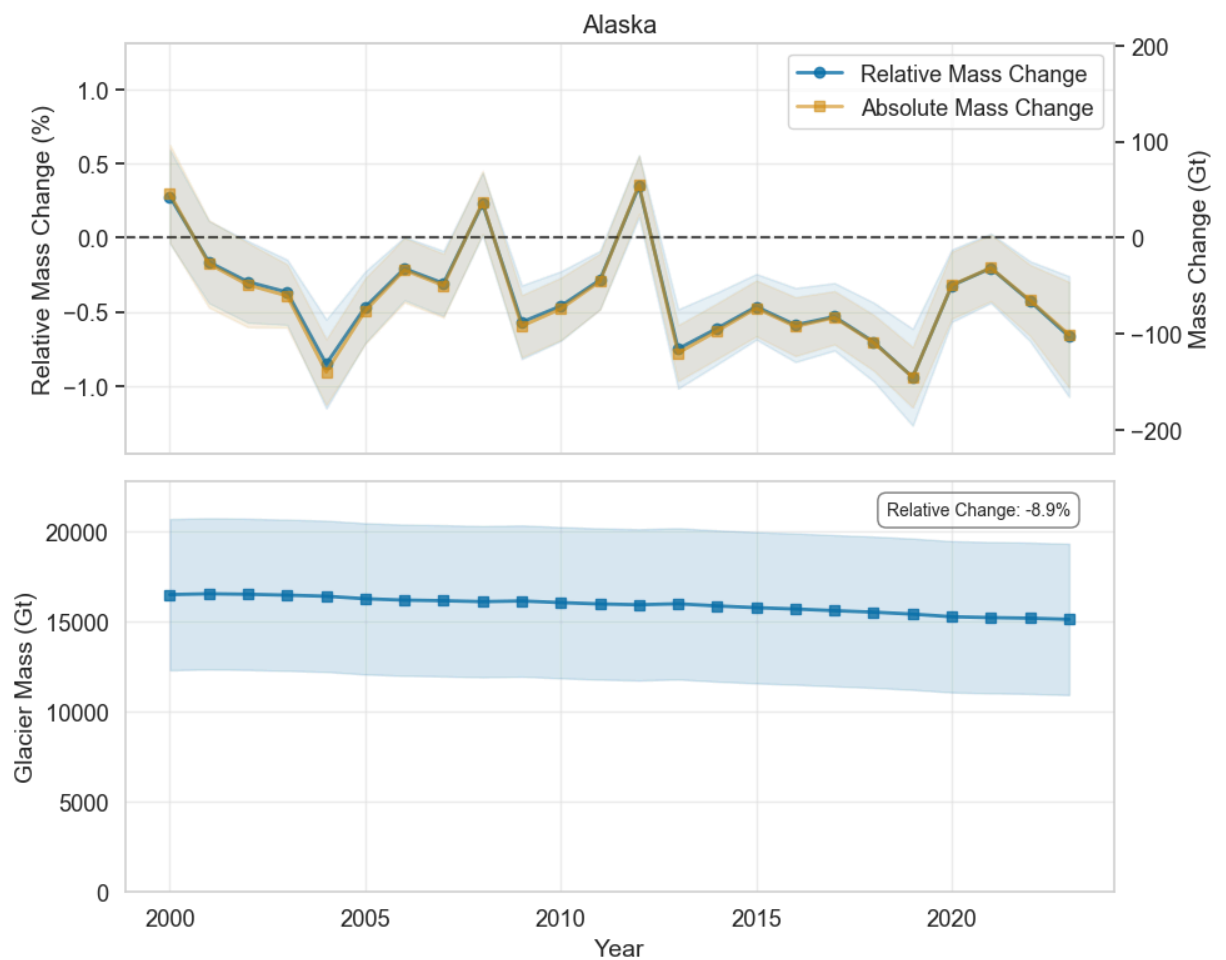
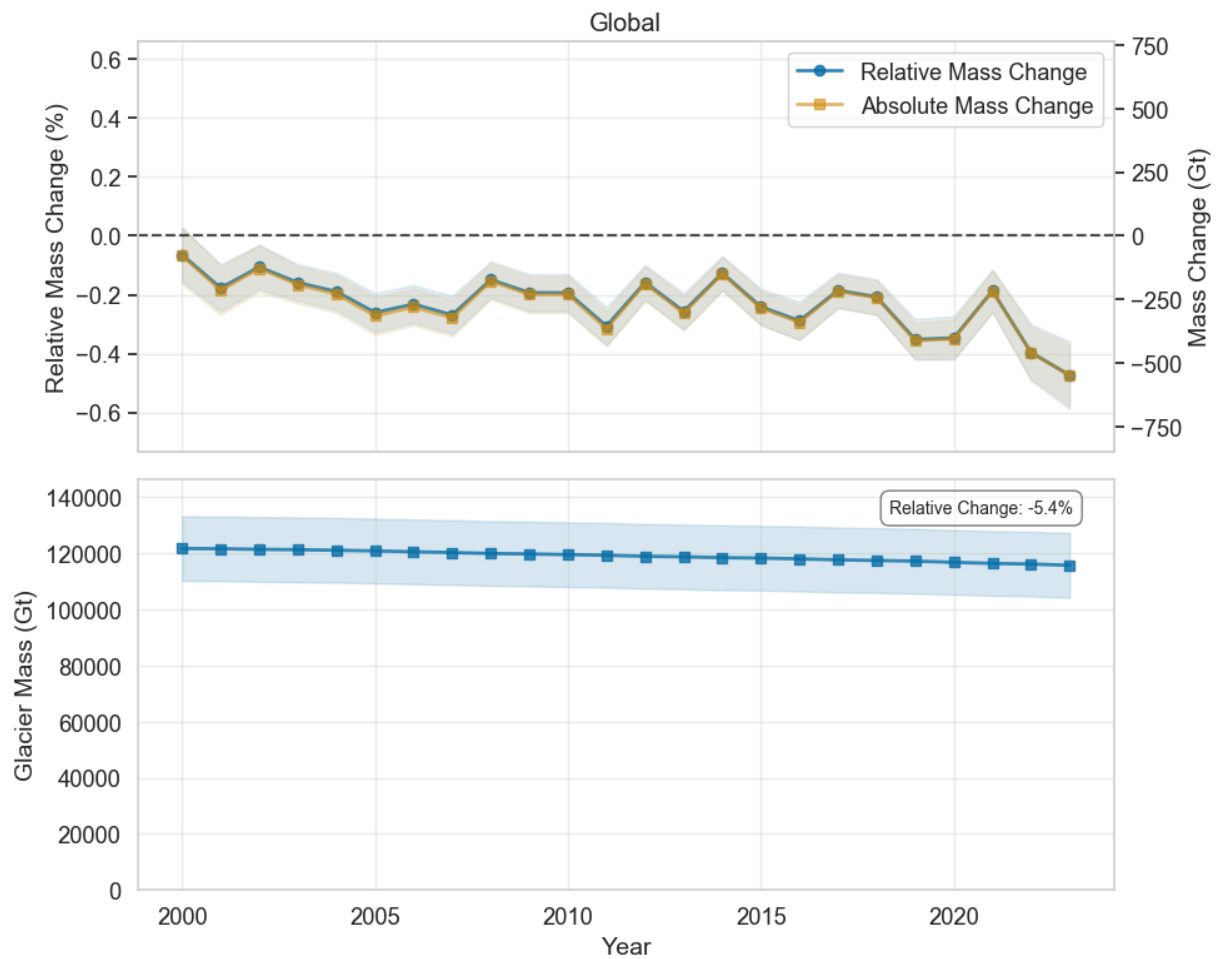
# Lower subplot: Glacier Mass (Gt)
ax_bottom.plot(x_mass, y_mass, marker='s', linestyle='-', alpha=0.8, linewidth=2, color=blue)
ax_bottom.fill_between(x_mass, y_mass - y_err_mass, y_mass + y_err_mass, alpha=0.15, color=blue)
y_upper_mass = np.nanmax(y_mass + y_err_mass)
ax_bottom.set_ylim(0, y_upper_mass * 1.1)

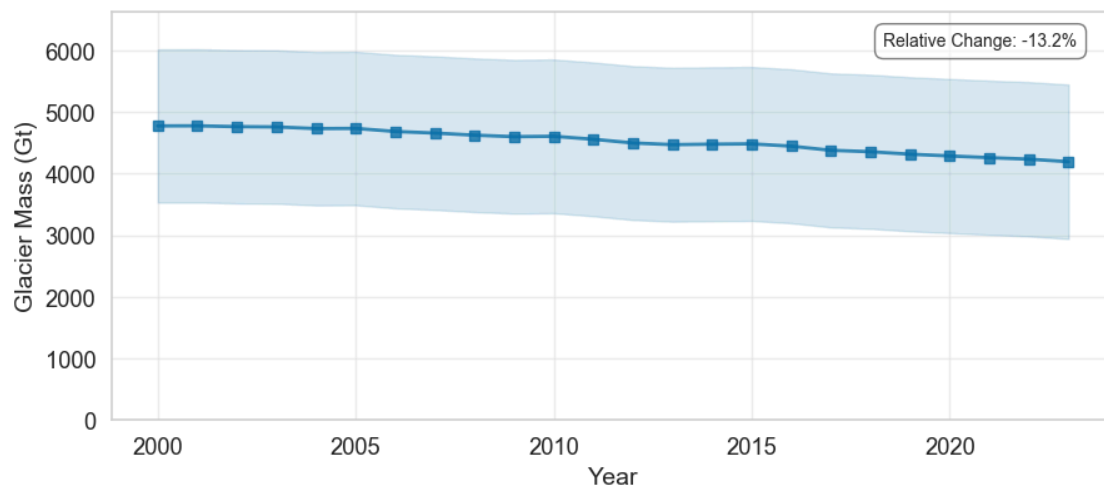
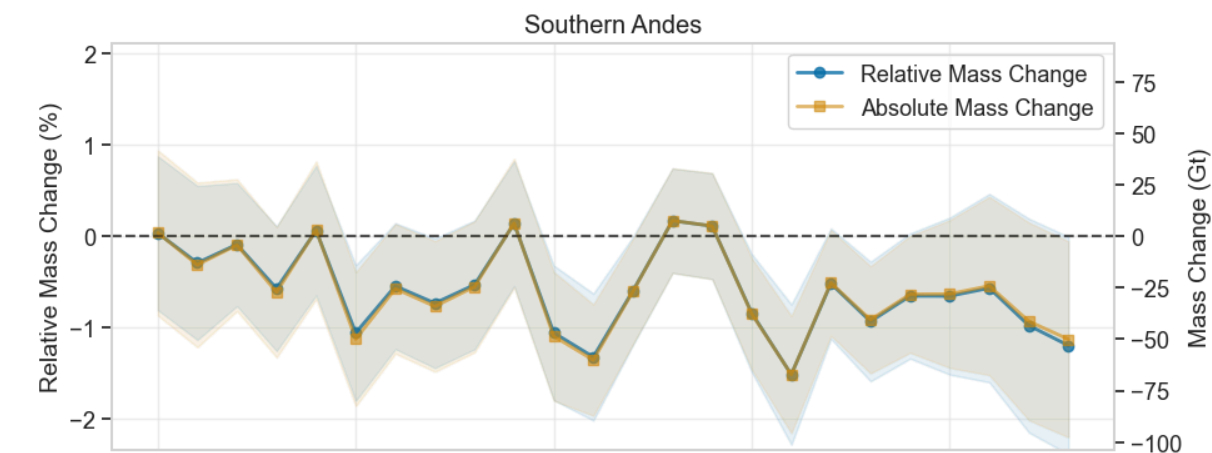
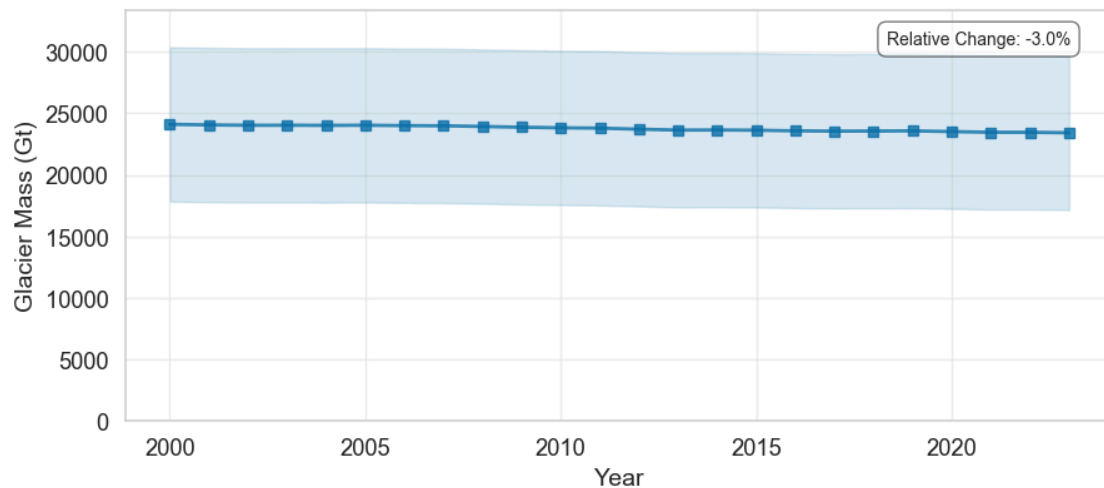
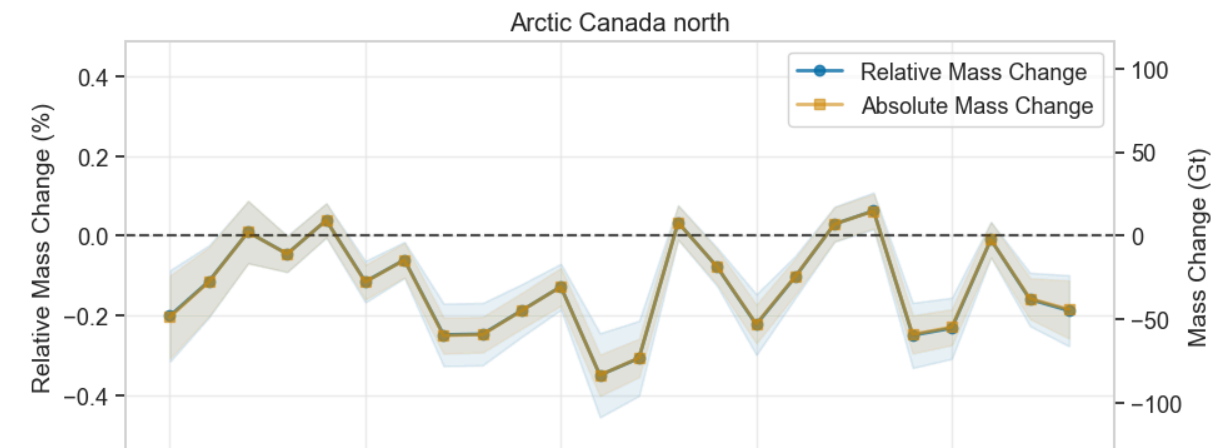
start_mass = region_data.iloc[0]['glacier_mass_at_start']
end_mass = region_data.iloc[-1]['glacier_mass_at_end']
overall_relative_change = ((end_mass - start_mass) / start_mass) * 100

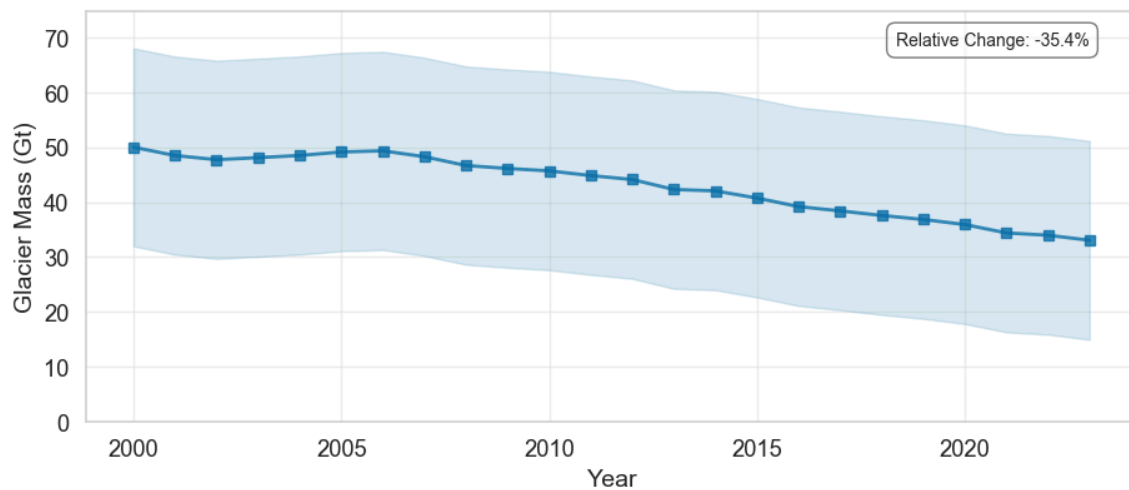
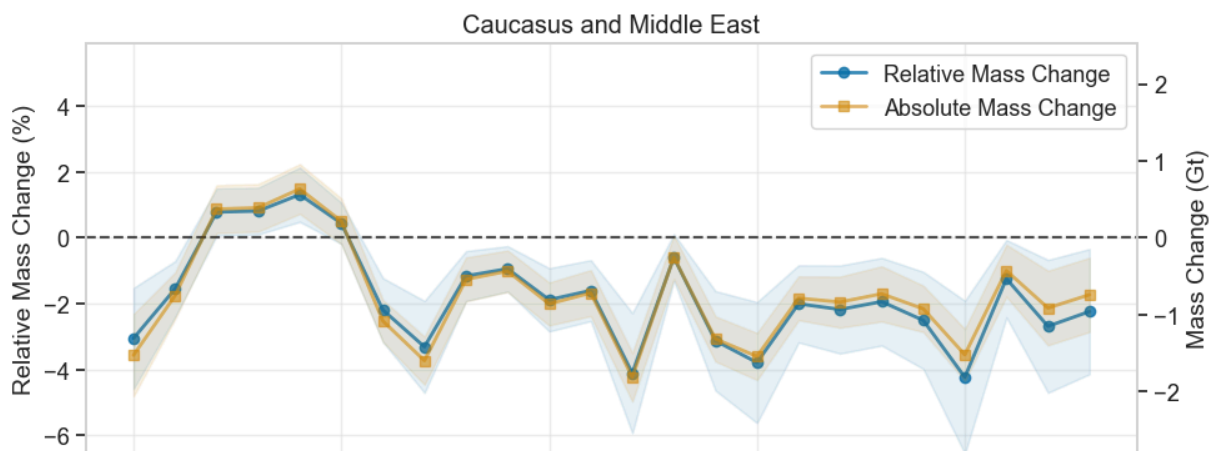
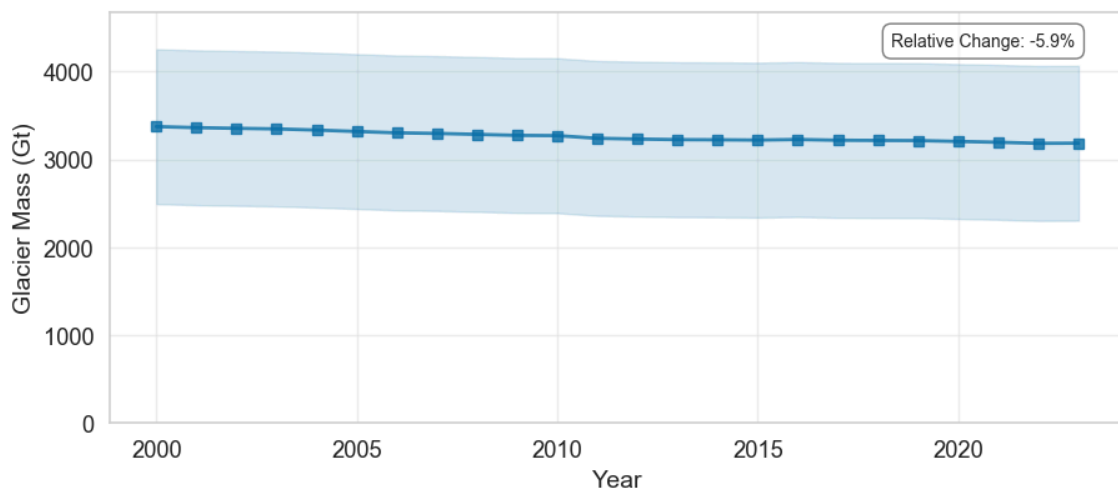
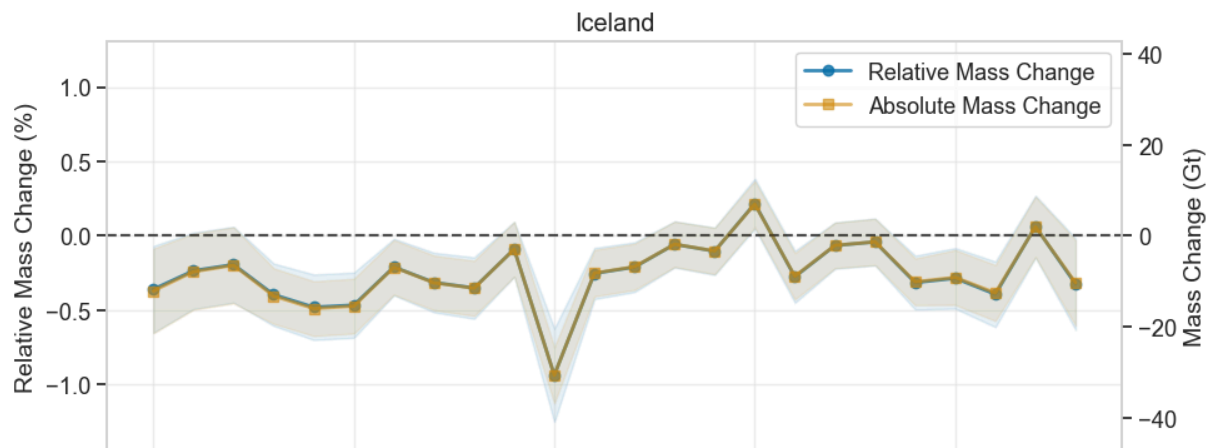
ax_bottom.text(0.77, 0.95, f"Relative Change: {overall_relative_change:.1f}%",
    transform=ax_bottom.transAxes, fontsize=10, verticalalignment='top',
    bbox=dict(
        boxstyle='round,pad=0.5',
        facecolor='white',
        edgecolor='black',
        linewidth=1,
        alpha=0.5))

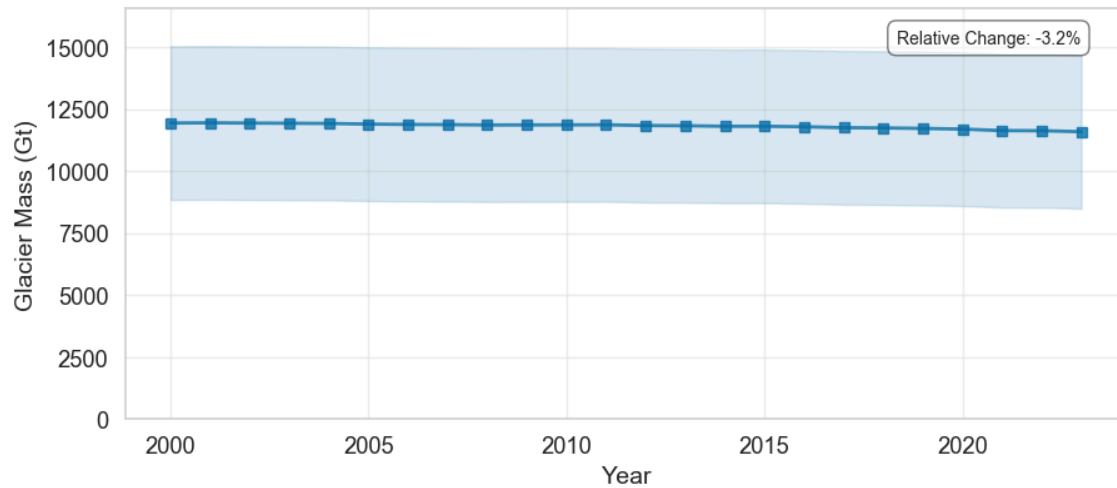
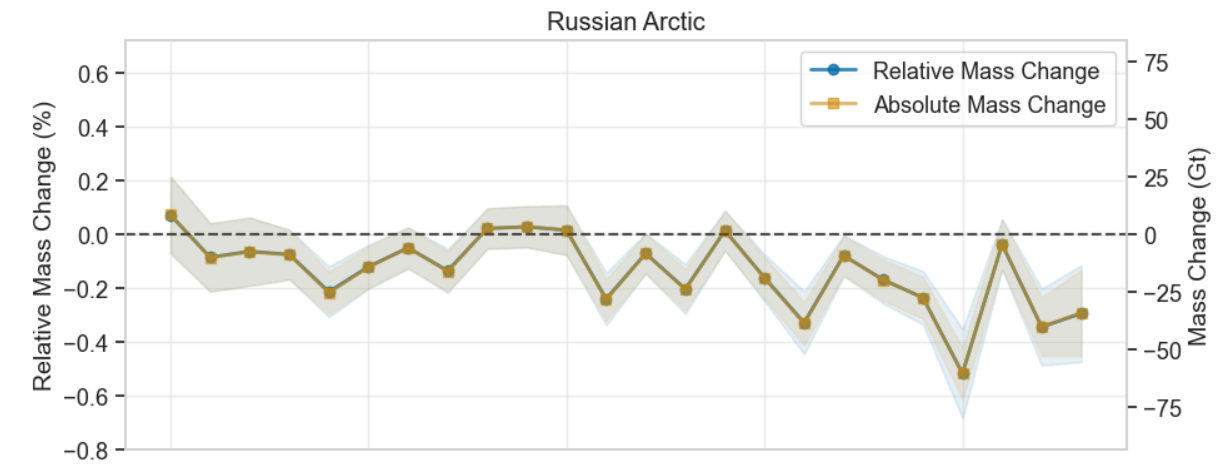
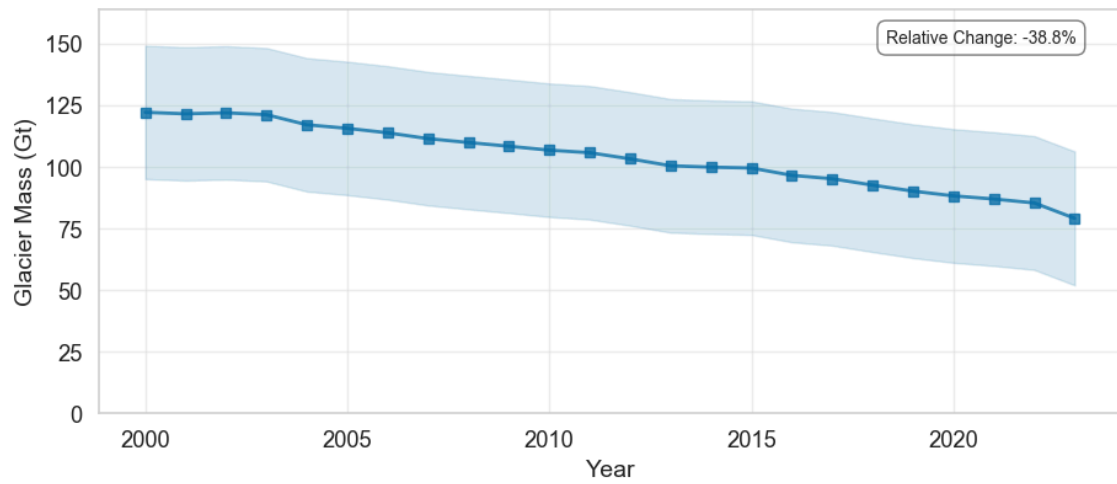
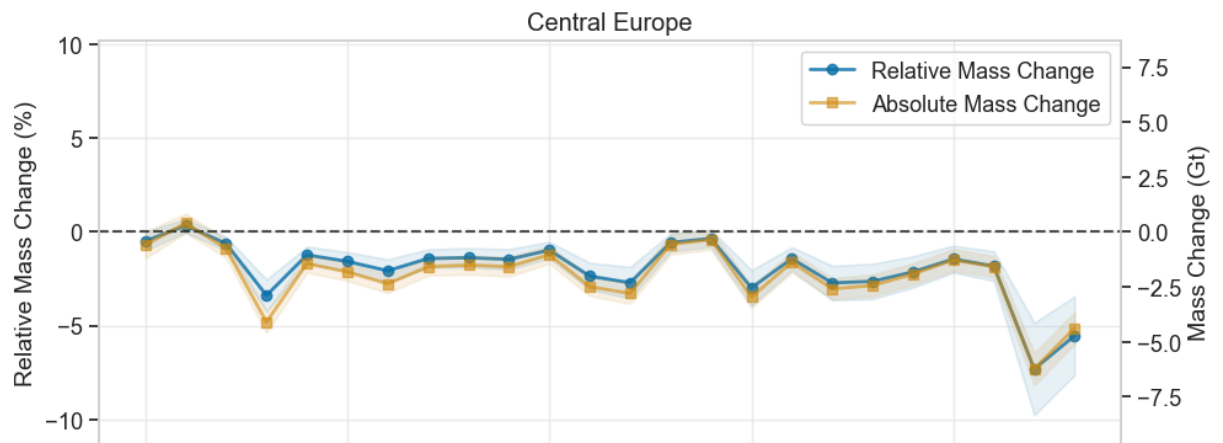
ax_bottom.set_ylabel('Glacier Mass (Gt)', fontsize=14)
ax_bottom.set_xlabel('Year', fontsize=14)
ax_bottom.grid(alpha=0.3)
plt.tight_layout()
plt.show()

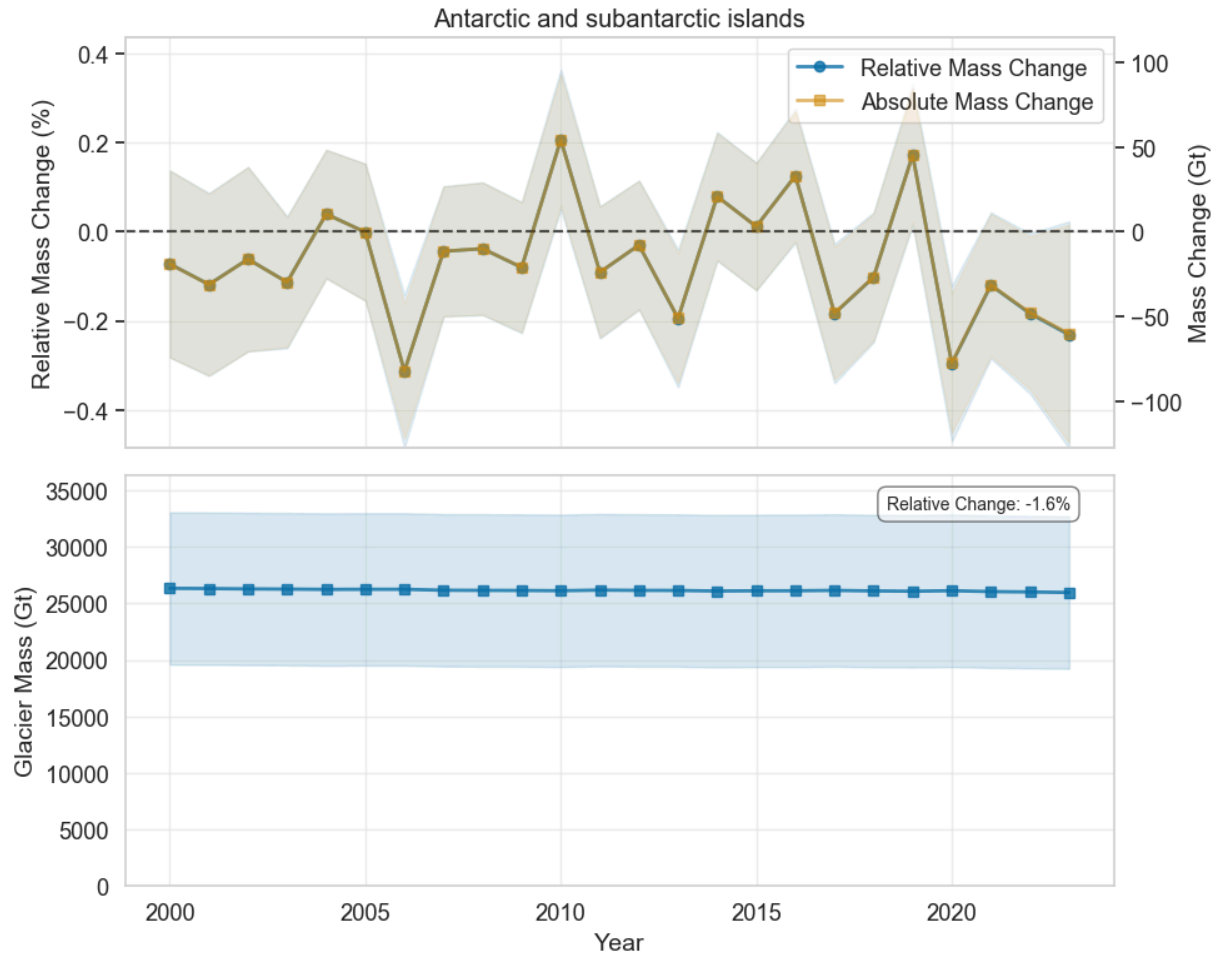
```











Top: Annual relative mass change (% left axis) and annual absolute mass change (Gt, right axis) for 2000–2023. Bottom: Time series of total glacier mass (Gt) showing the cumulative depletion over the study period.

We also see that with our calculations we reproduce the values of relative change as stated by Zemp et al. (2025).

```
In [8]: # Mann-Kendall trend tests for relative and absolute change
all_regions = sorted(results_df['region'].unique())

# Put Global first, then other regions
global_first = [
    r for r in all_regions if r.lower() == 'global'
] + [r for r in all_regions if r.lower() != 'global']

mk_summary = []
for region in global_first:
    region_data = results_df[results_df['region'] == region].sort_values('year_start')

    # Calculate overall change
    start_mass = region_data.iloc[0]['glacier_mass_at_start']
    end_mass = region_data.iloc[-1]['glacier_mass_at_end']
    overall_change_pct = ((end_mass - start_mass) / start_mass) * 100

    rel_result = mk.original_test(region_data['relative_change_pct'])
    abs_result = mk.original_test(region_data['annual_change_gt'])

    mk_summary.append({
        'Region': region,
        'Absolute change p-value': round(abs_result.p, 3),
        'Absolute change trend': abs_result.trend,
        'Relative change p-value': round(rel_result.p, 3),
        'Relative change trend': rel_result.trend,
        'Overall change (%)': round(overall_change_pct, 1)
    })

pd.DataFrame(mk_summary)
```

Out[8]:

	Region	Absolute change p-value	Absolute change trend	Relative change p-value	Relative change trend	Overall change (%)
0	Global	0.002	decreasing	0.002	decreasing	-5.4
1	Alaska	0.078	no trend	0.063	no trend	-8.9
2	Antarctic and subantarctic islands	0.503	no trend	0.472	no trend	-1.6
3	Arctic Canada north	0.823	no trend	0.710	no trend	-3.0
4	Arctic Canada south	0.901	no trend	0.901	no trend	-7.2
5	Caucasus and Middle East	0.264	no trend	0.031	decreasing	-35.4
6	Central Asia	0.078	no trend	0.056	no trend	-8.5
7	Central Europe	0.050	no trend	0.011	decreasing	-38.8
8	Greenland periphery	0.286	no trend	0.206	no trend	-6.8
9	Iceland	0.118	no trend	0.157	no trend	-5.9
10	Low latitudes	0.862	no trend	1.000	no trend	-22.7
11	New Zealand	0.244	no trend	0.063	no trend	-30.3
12	North Asia	0.286	no trend	0.087	no trend	-23.7
13	Russian Arctic	0.011	decreasing	0.011	decreasing	-3.2
14	Scandinavia	1.000	no trend	0.941	no trend	-14.6
15	South Asia east	0.673	no trend	0.472	no trend	-21.6
16	South Asia west	0.862	no trend	0.823	no trend	-5.2
17	Southern Andes	0.087	no trend	0.056	no trend	-13.2
18	Svalbard and Jan Mayen	0.005	decreasing	0.005	decreasing	-5.1
19	Western Canada and USA	0.056	no trend	0.024	decreasing	-23.4

Six regions show significant decreasing trends ($p < 0.05$) in relative change, while only three show significant decreasing trends in absolute change. This indicates that relative change metrics show an acceleration in mass loss rates compared to absolute values alone, particularly in smaller glacier regions already experiencing substantial depletion: Central Europe, Caucasus and Middle East and Western Canada and USA all lose more than 20% of their mass in this period.

So relative change matters especially for smaller glacier regions, while global conclusions are not fundamentally changed. Still, we see indications of an even worse trend for some regions when viewed in relative terms.

3. Glacier area calculation

In order to be able to convert units from gigatonnes to m w.e. and the other way around, Zemp et al. computes a yearly time series of regional glacier area from 2000 to 2023 for each glacier region (Zemp et al., 2025). For this computation they use a reference regional glacier area from the Randolph Glacier Inventory (RGI 6.0) (RGI Consortium, 2017). The RGI 6 inventory provides snapshot regional glacier areas for certain reference years (the specific year differs per region). The GlAMBIE team then accounts for glacier shrinkage by using the regional glacier area-change rates (% per year) compiled from IPCC AR5 extended with additional literature from Zemp et al. (IPCC, 2013; Zemp et al., 2019). Using these change rates, they correct the glacier area from the RGI 6 to the year 2000 and compute the yearly time series of regional glacier area (km^2) using the following formula:

$$(i) \text{Area}_t = \text{Area}_{reference} + (t - t_{reference}) * k * \text{Area}_{reference} \quad (1)$$

where:

$t = \text{year}$

$k = \text{glacier area change rate}$

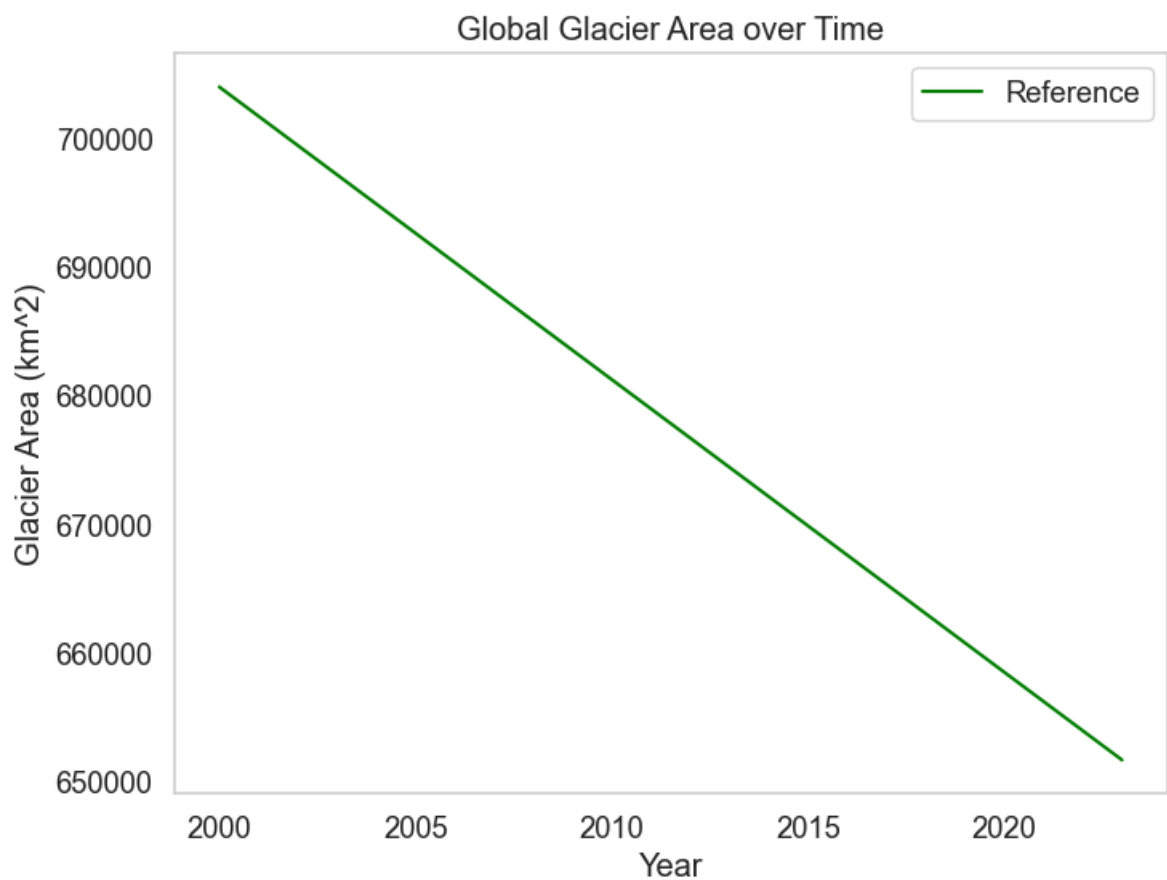
This results in a linear glacier area loss, as seen in the figure below:

```
In [9]: plt.figure(figsize = (8,6))

plt.plot(
    global_reference.start_dates,
    global_reference.glacier_area,
    color='green',
    label = 'Reference' )

plt.title('Global Glacier Area over Time')
plt.xlabel('Year')
plt.ylabel('Glacier Area (km^2)')

plt.legend()
plt.show()
```



RGI 6 vs RGI 7

In September 2023 an updated version of RGI 6 came out called RGI 7, with more accurate glacier area estimates (RGI Consortium, 2023). The GlaMBIE team mentions that they know it has been updated, but as it was updated after the launch of the model it was not used.

However, as the RGI 7 contains a more accurate estimation of the regional glacier areas than the RGI 6, we decided to run the model with the reference glacier areas of the RGI 7 and compare it to the results when using RGI 6 (reference run). Should there be a large difference in results, it could indicate that the results presented by Zemp et al. are not accurate and should not be used in further research when more up to date information is available. We ran the model with RGI 7 by changing the value of the RGI variable in the config file.

Analysis

The difference between the RGI 6 and RGI 7 values differs per region (see figures below). Because the reference value impacts the slope of the glacier area loss (see formula i), a smaller reference area changes both the intercept and the slope of the glacier area loss function.

```
In [10]: #Load in global dataset for RGI 7
global_rgi7 = pd.read_csv(os.path.join(rgi7_dir, '0_global.csv')).drop('Unnamed: 0', axis=1)

#Load in New Zealand data for the two runs
new_zealand_rgi6 = pd.read_csv(
    os.path.join(reference_run_dir, '18_new_zealand.csv')
).drop('Unnamed: 0', axis=1)
new_zealand_rgi7 = pd.read_csv(
    os.path.join(rgi7_dir, '18_new_zealand.csv')
).drop('Unnamed: 0', axis=1)

In [11]: #plot the glacier area over time for the two different runs
plt.figure(figsize = (12,5))

#plot the global glacier area of the two runs
plt.subplot(1,2,1)
plt.plot(global_reference.start_dates, global_reference.glacier_area, color='green', label = 'RGI 6' )
plt.plot(global_rgi7.start_dates, global_rgi7.glacier_area, color='blue', label = 'RGI 7')

plt.title('Global Glacier Area')
plt.xlabel('Year')
plt.ylabel('Glacier Area (km^2)')

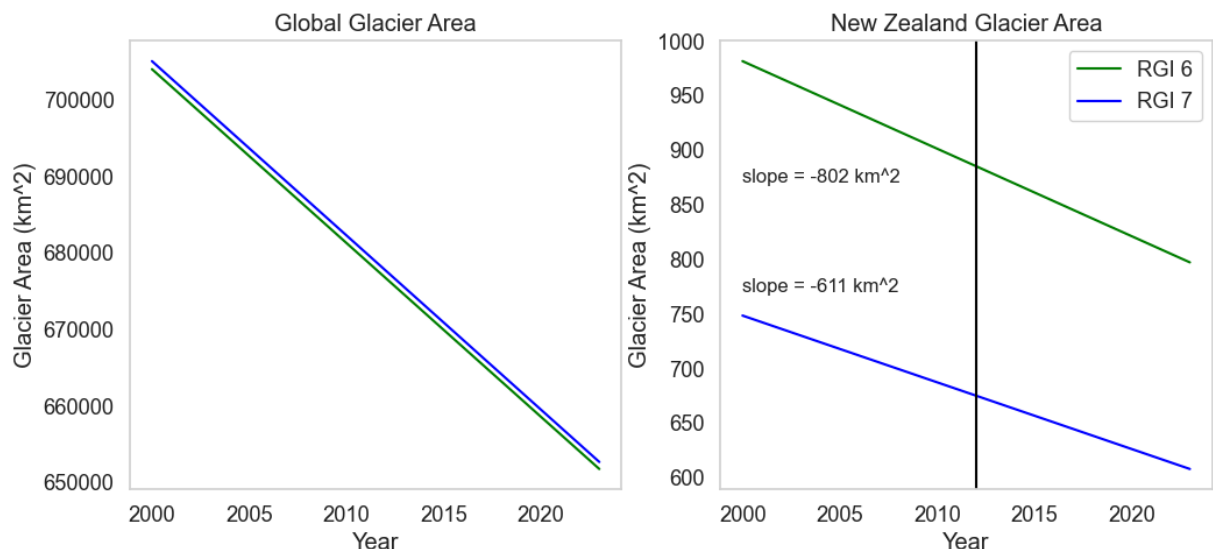
#plot the new zealand glacier area of the two runs
plt.subplot(1,2,2)
slope_6 = regions.REGIONS['new_zealand'].rgi6_area * regions.REGIONS['new_zealand'].area_change
slope_7 = regions.REGIONS['new_zealand'].rgi7_area * regions.REGIONS['new_zealand'].area_change

plt.plot(new_zealand_rgi6.start_dates, new_zealand_rgi6.glacier_area, color='green', label = 'RGI 6' )
plt.plot(new_zealand_rgi7.start_dates, new_zealand_rgi7.glacier_area, color='blue', label = 'RGI 7')

plt.axvline(x= 2012, color='black' )
plt.text(2000, 870, f"slope = {slope_6:.0f} km^2", fontsize=12)
plt.text(2000, 770, f"slope = {slope_7:.0f} km^2", fontsize=12)

plt.title('New Zealand Glacier Area')
plt.xlabel('Year')
plt.ylabel('Glacier Area (km^2)')
plt.legend()
plt.show

plt.legend()
plt.show()
```



The cumulative glacier area loss over time per region plots show that the difference between the two runs gets larger over time due to the aforementioned impact of RGI on the glacier loss slope (see figures below). Furthermore, the

difference between the two runs differs not only per region, but per unit as well, as it seems to have little impact on the m.w.e. results. On a global level, the difference between RGI 6 and RGI 7 is small (between the 2 and 3 gigatonnes), while on a regional level it can get as large as 22 gigatonnes (Southern Andes). The reason that the global results see little to no change, despite the differences on a regional level is that some regions see an increase and others a decrease in glacier area loss with the update, which evens out to almost no change on a global level.

In order to analyse the difference in results, we calculated the cumulative changes for both runs and for both units. We did this using the same function GlambIE uses for their calculations called "derivative_to_cumulative". The function can be found on the github in the file 'help_functions/plotting_functions.py'.

```
In [12]: for i in range(0, len(list_of_folders)):
          #define region name and path to results
          region = list_of_folders[i]

          #get the region name without the number
          region_name = region[region.find("_")+1:-4]

          #read in the data
          region_rgi6 = pd.read_csv(os.path.join(reference_run_dir, region)).drop('Unnamed: 0', axis=1)
          region_rgi7 = pd.read_csv(os.path.join(rgi7_dir, region)).drop('Unnamed: 0', axis=1)

          #calculate cumulative for rgi6
          cum_region_gt_6 = derivative_to_cumulative(
              region_rgi6.start_dates,
              region_rgi6.end_dates,
              region_rgi6.combined_gt)
          cum_region_gt_err_6 = derivative_to_cumulative(
              region_rgi6.start_dates,
              region_rgi6.end_dates,
              region_rgi6.combined_gt_errors,
              calculate_as_errors=True)

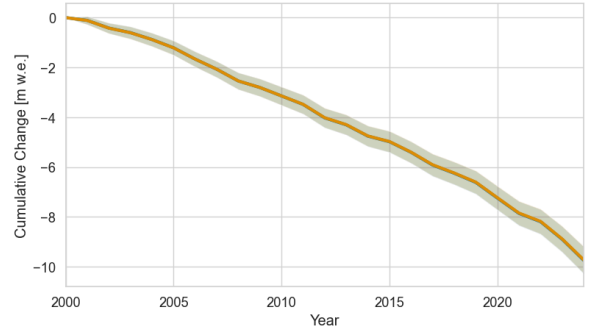
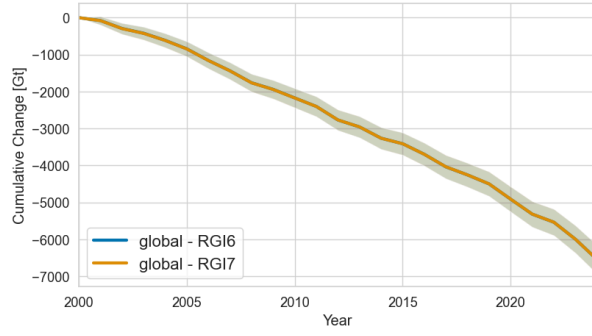
          cum_region_mwe_6 = derivative_to_cumulative(
              region_rgi6.start_dates,
              region_rgi6.end_dates,
              region_rgi6.combined_mwe)
          cum_region_mwe_err_6 = derivative_to_cumulative(
              region_rgi6.start_dates,
              region_rgi6.end_dates,
              region_rgi6.combined_mwe_errors,
              calculate_as_errors=True)

          #calculate cumulative for rgi7
          cum_region_gt_7 = derivative_to_cumulative(
              region_rgi7.start_dates,
              region_rgi7.end_dates,
              region_rgi7.combined_gt)
          cum_region_gt_err_7 = derivative_to_cumulative(
              region_rgi7.start_dates,
              region_rgi7.end_dates,
              region_rgi7.combined_gt_errors,
              calculate_as_errors=True)

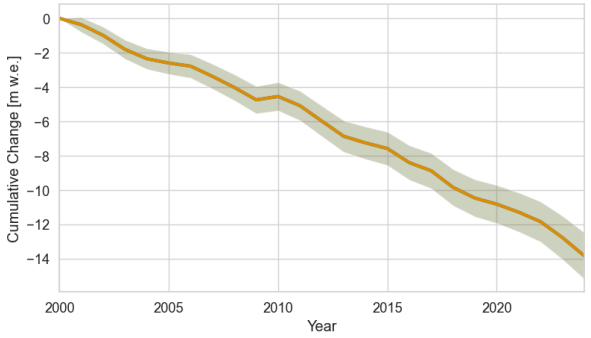
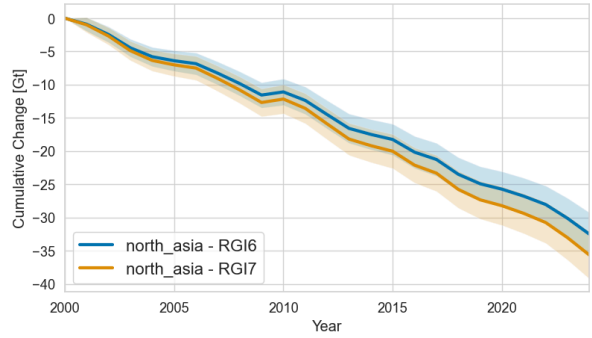
          cum_region_mwe_7 = derivative_to_cumulative(
              region_rgi7.start_dates,
              region_rgi7.end_dates,
              region_rgi7.combined_mwe)
          cum_region_mwe_err_7 = derivative_to_cumulative(
              region_rgi7.start_dates,
              region_rgi7.end_dates,
              region_rgi7.combined_mwe_errors,
              calculate_as_errors=True)

          #plot
          two_region_comparison_plot_2(
              region_name, 'RGI6', 'RGI7',
              cum_region_gt_6, cum_region_gt_err_6,
              cum_region_gt_7, cum_region_gt_err_7,
              cum_region_mwe_6, cum_region_mwe_err_6,
              cum_region_mwe_7, cum_region_mwe_err_7)
```

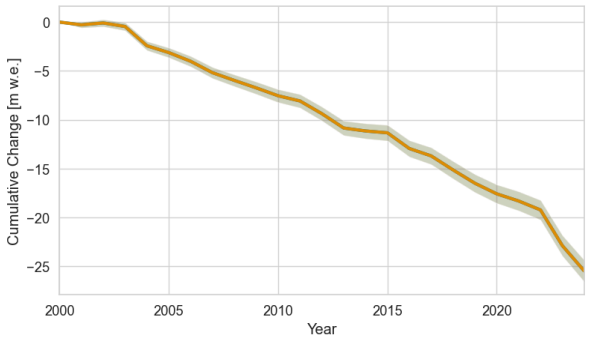
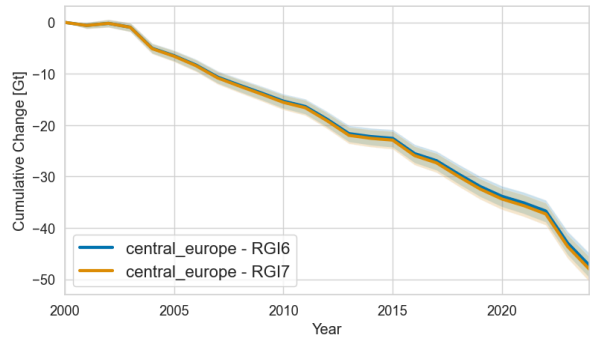
global: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



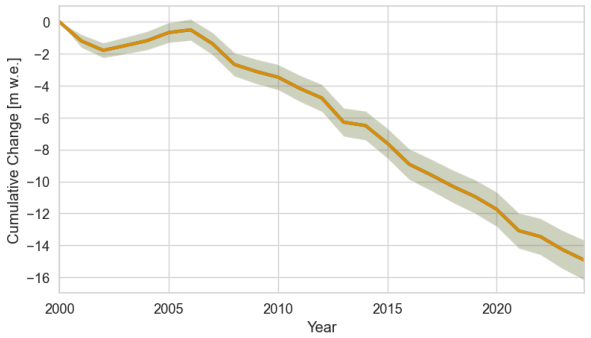
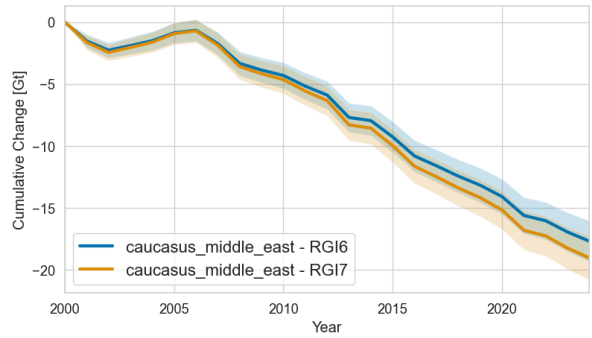
north_asia: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



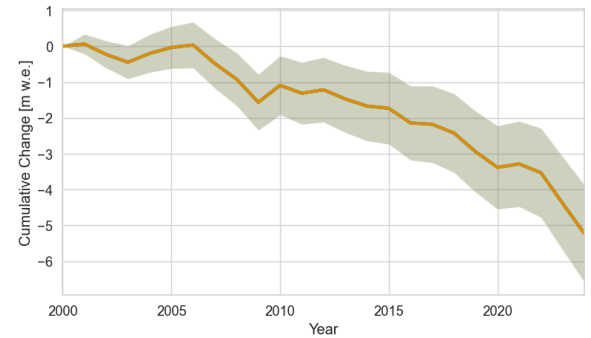
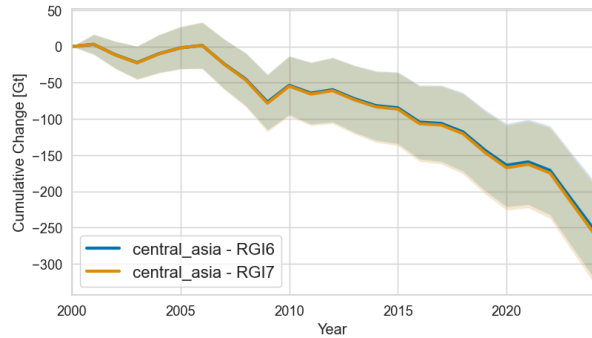
central_europe: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



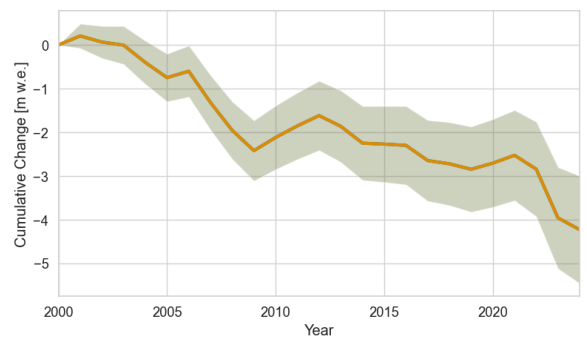
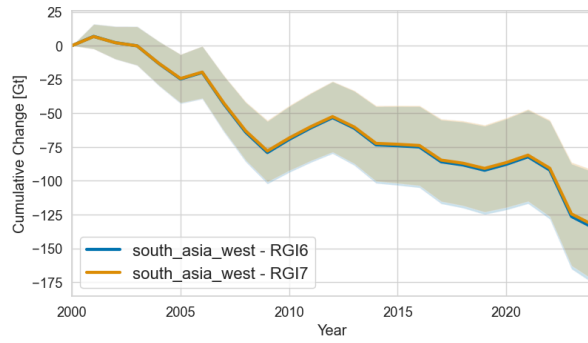
caucasus_middle_east: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



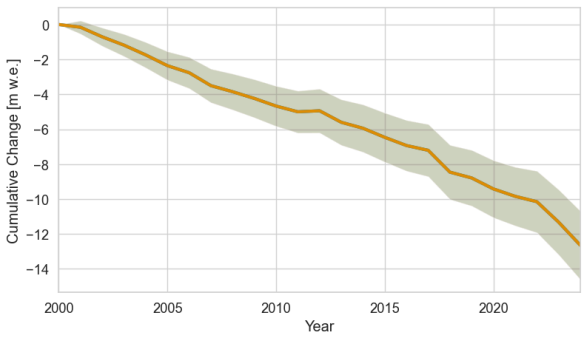
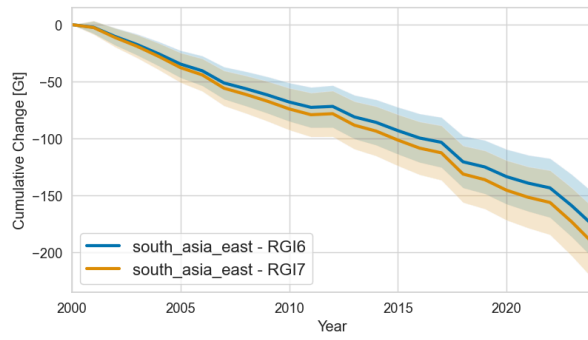
central_asia: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



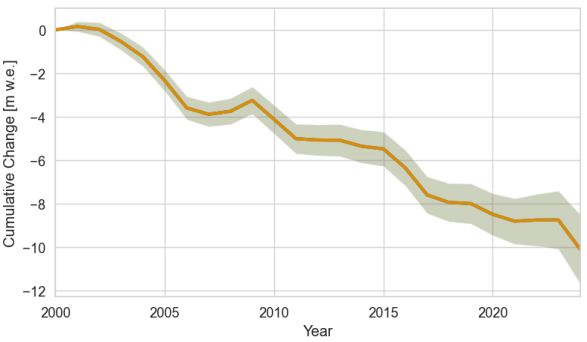
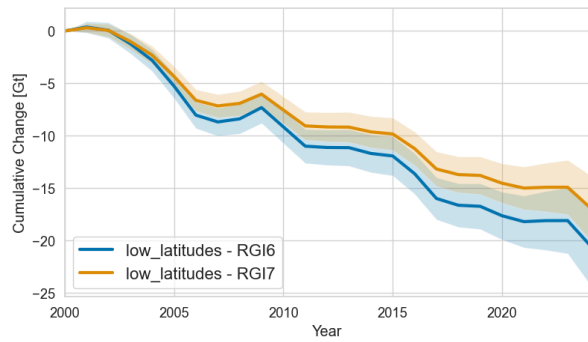
south_asia_west: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



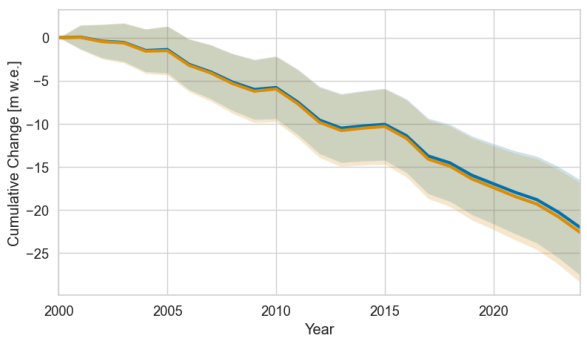
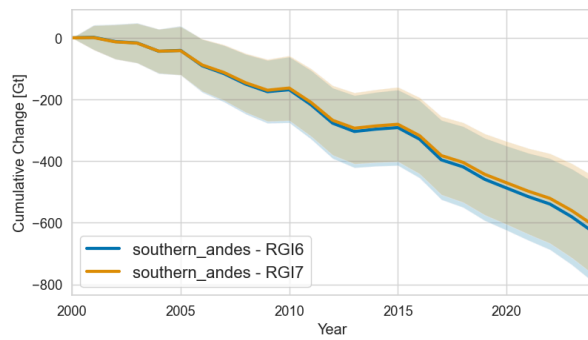
south_asia_east: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



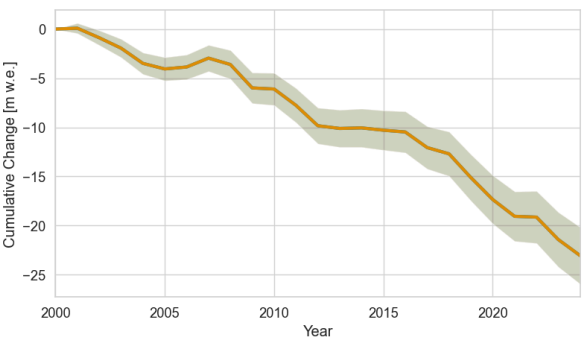
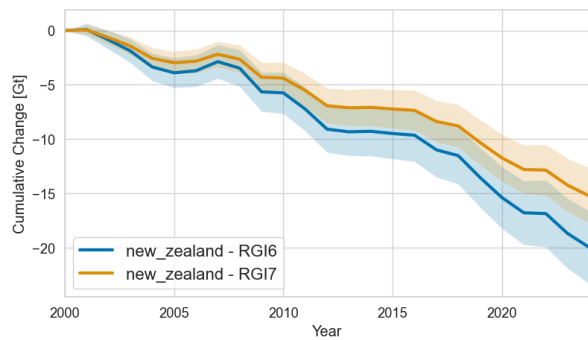
low_latitudes: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



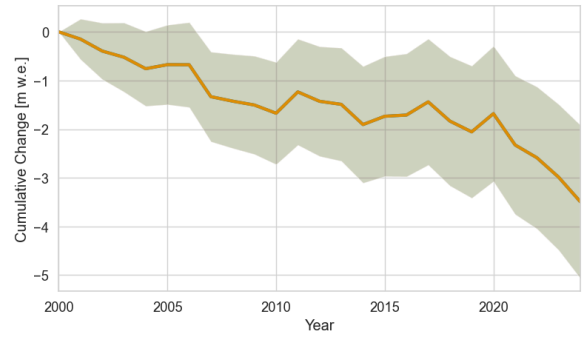
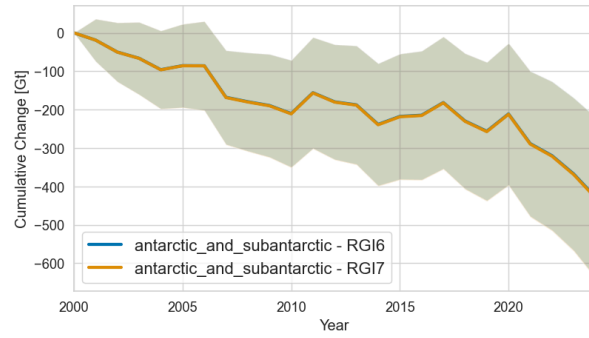
southern_andes: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



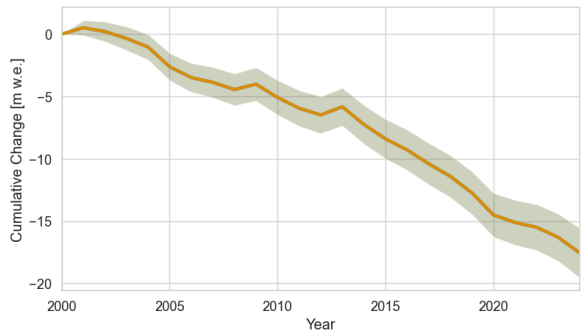
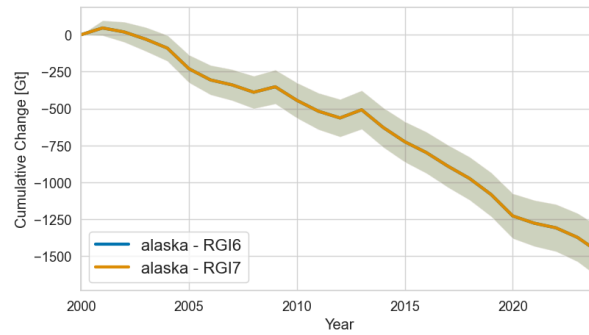
new_zealand: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



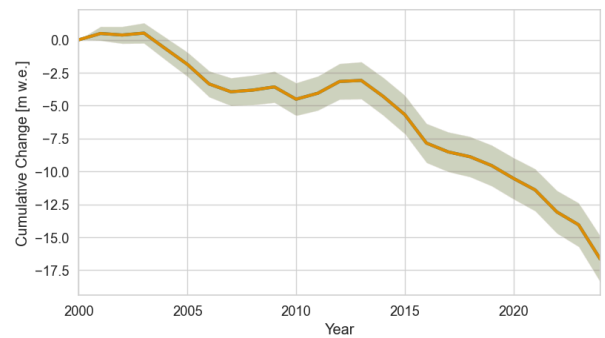
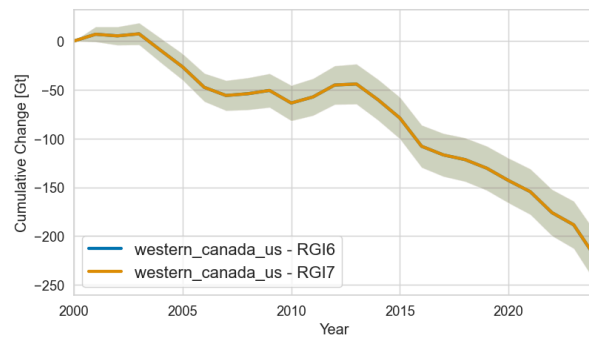
antarctic_and_subantarctic: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



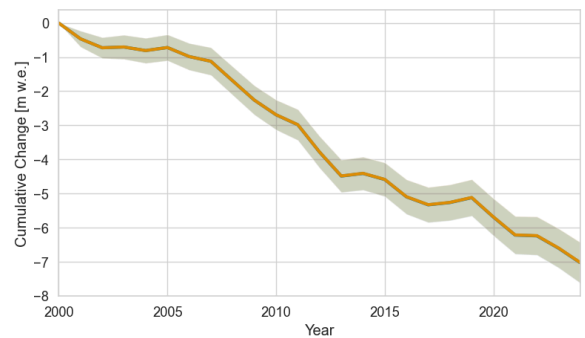
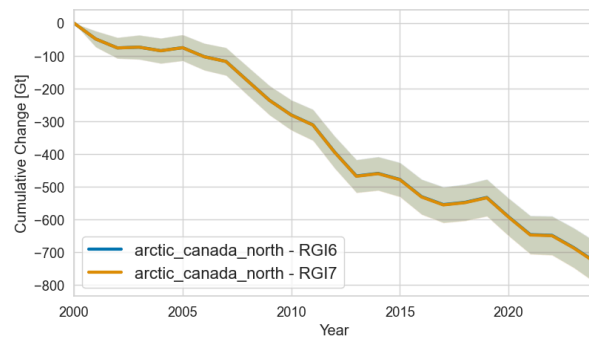
alaska: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



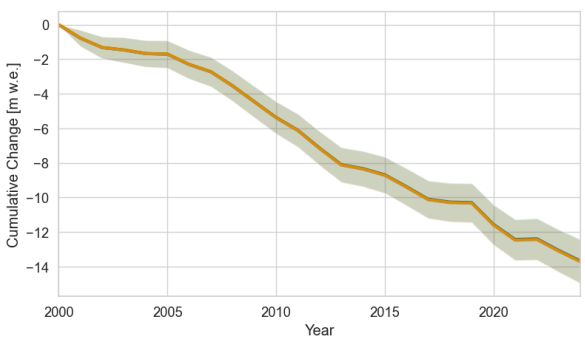
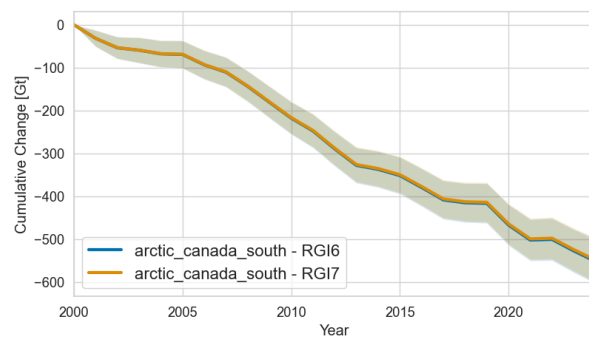
western_canada_us: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



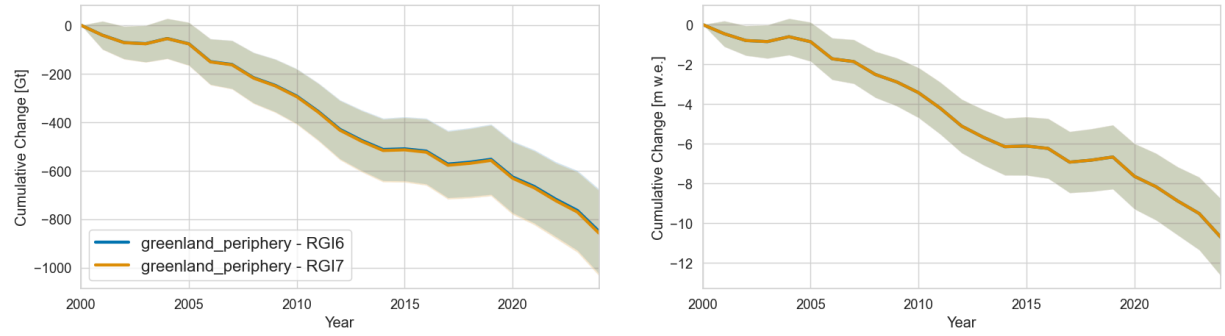
arctic_canada_north: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



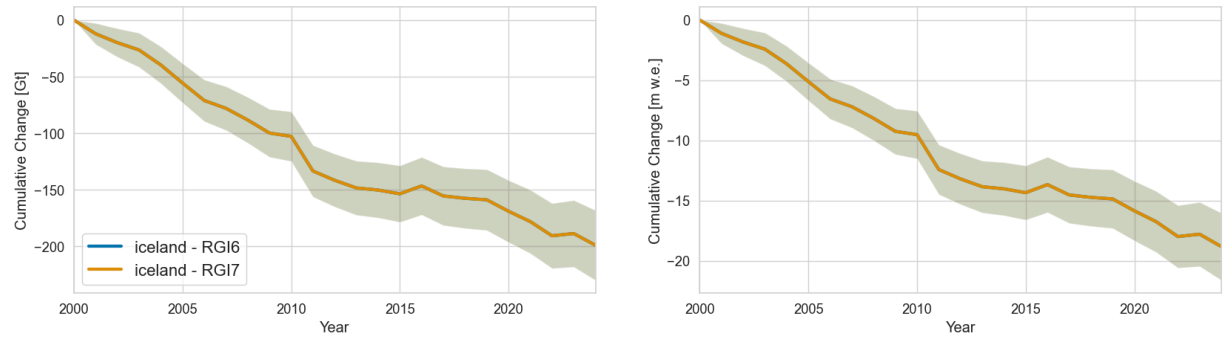
arctic_canada_south: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



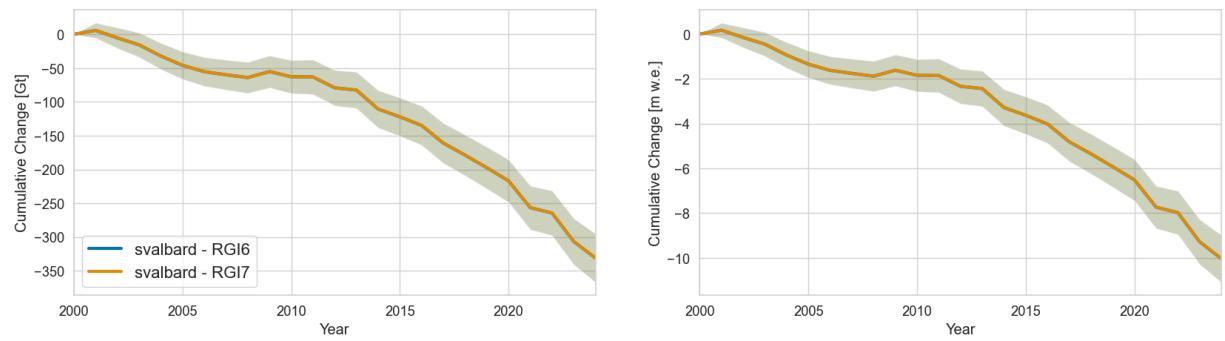
greenland_periphery: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



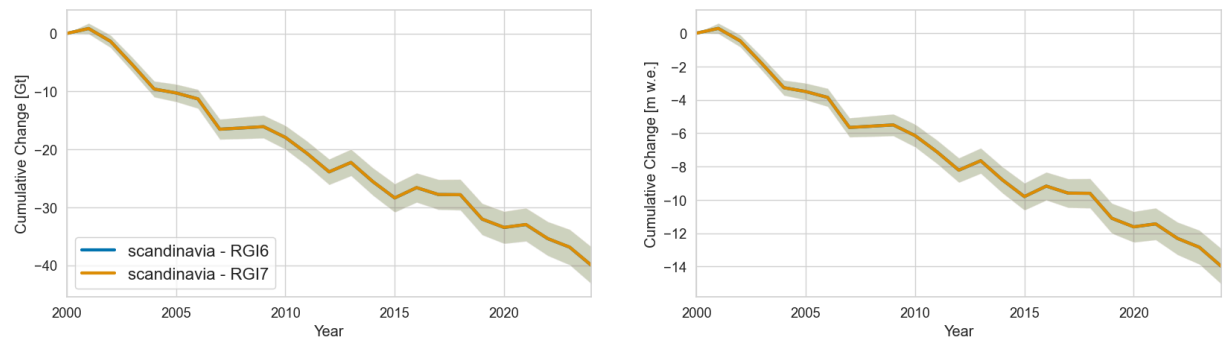
iceland: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



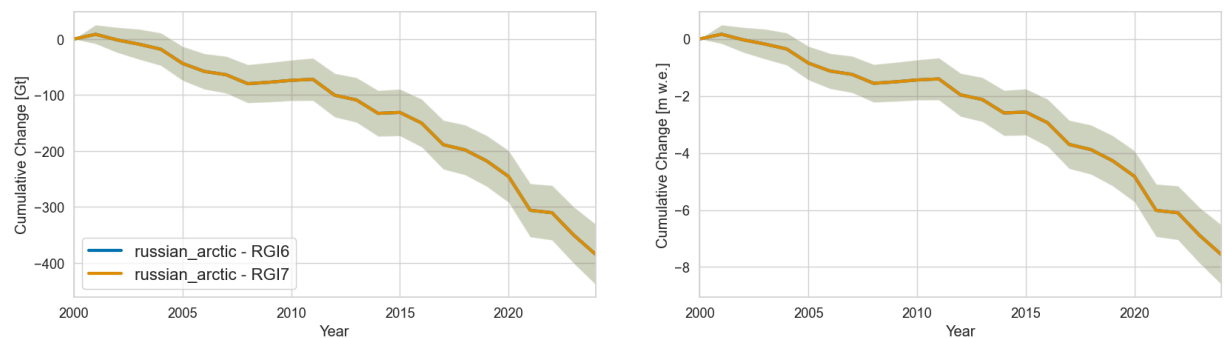
svalbard: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



scandinavia: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



russian_arctic: comparison RGI 6 and RGI 7: total ice loss, 2000 - 2023



In order to compare the differences per region more easily we computed the absolute difference (Gt), relative difference (%) and the Spearman correlation of the annual changes (in both gt and m.w.e.) of the two runs.

In [13]: *#calculate metrics per region*

```
#define empty lists
region_labels = []

rel_gt = []
abs_gt = []
corr_gt = []

rel_mwe = []
abs_mwe = []
corr_mwe = []

for i in range(0, len(list_of_folders)):
    #define region name and path to results
    region = list_of_folders[i]

    #get the region name without the number
    region_name = region[region.find("_")+1:-4]

    #read in the data
    region_rgi6 = pd.read_csv(os.path.join(reference_run_dir, region)).drop('Unnamed: 0', axis=1)
    region_rgi7 = pd.read_csv(os.path.join(rgi7_dir, region)).drop('Unnamed: 0', axis=1)

    #store region label
    region_labels.append(region_name)

    #merge datasets
    merged = pd.merge(
        region_rgi6,
        region_rgi7,
        on="start_dates",
        suffixes=("_run1", "_run2"),
        how="outer")

    #####
    #compute cumulative values

    #calculate cumulative values in gigatonnes
    cumul_rgi6_gt = merged["combined_gt_run1"].sum()
    cumul_rgi7_gt = merged["combined_gt_run2"].sum()

    #calculate cumulative values in mwe
    cumul_rgi6_mwe = merged["combined_mwe_run1"].sum()
    cumul_rgi7_mwe = merged["combined_mwe_run2"].sum()

    #####
    #compute metrics

    #calculate relative difference
    rel_diff_val_gt = (cumul_rgi7_gt - cumul_rgi6_gt) / abs(cumul_rgi6_gt) * 100.0
    rel_diff_val_mwe = (cumul_rgi7_mwe - cumul_rgi6_mwe) / abs(cumul_rgi6_mwe) * 100.0

    #calculate absolute difference
    abs_diff_val_gt = cumul_rgi7_gt - cumul_rgi6_gt
    abs_diff_val_mwe = cumul_rgi7_mwe - cumul_rgi6_mwe

    #calculate spearman correlation
    corr_val_gt, _ = spearmanr(merged["combined_gt_run1"], merged["combined_gt_run2"])
    corr_val_mwe, _ = spearmanr(merged["combined_mwe_run1"], merged["combined_mwe_run2"])

    #####
    #append to lists

    rel_gt.append(rel_diff_val_gt)
    abs_gt.append(abs_diff_val_gt)
    corr_gt.append(corr_val_gt)

    rel_mwe.append(rel_diff_val_mwe)
    abs_mwe.append(abs_diff_val_mwe)
    corr_mwe.append(corr_val_mwe)
```

```

In [14]: #plot for gt
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(13, max(6, len(region_labels) * 0.6)), sharey=True)

n = len(region_labels)
y_pos = np.arange(n)

# Panel 1: Global relative difference (%)
ax1.barh(y_pos, rel_gt)
ax1.set_yticks(y_pos)
ax1.set_yticklabels(region_labels, fontsize=12)
ax1.invert_yaxis()
ax1.set_xlabel("Relative Change (%)")
ax1.axvline(0, color="black", linewidth=0.8)
max_abs_rel = max(abs(v) for v in rel_gt) or 1
for i, val in enumerate(rel_gt):
    if val >= 0:
        ax1.text(val + 0.02 * max_abs_rel, i, f"{val:.3f}%", va="center", fontsize=10)
    else:
        ax1.text(val - 0.02 * max_abs_rel, i, f"{val:.3f}%", va="center", ha="right", fontsize=10)
pad = max_abs_rel * 0.20
ax1.set_xlim(min(rel_gt) - pad, max(rel_gt) + pad)

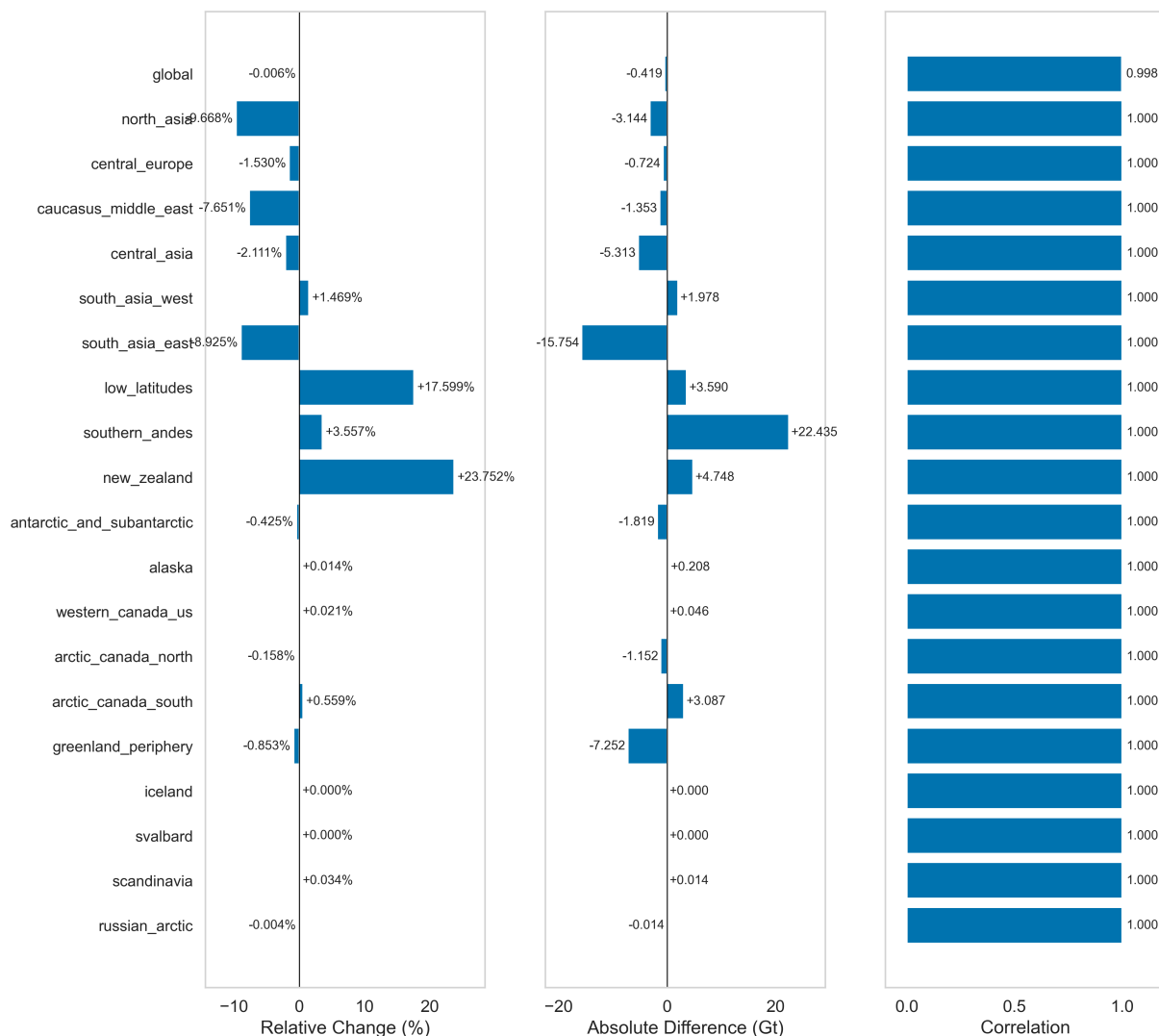
# Panel 2: Global absolute difference (Gt)
ax2.barh(y_pos, abs_gt)
ax2.set_xlabel("Absolute Difference (Gt)")
ax2.axvline(0, color="black", linewidth=0.8)
max_abs_diff = max(abs(v) for v in abs_gt) or 1
for i, val in enumerate(abs_gt):
    if val >= 0:
        ax2.text(val + 0.02 * max_abs_diff, i, f"{val:.3f}", va="center", fontsize=10)
    else:
        ax2.text(val - 0.02 * max_abs_diff, i, f"{val:.3f}", va="center", ha="right", fontsize=10)
pad = max_abs_diff * 0.30
ax2.set_xlim(min(abs_gt) - pad, max(abs_gt) + pad)

# Panel 3: Global correlation
ax3.barh(y_pos, corr_gt)
ax3.set_xlabel("Correlation")
ax3.set_xlim(-0.1, 1.2)
for i, val in enumerate(corr_gt):
    if val >= 0:
        ax3.text(val + 0.02, i, f"{val:.3f}", va="center", fontsize=10)
    else:
        ax3.text(val + 0.08, i, f"{val:.3f}", va="center", ha="right", fontsize=10)

fig.suptitle(f"Gigaton metrics RGI 6 vs RGI 7 for each region", fontsize=14)
fig.tight_layout()
fig.set_dpi(300)
plt.show()

```

Gigaton metrics RGI 6 vs RGI 7 for each region



```
In [15]: #plot for mwe
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(13, max(6, len(region_labels) * 0.6)), sharey=True)

n = len(region_labels)
y_pos = np.arange(n)

# Panel 1: Global relative difference (%)
ax1.barh(y_pos, rel_mwe)
ax1.set_yticks(y_pos)
ax1.set_yticklabels(region_labels, fontsize=12)
ax1.invert_yaxis()
ax1.set_xlabel("Relative Change (%)")
ax1.axvline(0, color="black", linewidth=0.8)
max_abs_rel = max(abs(v) for v in rel_mwe) or 1
for i, val in enumerate(rel_mwe):
    if val >= 0:
        ax1.text(val + 0.02 * max_abs_rel, i, f"+{val:.3f}%", va="center", fontsize=10)
    else:
        ax1.text(val - 0.02 * max_abs_rel, i, f"{val:.3f}%", va="center", ha="right", fontsize=10)
pad = max_abs_rel * 0.20
ax1.set_xlim(min(rel_mwe) - pad, max(rel_mwe) + pad)

# Panel 2: Global absolute difference (mwe)
ax2.barh(y_pos, abs_mwe)
ax2.set_xlabel("Absolute Difference (mwe)")
ax2.axvline(0, color="black", linewidth=0.8)
max_abs_diff = max(abs(v) for v in abs_mwe) or 1
for i, val in enumerate(abs_mwe):
    if val >= 0:
        ax2.text(val + 0.02 * max_abs_diff, i, f"+{val:.3f}", va="center", fontsize=10)
    else:
        ax2.text(val - 0.02 * max_abs_diff, i, f"{val:.3f}", va="center", ha="right", fontsize=10)
pad = max_abs_diff * 0.30
```

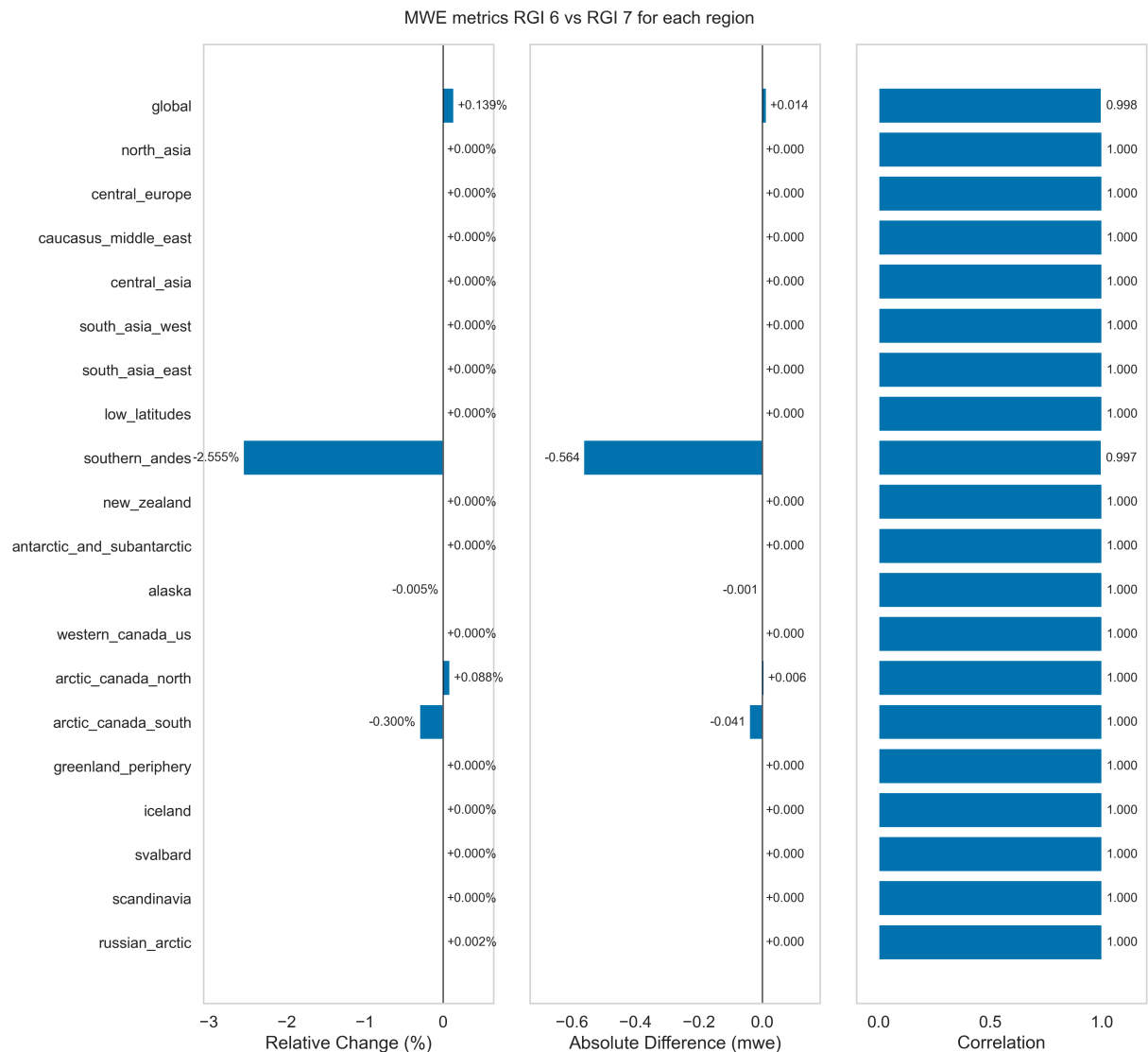
```

ax2.set_xlim(min(abs_mwe) - pad, max(abs_mwe) + pad)

# Panel 3: Global correlation
ax3.barh(y_pos, corr_mwe)
ax3.set_xlabel("Correlation")
ax3.set_xlim(-0.1, 1.2)
for i, val in enumerate(corr_mwe):
    if val >= 0:
        ax3.text(val + 0.02, i, f"{val:.3f}", va="center", fontsize=10)
    else:
        ax3.text(val + 0.08, i, f"{val:.3f}", va="center", ha="right", fontsize=10)

fig.suptitle(f"MWE metrics RGI 6 vs RGI 7 for each region", fontsize=14)
fig.tight_layout()
fig.set_dpi(300)
plt.show()

```



These barcharts show the relative and absolute difference between the cumulative glacier mass changes of the two runs per region, as well as the spearman correlation. Once again, we see the difference in results per region and unit.

The barcharts confirm that the RGI version has no significant impact on the modelled glacier loss in m.w.e. The Southern Andes is the only region with a relative difference above 0.3%. However, even in that region, the relative and absolute difference are still only 2.5% and -0.56 m.w.e. While these values are not large, it is still significantly larger than in the other regions. This could be due the large absolute difference in gigatonnes for this region.

While the RGI version seems to have no significant impact on modelled m.w.e. glacier loss, the same can not be said for the modelled glacier loss in gigatonnes. The absolute difference between gigatonnes in change differs from -15 (South Asia East) to 22 (Southern Andes) and the relative difference ranges from -9.7% (North Asia) to 23.7% (New Zealand). For all regions and both metrics, the spearman correlation is close to 1, indicating that, while the values might be different, the patterns of glacier loss over time are the same.

As such, we can conclude that, while the RGI version does not have a large impact on m w.e., it does have a significant impact on the glacier ice loss in gigatonnes for multiple regions. While it is not a fault of the authors, as RGI 7 was not yet published when GlaMBIE went live, it is important that the model is updated to be using RGI 7 instead of RGI 6.

Non-Static area change rate

As stated above, Zemp et al. (2025) use a static glacier area change rate for computing the yearly time series of regional glacier area (km^2). As a result, the computed glacier area loss is linear.

Whether or not glacier area loss is linear or accelerating is currently not known, as most researchers focus on mass and/or volume loss and not on area loss. However, previous research has indicated that both glacier mass and glacier volume loss are accelerating, making us skeptical of the assumed linearity of glacier area loss (Barry, 2006; Hugonnet et al., 2021; Fox-Kemper et al., 2021). As such, we set out to change the static glacier area loss rate into an accelerating one.

Model

In order to test how an accelerating area loss rate would impact the model results, two additional models were ran.

For the model called 2023, we made sure that the model matches with the reference model at the beginning and the end of the time series, and that the slope of the model matched the in the middle (see figure below). This way, any difference in model results would be due to the non-linear slope and not due to a difference in final glacier area.

The second model, model 2012, meets the reference model in the middle of the time series, and has the same slope as the reference a quarter into the reference. This one was picked because the IPCC report that GlaMBIE gets the glacier area change rates from was published in 2013 and, as such, the rates might be more realistic in the beginning of the time series as opposed to the end.

```
In [16]: global_reference = pd.read_csv(
        os.path.join(reference_run_dir, '0_global.csv')
    ).drop('Unnamed: 0', axis=1)
global_2012 = pd.read_csv(os.path.join(d_dir_2012, '0_global.csv')).drop('Unnamed: 0', axis=1)
global_2023 = pd.read_csv(os.path.join(d_dir_2023, '0_global.csv')).drop('Unnamed: 0', axis=1)
```

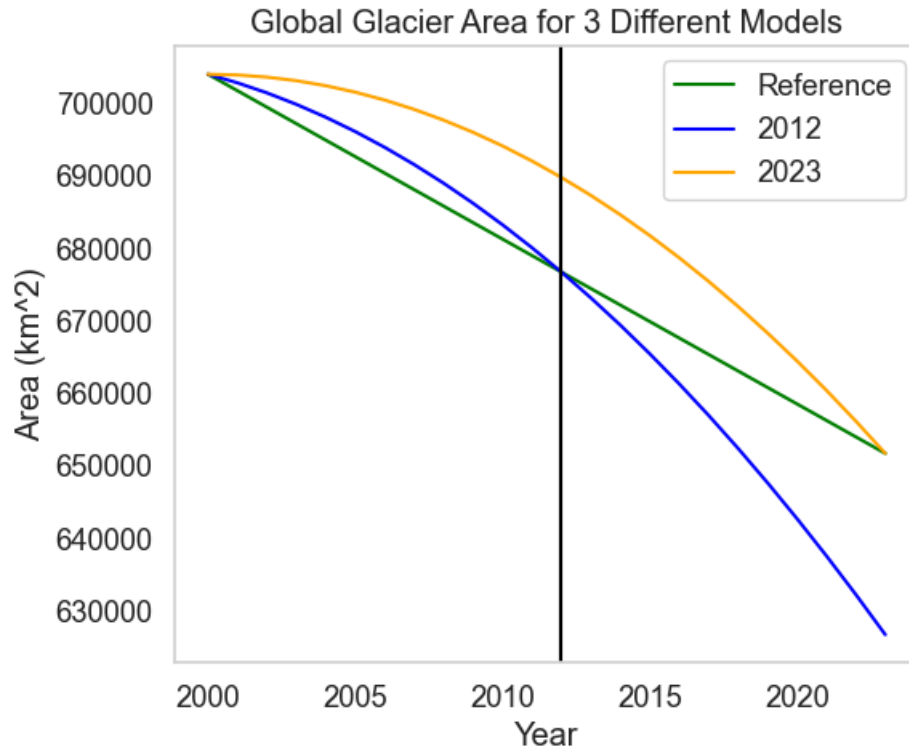
```
In [17]: #plot the glacier area over time for the two different runs
plt.figure(figsize = (6,5))

plt.plot(
    global_reference.start_dates,
    global_reference.glacier_area,
    color='green',
    label = 'Reference'
)
plt.plot(global_2012.start_dates, global_2012.glacier_area, color='blue', label = '2012')
plt.plot(global_2023.start_dates, global_2023.glacier_area, color='orange', label = '2023')

plt.axvline(x= 2012, color='black' )

plt.title('Global Glacier Area for 3 Different Models')

plt.xlabel('Year')
plt.ylabel('Area (km^2)')
plt.legend()
plt.show()
```



These model shapes were achieved using formula (ii), where we added an additional term to the model that ensured that the new glacier area would be the same as the reference at the year 2000 and the year M . Furthermore, by making the c parameter equal to the glacier area change rate multiplied by the reference area and divided by 23 (difference between end and beginning of the time series), we ensured that the glacier area would never increase.

$$(ii) \text{Area}_t = \text{Area}_{reference} + (t - t_{reference}) * k * \text{Area}_{reference} + c * (t - 2000)(t - M)$$

$$c = \frac{k * \text{Area}_{reference}}{23}$$

where:

t = year

k = glacier area change rate

M = year we want the glacier area to meet the reference area (2012 or 2023)

In order to implement these changes in the model, we changed the `get_adjusted_area()` function as defined in the file `glambie/const/regions.py`.

The function used for the 2012 model can be found below and in the file `glambie/const/regions_2012.py`. The function used for the 2023 model is the same as the one for the 2012 model, with the exception of the variable 'model_year' and can be found in the file `glambie/const/regions_2023.py`. For the 2023 model, the variable 'model_year' equals 2023.5 instead of 2012.5.

```
In [18]: #the original area calculation, it can also be found in the file glambie/const/regions.py
def get_adjusted_area(
    self, start_date: float, end_date: float, rgi_area_version: int
) -> float:
    """
    Calculates the area of a region adjusted with the area change to the given time period

    Parameters
    -----
    start_date : float
        start date of time period to be considered
    end_date : float
        end date of time period to be considered
    rgi_area_version : int, optional
        version of RGI area, currently implemented are 5, 6 and 7
```



```

Returns
-----
float
    adjusted area to time period given in km2
"""

# get the average between the start date and end date of the period
# (so halfway through the year)
time_period_mean = (start_date + end_date) / 2

# retrieve reference glacier area based on the value of the RGI variable
static_area = getattr(self, "rgi{}_area".format(str(rgi_area_version)))

#calculate the adjusted area
adjusted_area = (
    static_area
    + (time_period_mean - self.area_change_reference_year)
    * (self.area_change / 100)
    * static_area
)
return adjusted_area

```

```

In [19]: #the adjusted area calculation for model 2012,
#it can also be found in the file glambie/const/regions_2012.py
#the adjusted area calculation for model 2023 is the same
#except for the value of model_year
#and can be found in the file glambie/const/regions_2023.py

def get_adjusted_area(
    self, start_date: float, end_date: float, rgi_area_version: int
) -> float:
    """
    Calculates the area of a region adjusted with the area change to the given time period

    Parameters
    -----
    start_date : float
        start date of time period to be considered
    end_date : float
        end date of time period to be considered
    rgi_area_version : int, optional
        version of RGI area, currently implemented are 5, 6 and 7

    Returns
    -----
    float
        adjusted area to time period given in km2
    """
    # get the average between the start date and end date of the period
    # (so halfway through the year)
    time_period_mean = (start_date + end_date) / 2

    # retrieve reference glacier area based on the value of the RGI variable
    static_area = getattr(self, "rgi{}_area".format(str(rgi_area_version)))

    #define meeting years ( + 0.5)
    start_year = 2000.5
    model_year = 2012.5 #for model 2023 change this value to 2023.5

    #retrieve static area change rate and divide it by 100
    area_change2 = (self.area_change / 100)

    #calculate c
    c = (area_change2 * static_area)/23

    # calculate glacier area
    adjusted_area = (
        static_area
        + (time_period_mean - self.area_change_reference_year)
        * area_change2
        * static_area
        + c * (time_period_mean - start_year)*(time_period_mean - model_year)
    )

```

```
return adjusted_area
```

Analysis

We looked at multiple values in order to properly compare the differences between the model runs. We plotted the cumulative glacier ice loss for all three models in the same plot in order to see how any differences developed over time.

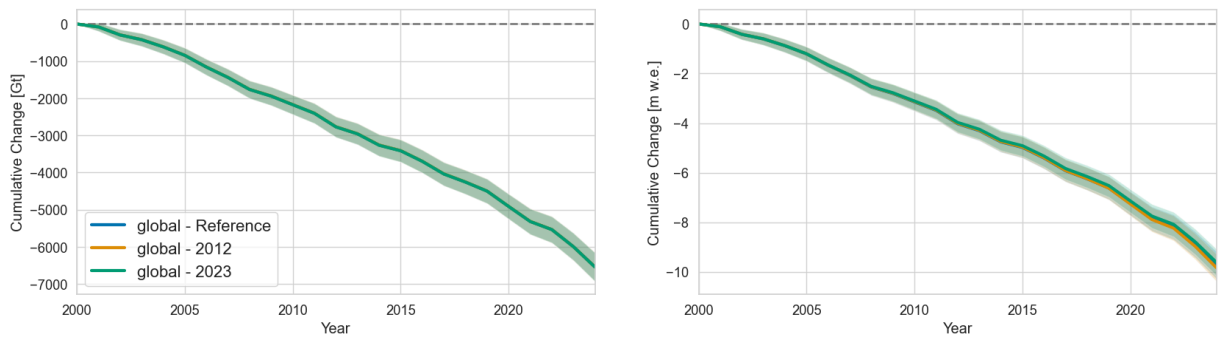
```
In [20]: for i in range(0, len(list_of_folders)):
#define region name and path to results
region = list_of_folders[i]

#get the region name without the number
region_name = region[region.find("_")+1:-4]

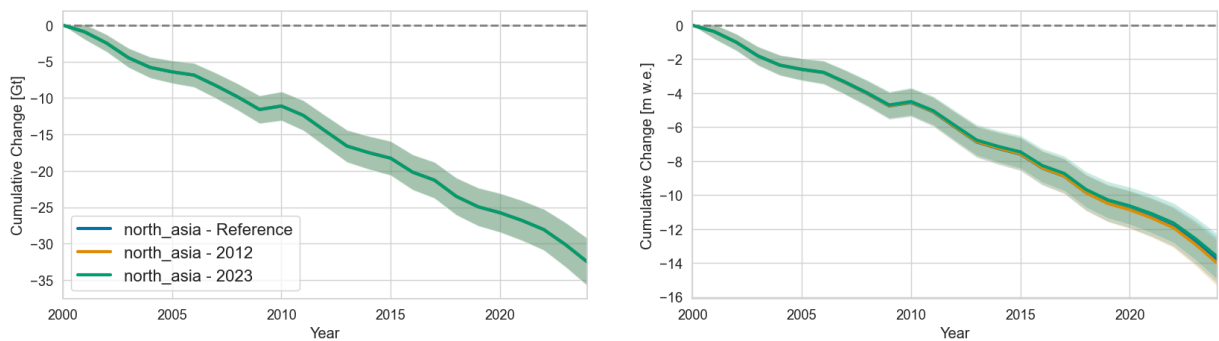
#read in the data
region_normal = pd.read_csv(os.path.join(reference_run_dir, region)).drop('Unnamed: 0', axis=1)
region_2012 = pd.read_csv(os.path.join(d_dir_2012, region)).drop('Unnamed: 0', axis=1)
region_2023 = pd.read_csv(os.path.join(d_dir_2023, region)).drop('Unnamed: 0', axis=1)

#plot
three_runs_cumulative_comparison_plot(
    region_name,
    'Reference', '2012', '2023',
    region_normal, region_2012, region_2023)
```

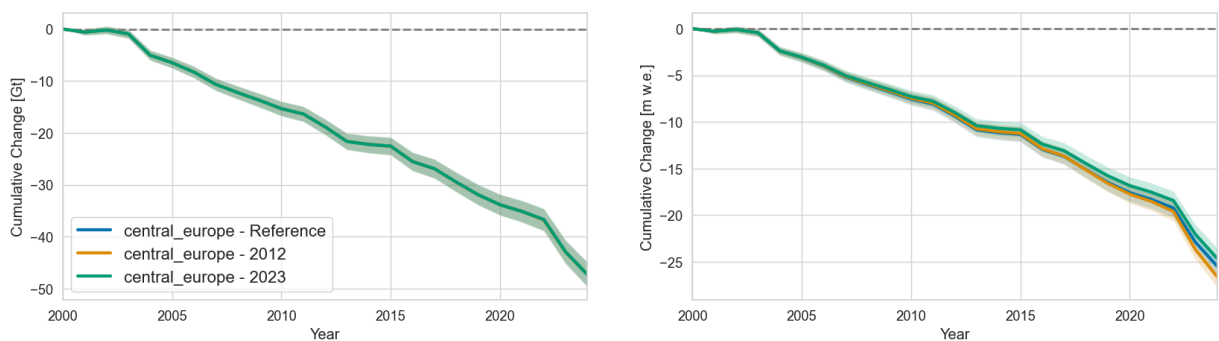
global cumulative ice loss predicted by models: Reference, 2012 and 2023



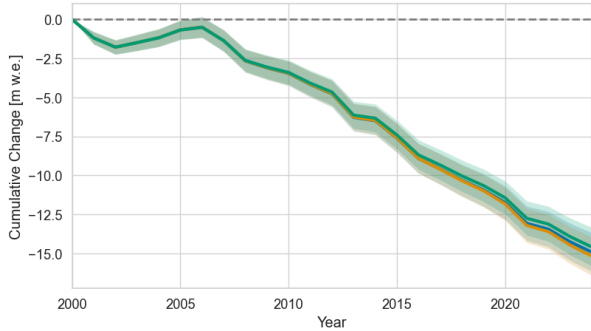
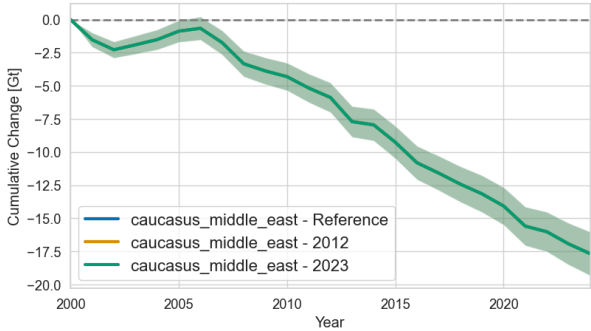
north_asia cumulative ice loss predicted by models: Reference, 2012 and 2023



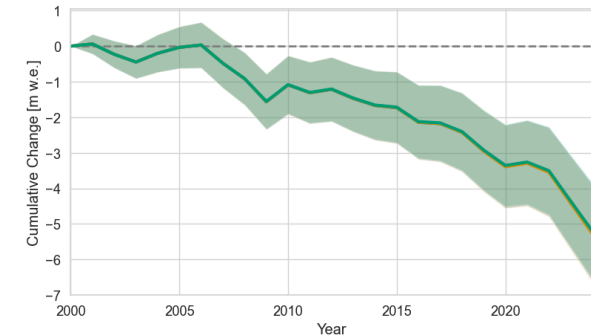
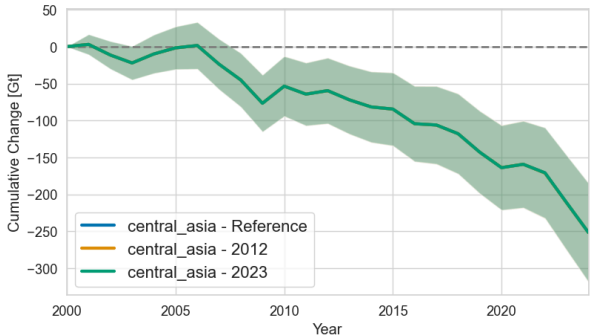
central_europe cumulative ice loss predicted by models: Reference, 2012 and 2023



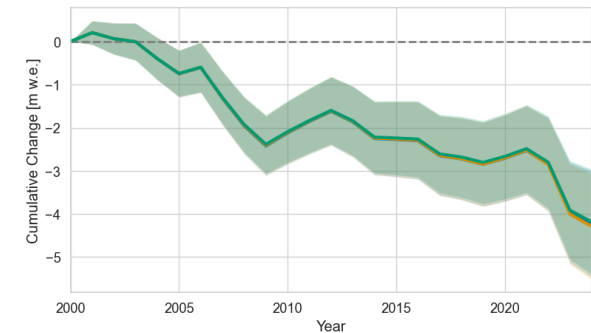
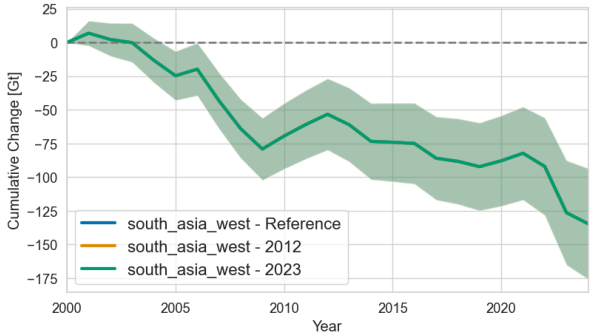
caucasus_middle_east cumulative ice loss predicted by models: Reference, 2012 and 2023



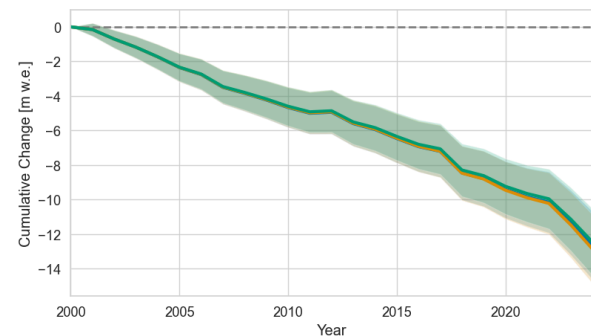
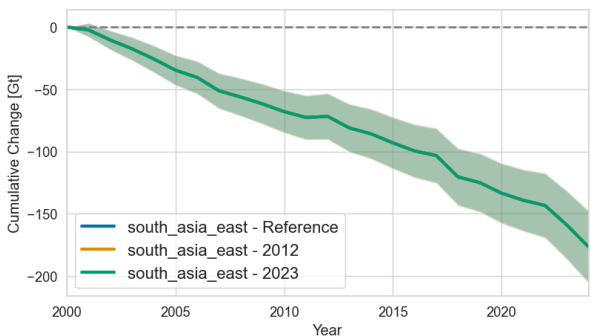
central_asia cumulative ice loss predicted by models: Reference, 2012 and 2023



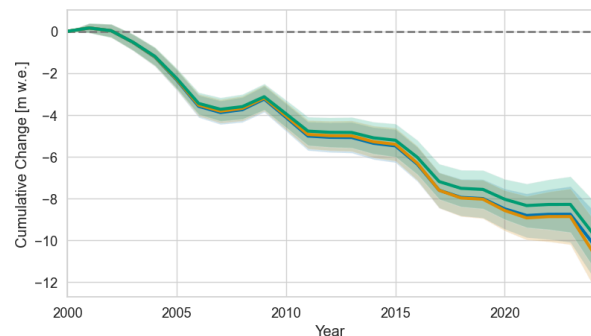
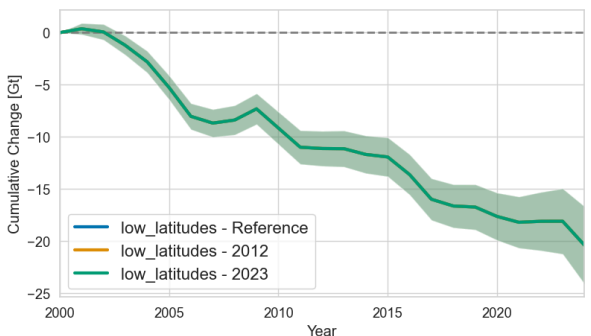
south_asia_west cumulative ice loss predicted by models: Reference, 2012 and 2023



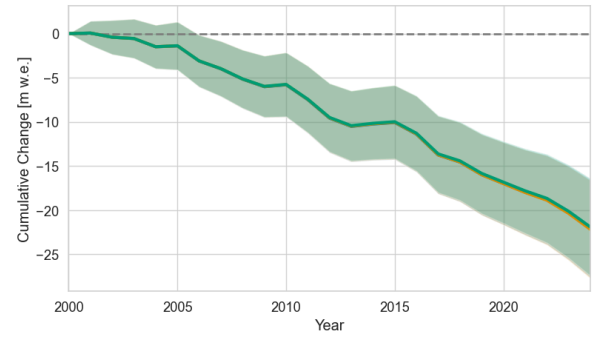
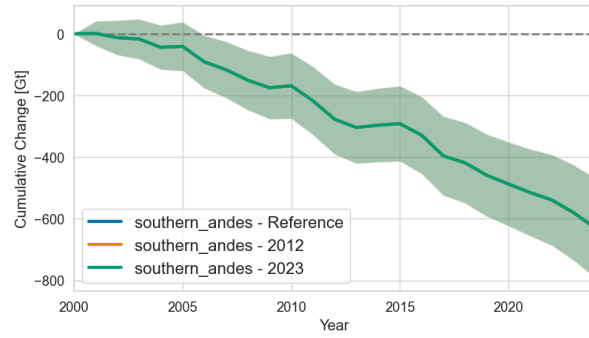
south_asia_east cumulative ice loss predicted by models: Reference, 2012 and 2023



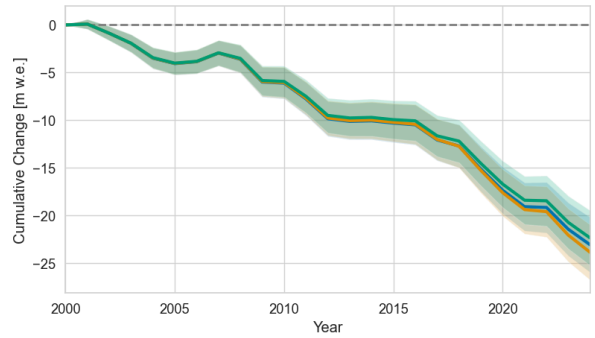
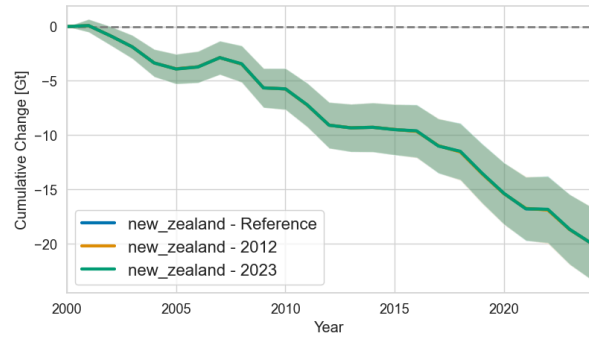
low_latitudes cumulative ice loss predicted by models: Reference, 2012 and 2023



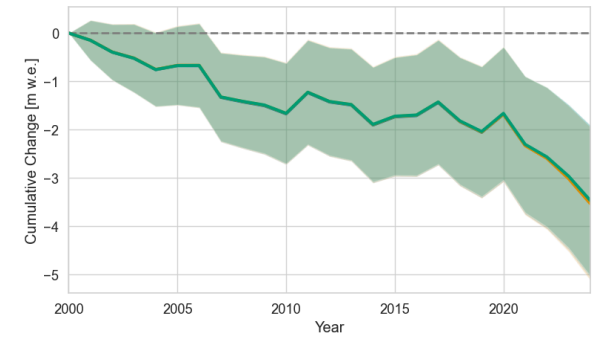
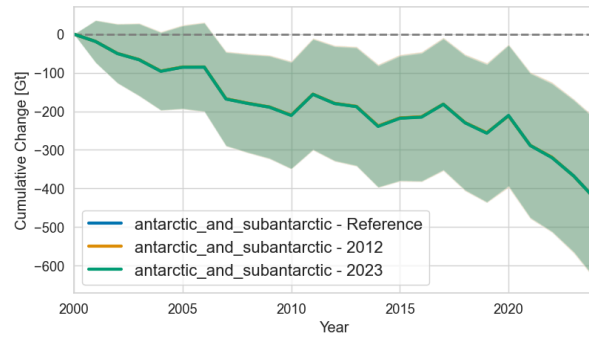
southern_andes cumulative ice loss predicted by models: Reference, 2012 and 2023



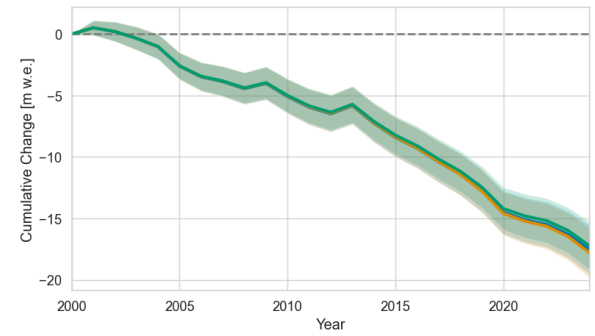
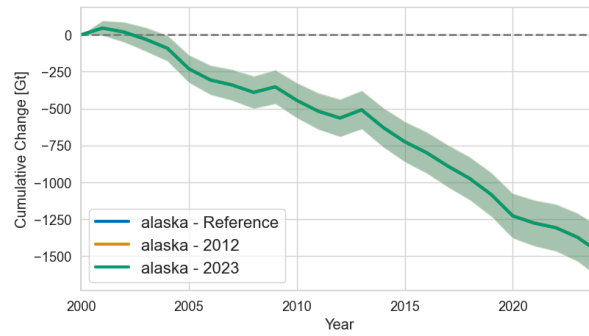
new_zealand cumulative ice loss predicted by models: Reference, 2012 and 2023



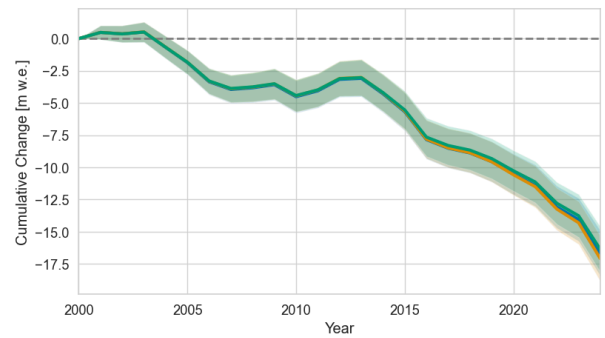
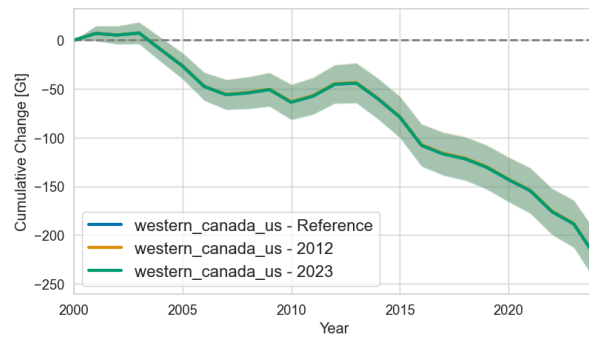
antarctic_and_subantarctic cumulative ice loss predicted by models: Reference, 2012 and 2023



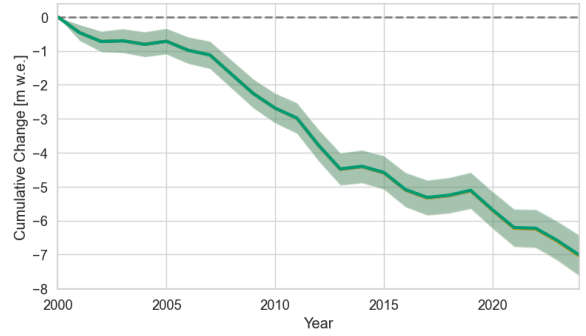
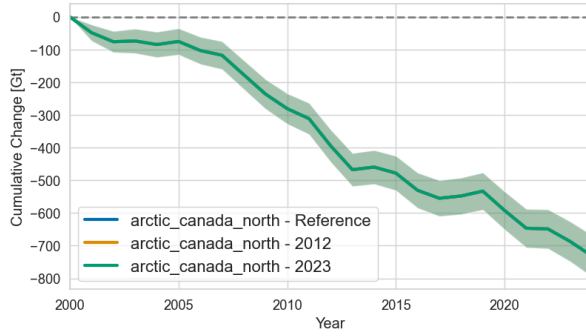
alaska cumulative ice loss predicted by models: Reference, 2012 and 2023



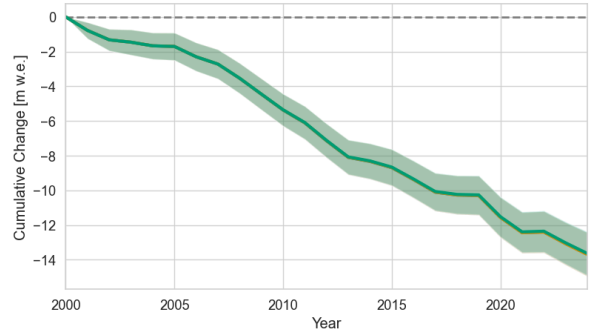
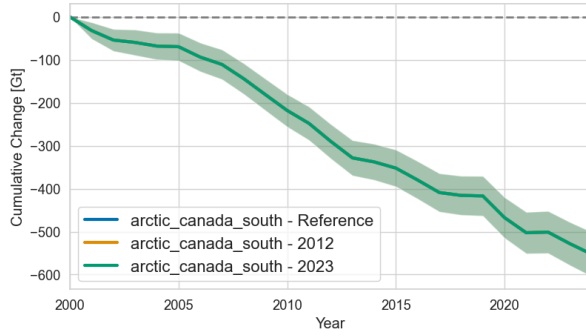
western_canada_us cumulative ice loss predicted by models: Reference, 2012 and 2023



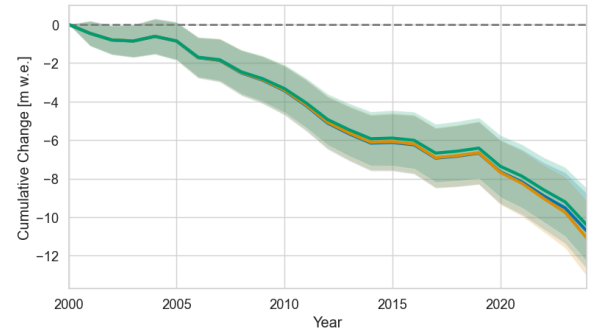
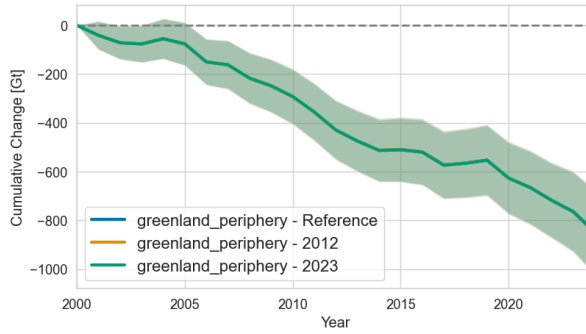
arctic_canada_north cumulative ice loss predicted by models: Reference, 2012 and 2023



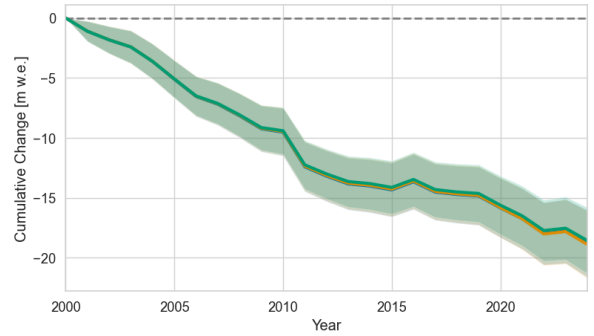
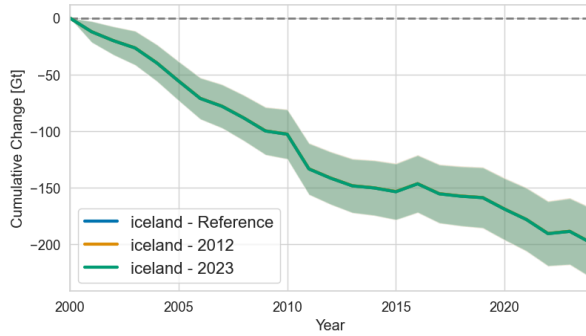
arctic_canada_south cumulative ice loss predicted by models: Reference, 2012 and 2023



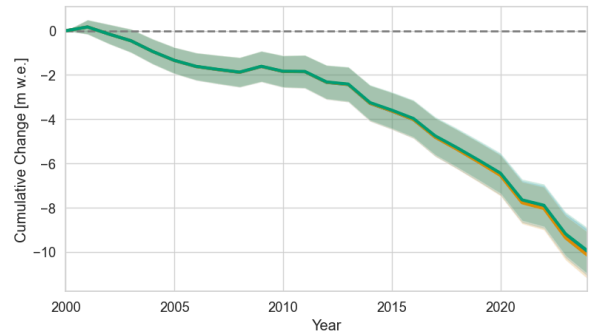
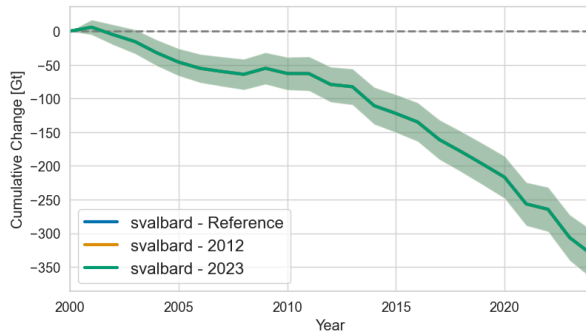
greenland_periphery cumulative ice loss predicted by models: Reference, 2012 and 2023

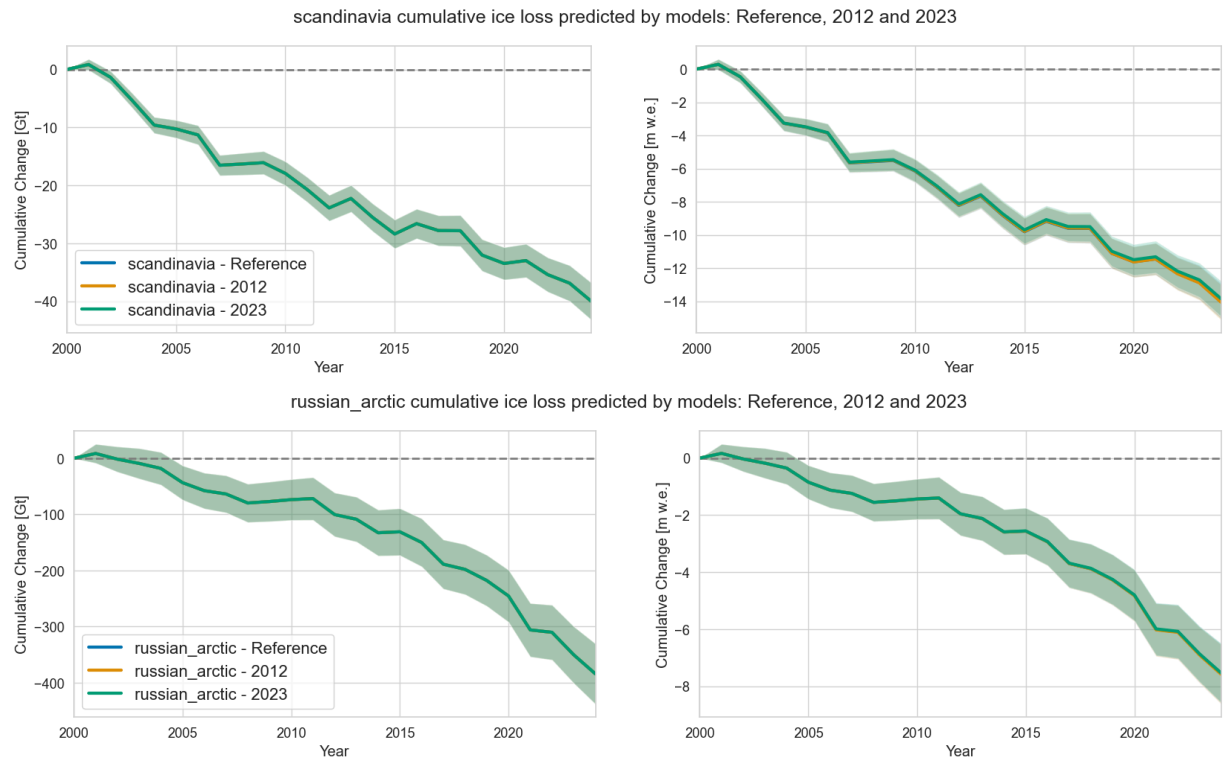


iceland cumulative ice loss predicted by models: Reference, 2012 and 2023



svalbard cumulative ice loss predicted by models: Reference, 2012 and 2023





Looking at the cumulative glacier area loss over time run comparison plots, the shape of the glacier area change seems to have surprisingly little impact. It seems to have little to no impact on glacier loss in gigatonnes and while it does seem to have an impact on glacier loss in m.w.e., this impact is small and within the margin of error of the reference run, both on a regional and on a global level, with one exception. Only the region of central Europe had the 2012 model breach the projected uncertainty of the reference model in the last 2 years of the time series.

As the differences are small, we decided to calculate the relative difference between models 2012 and 2023 and the reference model over time in order to better assess how the difference in glacier ice loss develops over time.

```
In [21]: #define function that calculates the relative difference time series
def compute_relative_diff_timeseries(df1, df2):
    merged = pd.merge(df1, df2, on="start_dates", suffixes=("_run1", "_run2"), how="outer")

    gt_run1 = merged["combined_gt_run1"]
    gt_run2 = merged["combined_gt_run2"]
    mwe_run1 = merged["combined_mwe_run1"]
    mwe_run2 = merged["combined_mwe_run2"]

    rel_diff_gt = (gt_run2 - gt_run1) / abs(gt_run1) * 100.0
    rel_diff_mwe = (mwe_run2 - mwe_run1) / abs(mwe_run1) * 100.0

    rel_diff_df = pd.DataFrame({
        'start_dates': merged['start_dates'],
        'gt_difference': rel_diff_gt,
        'mwe_difference': rel_diff_mwe
    })
    return rel_diff_df
```

```
In [22]: #plot the relative difference of the models compared to the reference models
#over time for each region
for i in range(0, len(list_of_folders)):
    #define region name and path to results
    region = list_of_folders[i]

    #get the region name without the number
    region_name = region[region.find("_")+1:-4]

    #Load in the data
    region_normal_df = pd.read_csv(os.path.join(reference_run_dir, region)).drop('Unnamed: 0', axis=1)
    region_2012_df = pd.read_csv(os.path.join(d_dir_2012, region)).drop('Unnamed: 0', axis=1)
    region_2023_df = pd.read_csv(os.path.join(d_dir_2023, region)).drop('Unnamed: 0', axis=1)

    #compute metrics & store values 2012
```

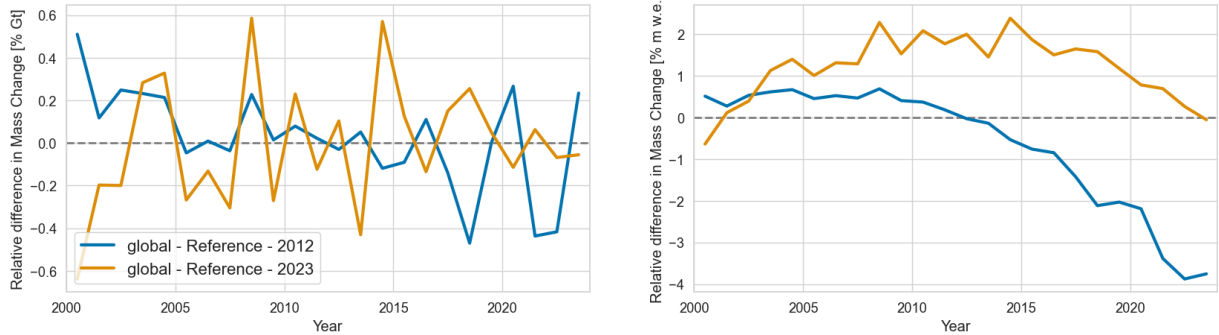
```

rel_diff_df_2012 = compute_relative_diff_timeseries(region_normal_df, region_2012_df)
rel_diff_df_2023 = compute_relative_diff_timeseries(region_normal_df, region_2023_df)

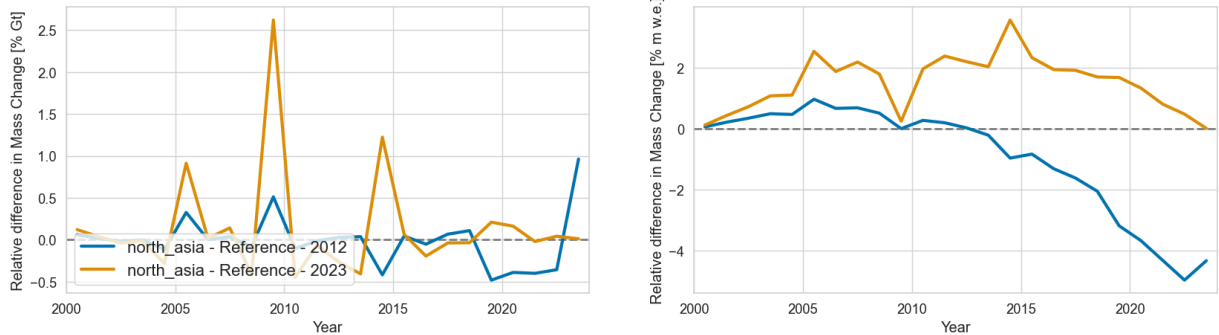
#plot
three_runs_relative_difference_plot(
    region_name,
    'Reference - 2012', 'Reference - 2023',
    rel_diff_df_2012, rel_diff_df_2023)

```

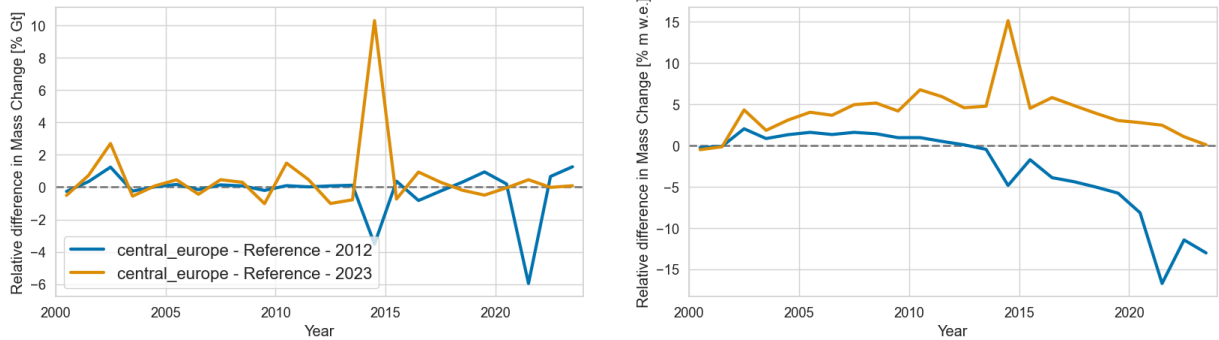
global: Relative difference in ice loss between runs



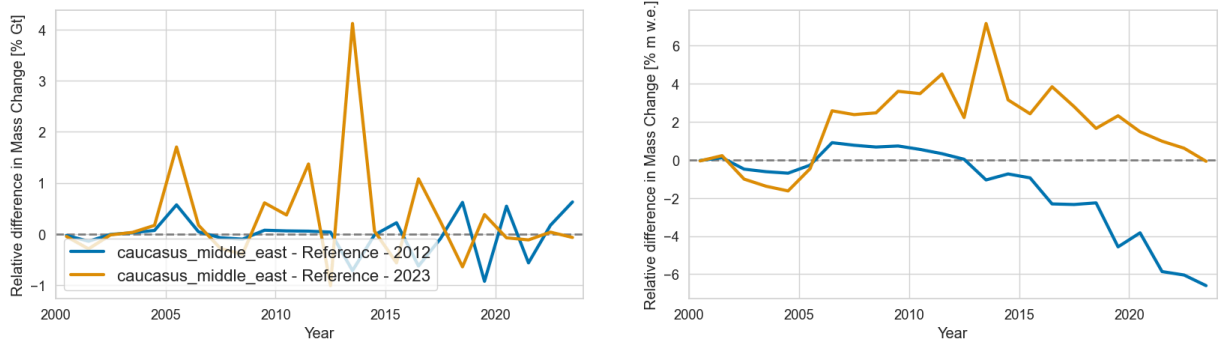
north_asia: Relative difference in ice loss between runs



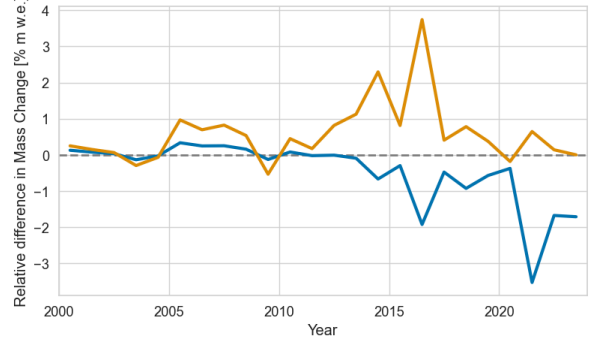
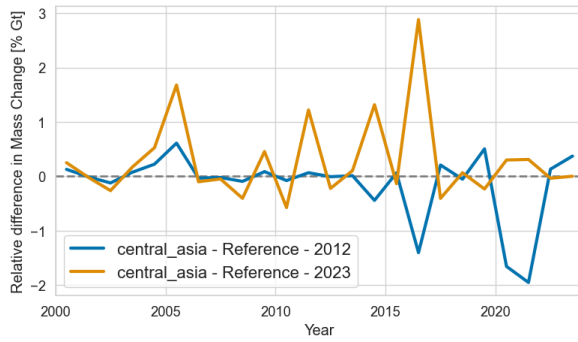
central_europe: Relative difference in ice loss between runs



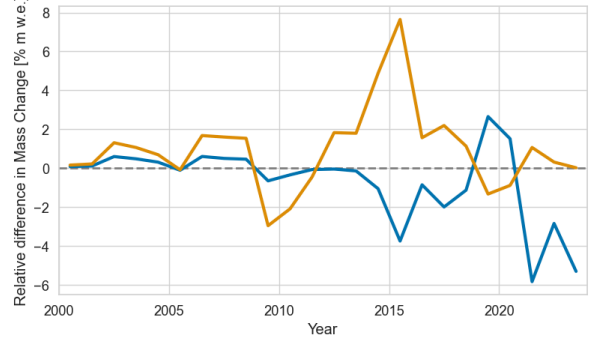
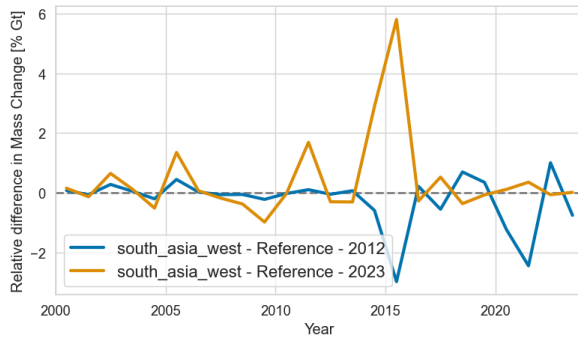
caucasus_middle_east: Relative difference in ice loss between runs



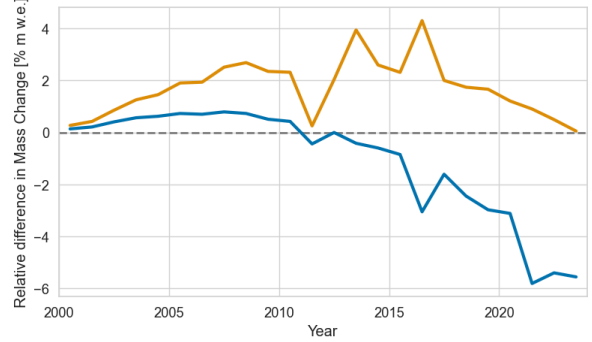
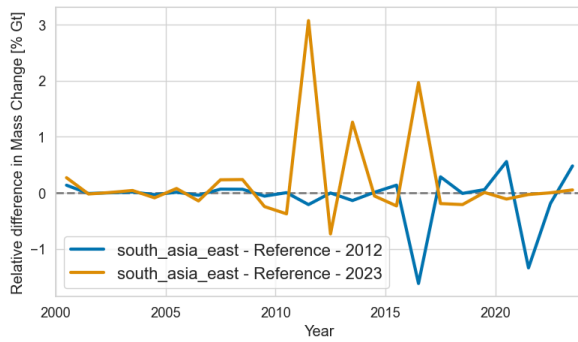
central_asia: Relative difference in ice loss between runs



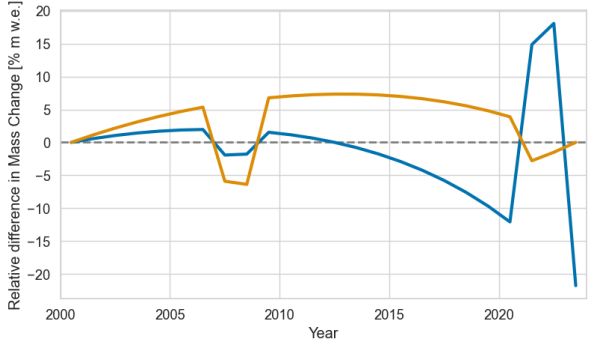
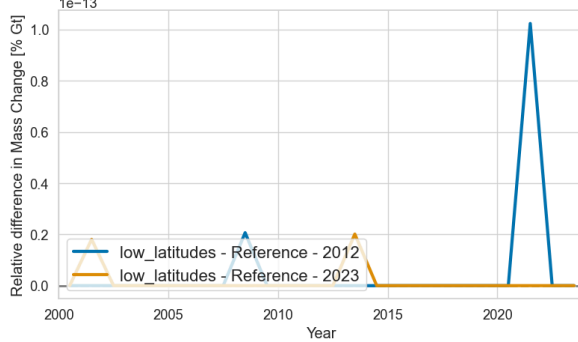
south_asia_west: Relative difference in ice loss between runs



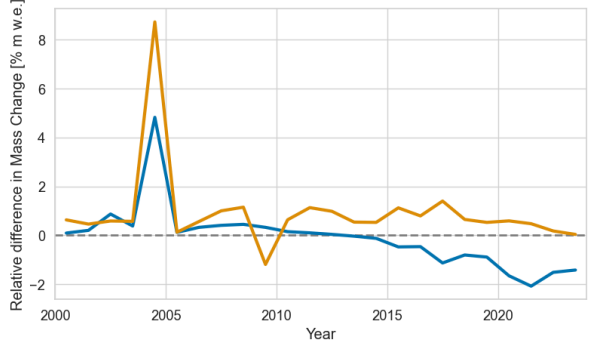
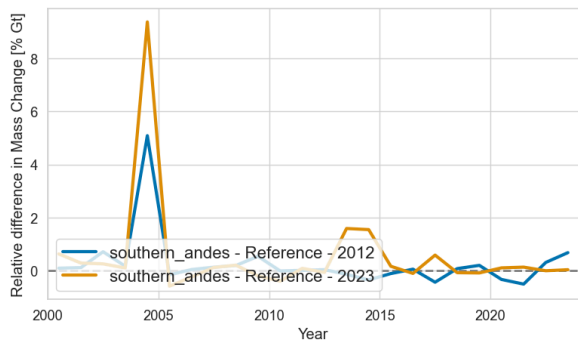
south_asia_east: Relative difference in ice loss between runs

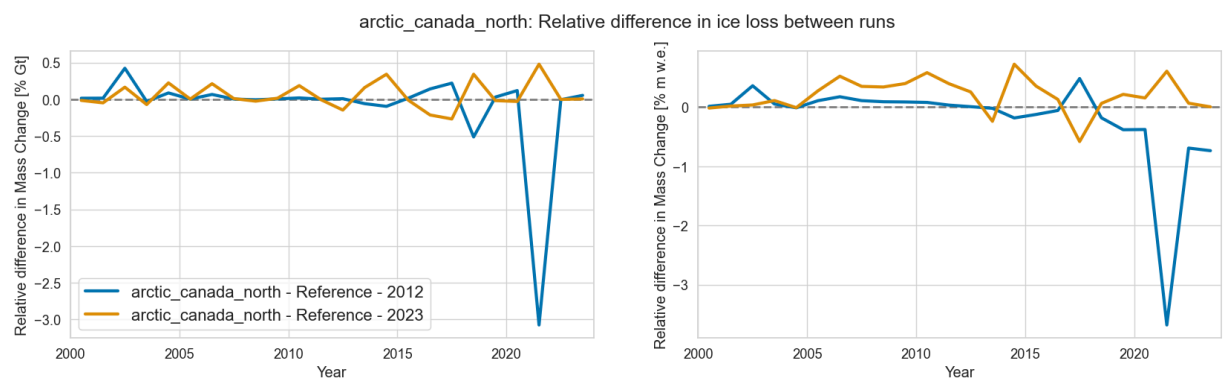
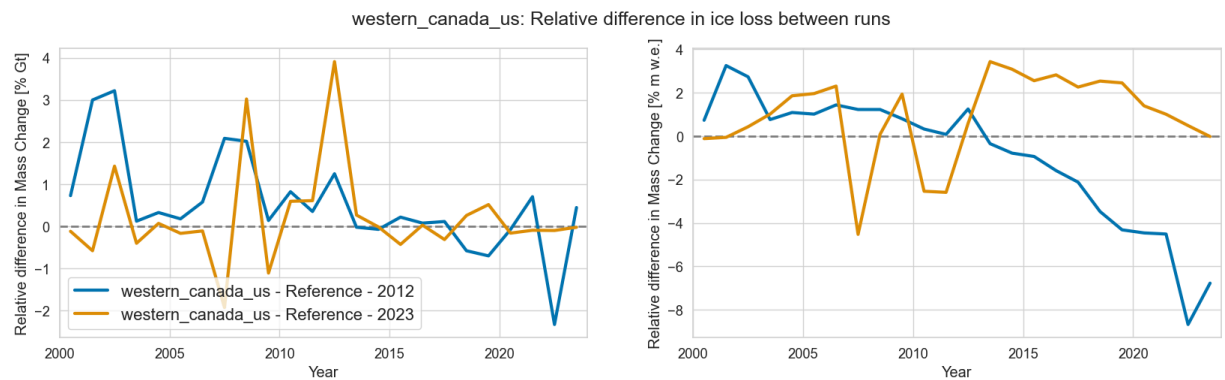
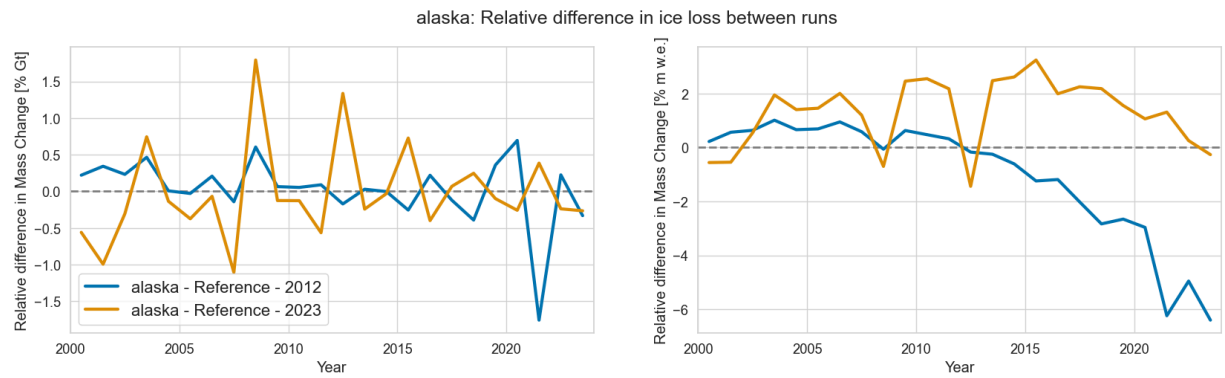
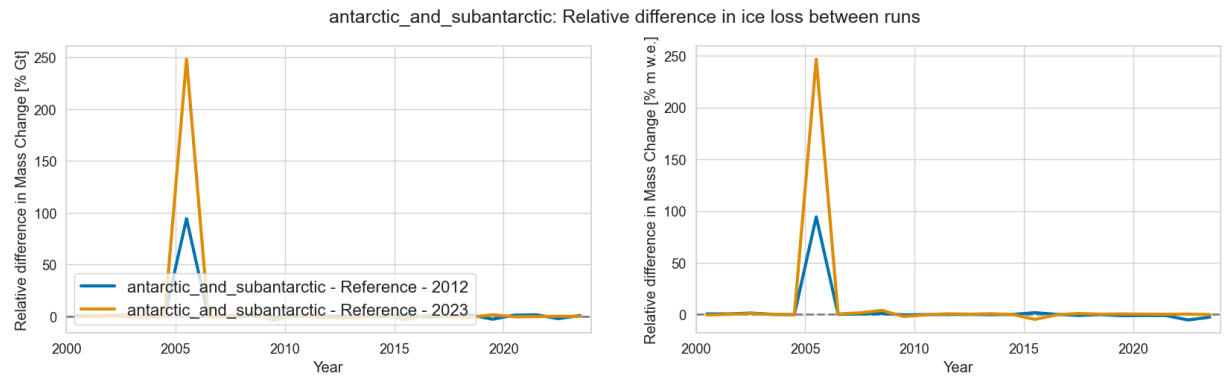
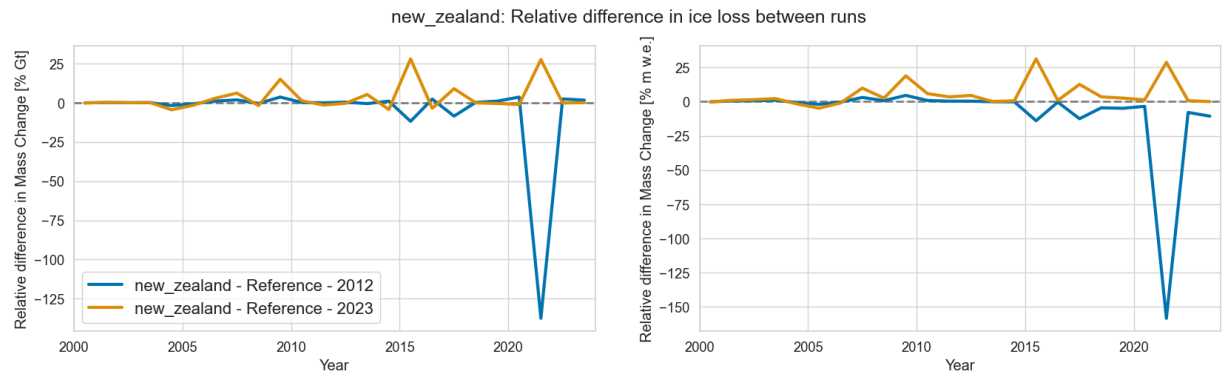


low_latitudes: Relative difference in ice loss between runs

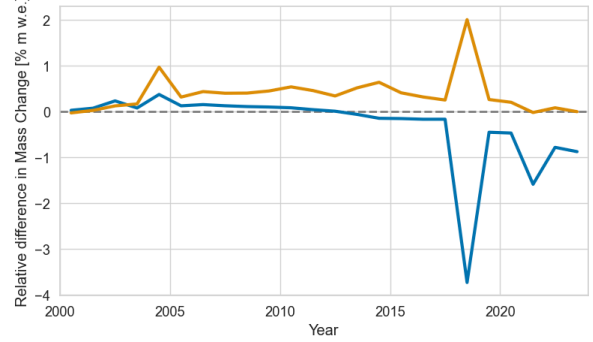
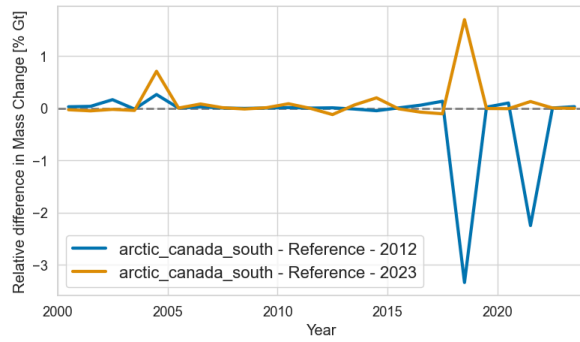


southern_andes: Relative difference in ice loss between runs

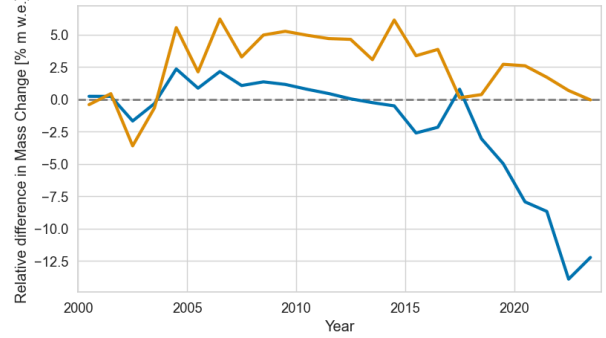
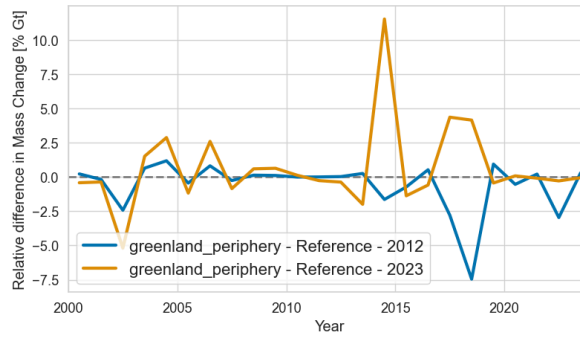




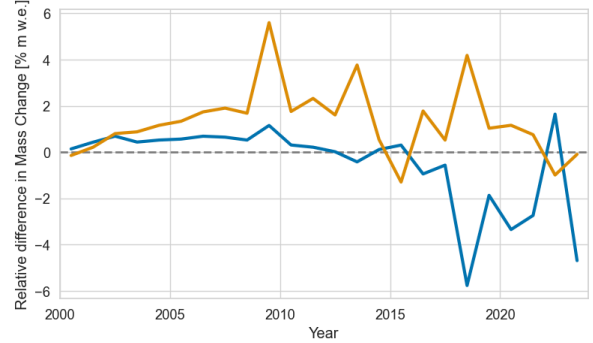
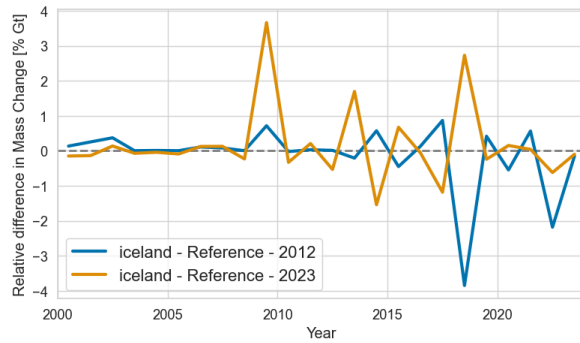
arctic_canada_south: Relative difference in ice loss between runs



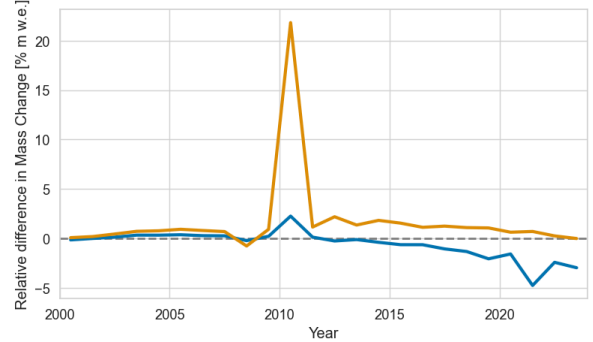
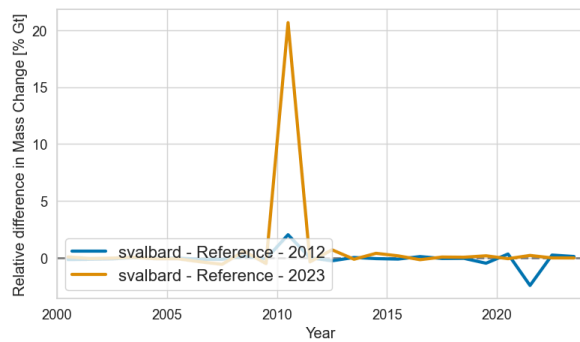
greenland_periphery: Relative difference in ice loss between runs



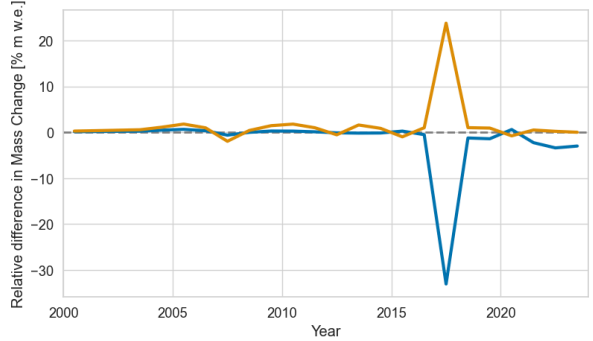
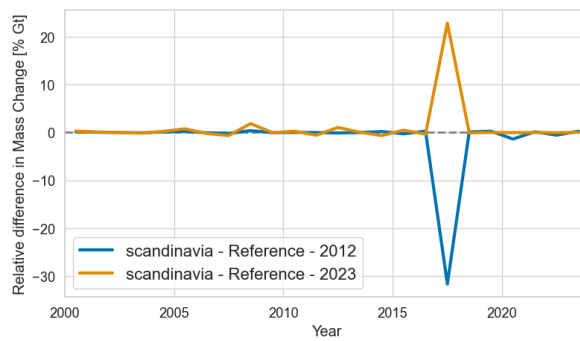
iceland: Relative difference in ice loss between runs

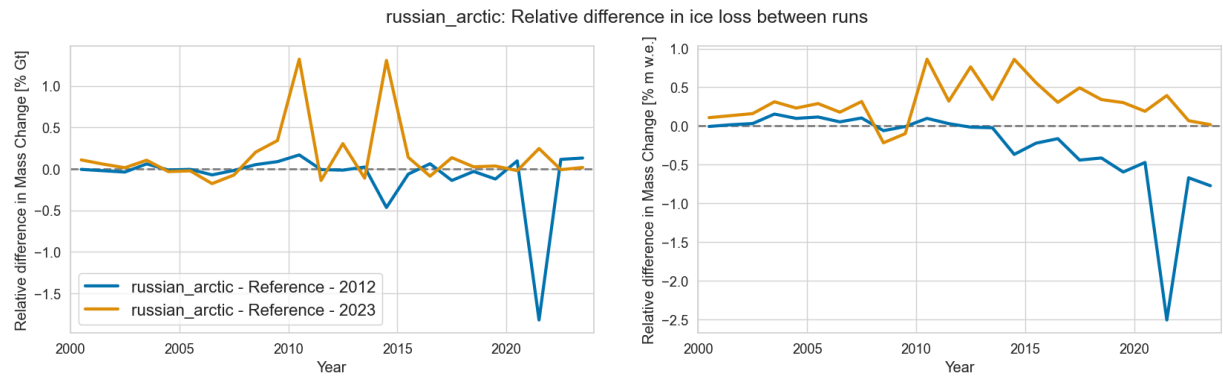


svalbard: Relative difference in ice loss between runs



scandinavia: Relative difference in ice loss between runs





The relative difference plots show the difference between the effect of the different glacier area loss calculations on the different units, both on a global and on a regional level. The global relative difference plot shows this difference clearly. For gt, the relative difference stays low (globally below 0.6%) and there does not seem to be a pattern in the development of the relative difference over time. When we look at the relative difference in modelled global glacier ice loss in m.w.e., however, we see a clear pattern in the development of the relative difference over time, one that mimicks the shape of the glacier area time series.

This trend is also visible on a regional level. Most regions show the same trend over time with perhaps one or two outliers caused by an unusually low glacier ice loss (close to 0) in that particular year. Some regions have the relative difference flip from negative to positive or the other way around, most likely those regions saw a glacier ice gain instead of a loss in these years.

In order to compare the differences per region more easily we computed the absolute difference (Gt), relative difference (%) and the Spearman correlation of the annual changes (in both gt and m.w.e.) of the two models in comparison to the reference model.

```
In [23]: #define functions to calculate the absolute difference, relative difference
#and spearman correlation

#for gt
def compute_all_metrics_gt(df1, df2):
    #merge datasets & calculate final changes
    merged = pd.merge(df1, df2, on="start_dates", suffixes=("_run1", "_run2"), how="outer")
    cumul_run1 = merged["combined_gt_run1"].sum()
    cumul_run2 = merged["combined_gt_run2"].sum()

    #compute relative difference
    if abs(cumul_run1) > 0.01:
        rel_diff = (cumul_run2 - cumul_run1) / abs(cumul_run1) * 100.0
    else:
        rel_diff = 0.0

    #compute absolute difference
    abs_diff = cumul_run2 - cumul_run1

    #compute spearman correlation
    corr, _ = spearmanr(merged["combined_gt_run1"], merged["combined_gt_run2"])

    return rel_diff, abs_diff, corr

#for mwe
def compute_all_metrics_mwe(df1, df2):
    #merge datasets & calculate final changes
    merged = pd.merge(df1, df2, on="start_dates", suffixes=("_run1", "_run2"), how="outer")
    cumul_run1 = merged["combined_mwe_run1"].sum()
    cumul_run2 = merged["combined_mwe_run2"].sum()

    #compute relative difference
    if abs(cumul_run1) > 0.01:
        rel_diff = (cumul_run2 - cumul_run1) / abs(cumul_run1) * 100.0
    else:
        rel_diff = 0.0

    #compute absolute difference
    abs_diff = cumul_run2 - cumul_run1
```

```

#compute spearman correlation
corr, _ = spearmanr(merged["combined_mwe_run1"], merged["combined_mwe_run2"])

return rel_diff, abs_diff, corr

```

```

In [24]: #calculate difference metrics
#####

#define empty lists
region_labels = []

#for 2012
rel_2012_gt = []
abs_2012_gt = []
corr_2012_gt = []

rel_2012_mwe = []
abs_2012_mwe = []
corr_2012_mwe = []

#for 2023
rel_2023_gt = []
abs_2023_gt = []
corr_2023_gt = []

rel_2023_mwe = []
abs_2023_mwe = []
corr_2023_mwe = []

#fill lists with for loop
for i in range(0, len(list_of_folders)):
    #define region name and path to results
    region = list_of_folders[i]

    #get the region name without the number
    region_name = region[region.find("_")+1:-4]

    #Load in the data
    region_normal_df = pd.read_csv(os.path.join(reference_run_dir, region)).drop('Unnamed: 0', axis=1)
    region_2012_df = pd.read_csv(os.path.join(d_dir_2012, region)).drop('Unnamed: 0', axis=1)
    region_2023_df = pd.read_csv(os.path.join(d_dir_2023, region)).drop('Unnamed: 0', axis=1)

    #store region label
    region_labels.append(region_name)

    #compute metrics & store values gt 2012
    rel_diff_val_2012_gt, abs_diff_val_2012_gt, corr_val_2012_gt = compute_all_metrics_gt(
        region_normal_df, region_2012_df
    )

    rel_2012_gt.append(rel_diff_val_2012_gt)
    abs_2012_gt.append(abs_diff_val_2012_gt)
    corr_2012_gt.append(corr_val_2012_gt)

    #compute metrics & store values mwe 2012
    rel_diff_val_2012_mwe, abs_diff_val_2012_mwe, corr_val_2012_mwe = compute_all_metrics_mwe(
        region_normal_df, region_2012_df
    )

    rel_2012_mwe.append(rel_diff_val_2012_mwe)
    abs_2012_mwe.append(abs_diff_val_2012_mwe)
    corr_2012_mwe.append(corr_val_2012_mwe)

    #compute metrics & store values gt 2023
    rel_diff_val_2023_gt, abs_diff_val_2023_gt, corr_val_2023_gt = compute_all_metrics_gt(
        region_normal_df, region_2023_df
    )

    rel_2023_gt.append(rel_diff_val_2023_gt)
    abs_2023_gt.append(abs_diff_val_2023_gt)
    corr_2023_gt.append(corr_val_2023_gt)

    #compute metrics & store values mwe 2023
    rel_diff_val_2023_mwe, abs_diff_val_2023_mwe, corr_val_2023_mwe = compute_all_metrics_mwe(
        region_normal_df, region_2023_df
    )

```

```
)

rel_2023_mwe.append(rel_diff_val_2023_mwe)
abs_2023_mwe.append(abs_diff_val_2023_mwe)
corr_2023_mwe.append(corr_val_2023_mwe)
```

```
In [25]: #plot for gt
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(16, max(6, len(region_labels) * 0.6)), sharey=True)

n = len(region_labels)
y_pos = np.arange(n)

w= 0.4

#####
# Panel 1: Global relative difference (%)
ax1.barh(y_pos - w/2, rel_2012_gt, height=w, label='2012')
ax1.barh(y_pos + w/2, rel_2023_gt, height=w, label='2023')

ax1.set_yticks(y_pos)
ax1.set_yticklabels(region_labels, fontsize=12)
ax1.invert_yaxis()
ax1.set_xlabel("Relative Change (%)")
ax1.axvline(0, color="black", linewidth=0.8)

#text 2012
max_abs_rel_2012 = max(abs(v) for v in rel_2012_gt) or 1
for i, val in enumerate(rel_2012_gt):
    if val >= 0:
        ax1.text(
            val + 0.02 * max_abs_rel_2012, i - w/2,
            f"+{val:.2f}%", va="center", fontsize=10
        )
    else:
        ax1.text(
            val - 0.02 * max_abs_rel_2012, i - w/2,
            f"{val:.2f}%", va="center", ha="right", fontsize=10
        )

#text 2023
max_abs_rel_2023 = max(abs(v) for v in rel_2023_gt) or 1
for i, val in enumerate(rel_2023_gt):
    if val >= 0:
        ax1.text(
            val + 0.02 * max_abs_rel_2023, i + w/2,
            f"+{val:.2f}%", va="center", fontsize=10
        )
    else:
        ax1.text(
            val - 0.02 * max_abs_rel_2023, i + w/2,
            f"{val:.2f}%", va="center", ha="right", fontsize=10
        )

pad = max(max_abs_rel_2012, max_abs_rel_2023)* 0.20
ax1.set_xlim(min(rel_2012_gt + rel_2023_gt) - pad, max(rel_2012_gt + rel_2023_gt) + pad)

#####
# Panel 2: Global absolute difference (Gt)
ax2.barh(y_pos - w/2, abs_2012_gt, height=w, label='2012')
ax2.barh(y_pos + w/2, abs_2023_gt, height=w, label='2023')

ax2.set_xlabel("Absolute Difference (Gt)")
ax2.axvline(0, color="black", linewidth=0.8)

#text 2012
max_abs_diff_2012 = max(abs(v) for v in abs_2012_gt) or 1
for i, val in enumerate(abs_2012_gt):
    if val >= 0:
        ax2.text(
            val + 0.02 * max_abs_diff_2012, i - w/2,
            f"+{val:.2f}", va="center", fontsize=10
        )
    else:
        ax2.text(
            val - 0.02 * max_abs_diff_2012, i - w/2,
```

```

        f"{val:.2f}", va="center", ha="right", fontsize=10
    )

#text 2023
max_abs_diff_2023 = max(abs(v) for v in abs_2023_gt) or 1
for i, val in enumerate(abs_2023_gt):
    if val >= 0:
        ax2.text(
            val + 0.02 * max_abs_diff_2023, i + w/2,
            f"{val:.2f}", va="center", fontsize=10
        )
    else:
        ax2.text(
            val - 0.02 * max_abs_diff_2023, i + w/2,
            f"{val:.2f}", va="center", ha="right", fontsize=10
        )

pad = max(max_abs_diff_2012, max_abs_diff_2023) * 0.30
ax2.set_xlim(min(abs_2012_gt + abs_2023_gt) - pad, max(abs_2012_gt + abs_2023_gt) + pad)

#####3
# Panel 3: Global correlation
ax3.barh(y_pos - w/2, corr_2012_gt, height=w, label='2012')
ax3.barh(y_pos + w/2, corr_2023_gt, height=w, label='2023')

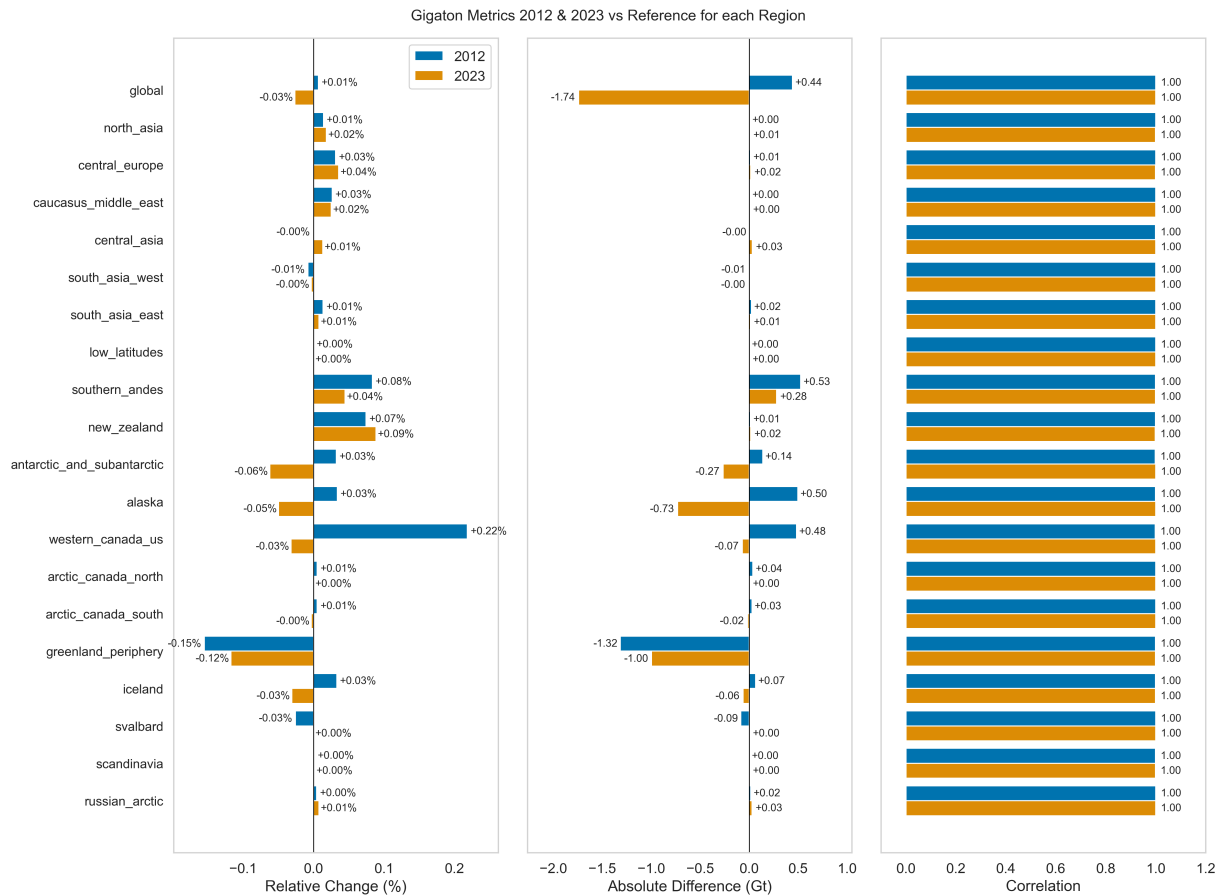
ax3.set_xlabel("Correlation")
ax3.set_xlim(-0.1, 1.2)

#text 2012
for i, val in enumerate(corr_2012_gt):
    if val >= 0:
        ax3.text(val + 0.02, i - w/2, f"{val:.2f}", va="center", fontsize=10)
    else:
        ax3.text(val + 0.08, i - w/2, f"{val:.2f}", va="center", ha="right", fontsize=10)

#text 2023
for i, val in enumerate(corr_2023_gt):
    if val >= 0:
        ax3.text(val + 0.02, i + w/2, f"{val:.2f}", va="center", fontsize=10)
    else:
        ax3.text(val + 0.08, i + w/2, f"{val:.2f}", va="center", ha="right", fontsize=10)

fig.suptitle(f"Gigaton Metrics 2012 & 2023 vs Reference for each Region", fontsize=14)
fig.tight_layout()
fig.set_dpi(300)
ax1.legend()
plt.show()

```



```
In [26]: #plot for mwe
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(16, max(6, len(region_labels) * 0.6)), sharey=True)

n = len(region_labels)
y_pos = np.arange(n)

w = 0.4

#####
# Panel 1: Global relative difference (%)
ax1.barh(y_pos - w/2, rel_2012_mwe, height=w, label='2012')
ax1.barh(y_pos + w/2, rel_2023_mwe, height=w, label='2023')

ax1.set_yticks(y_pos)
ax1.set_yticklabels(region_labels, fontsize=12)
ax1.invert_yaxis()
ax1.set_xlabel("Relative Change (%)")
ax1.axvline(0, color="black", linewidth=0.8)

#text 2012
max_abs_rel_2012 = max(abs(v) for v in rel_2012_mwe) or 1
for i, val in enumerate(rel_2012_mwe):
    if val >= 0:
        ax1.text(
            val + 0.02 * max_abs_rel_2012, i - w/2,
            f"+{val:.2f}%", va="center", fontsize=10
        )
    else:
        ax1.text(
            val - 0.02 * max_abs_rel_2012, i - w/2,
            f"{val:.2f}%", va="center", ha="right", fontsize=10
        )

#text 2023
max_abs_rel_2023 = max(abs(v) for v in rel_2023_mwe) or 1
for i, val in enumerate(rel_2023_mwe):
    if val >= 0:
        ax1.text(
            val + 0.02 * max_abs_rel_2023, i + w/2,
            f"+{val:.2f}%", va="center", fontsize=10
        )
```

```

else:
    ax1.text(
        val - 0.02 * max_abs_rel_2023, i + w/2,
        f"{val:.2f}%", va="center", ha="right", fontsize=10
    )

pad = max(max_abs_rel_2012, max_abs_rel_2023) * 0.20
ax1.set_xlim(min(rel_2012_mwe + rel_2023_mwe) - pad, max(rel_2012_mwe + rel_2023_mwe) + pad)

#####
# Panel 2: Global absolute difference (mwe)
ax2.barh(y_pos - w/2, abs_2012_mwe, height=w, label='2012')
ax2.barh(y_pos + w/2, abs_2023_mwe, height=w, label='2023')

ax2.set_xlabel("Absolute Difference (m w.e.)")
ax2.axvline(0, color="black", linewidth=0.8)

#text 2012
max_abs_diff_2012 = max(abs(v) for v in abs_2012_mwe) or 1
for i, val in enumerate(abs_2012_mwe):
    if val >= 0:
        ax2.text(
            val + 0.02 * max_abs_diff_2012, i - w/2,
            f"{val:.2f}", va="center", fontsize=10
        )
    else:
        ax2.text(
            val - 0.02 * max_abs_diff_2012, i - w/2,
            f"{val:.2f}", va="center", ha="right", fontsize=10
        )

#text 2023
max_abs_diff_2023 = max(abs(v) for v in abs_2023_mwe) or 1
for i, val in enumerate(abs_2023_mwe):
    if val >= 0:
        ax2.text(
            val + 0.02 * max_abs_diff_2023, i + w/2,
            f"{val:.2f}", va="center", fontsize=10
        )
    else:
        ax2.text(
            val - 0.02 * max_abs_diff_2023, i + w/2,
            f"{val:.2f}", va="center", ha="right", fontsize=10
        )

pad = max(max_abs_diff_2012, max_abs_diff_2023) * 0.30
ax2.set_xlim(min(abs_2012_mwe + abs_2023_mwe) - pad, max(abs_2012_mwe + abs_2023_mwe) + pad)

##### 3
# Panel 3: Global correlation
ax3.barh(y_pos - w/2, corr_2012_mwe, height=w, label='2012')
ax3.barh(y_pos + w/2, corr_2023_mwe, height=w, label='2023')

ax3.set_xlabel("Correlation")
ax3.set_xlim(-0.1, 1.2)

#text 2012
for i, val in enumerate(corr_2012_mwe):
    if val >= 0:
        ax3.text(val + 0.02, i - w/2, f"{val:.2f}", va="center", fontsize=10)
    else:
        ax3.text(val + 0.08, i - w/2, f"{val:.2f}", va="center", ha="right", fontsize=10)

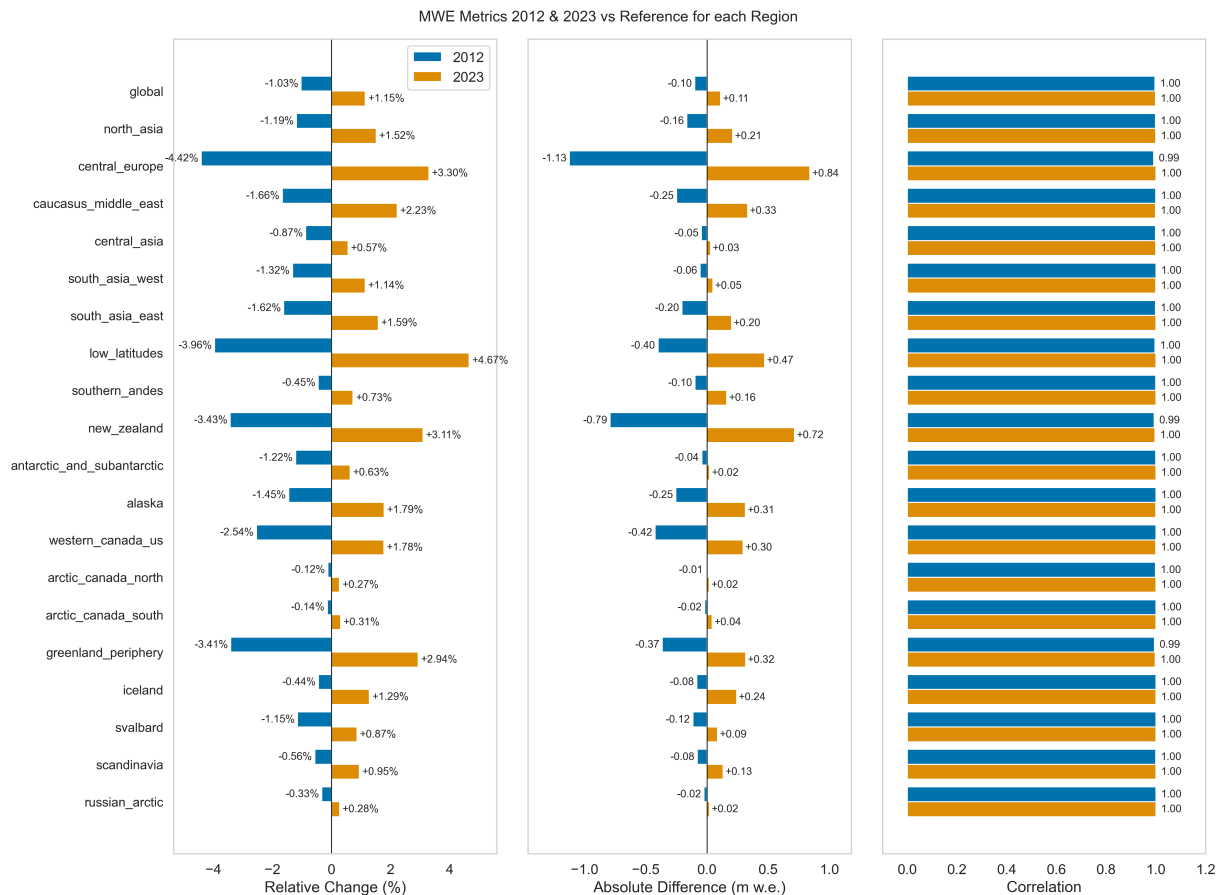
#text 2023
for i, val in enumerate(corr_2023_mwe):
    if val >= 0:
        ax3.text(val + 0.02, i + w/2, f"{val:.2f}", va="center", fontsize=10)
    else:
        ax3.text(val + 0.08, i + w/2, f"{val:.2f}", va="center", ha="right", fontsize=10)

fig.suptitle(f"MWE Metrics 2012 & 2023 vs Reference for each Region", fontsize=14)
fig.tight_layout()
fig.set_dpi(300)

```



```
ax1.legend()
plt.show()
```



Looking at the comparison between the final relative and absolute difference of the two models compared to the reference model on a regional level, the difference per region and unit is very clear. The accelerating glacier area loss of both models 2012 and 2023 has minimal impact on the results in gigatonnes. The relative difference is 0.2% at most (western Canada and us), with an absolute difference of maximum 1.74 gigatonnes (global).

When it comes to the m.w.e. results, the accelerating glacier area loss of both models 2012 and 2023 has a larger impact. Model 2023 has a positive final relative and absolute difference in glacier ice loss (meaning that the total ice loss was less than for the reference) for all regions, model 2012 has a negative final relative and absolute difference in glacier ice loss all regions. The region with the largest relative and absolute differences between model 2012 and the reference model is central Europe, with a relative difference of -4.4% and an absolute difference of -1.13 m.w.e.. Central Europe also has the largest absolute difference for model 2023. The region with the largest relative difference for model 2023 is the region low latitudes, with a relative difference of 4.7%. While the differences are larger for m.w.e. than for gt, they are all within 5% and thus still small. Furthermore, as stated before, almost all of them are within the uncertainty of the reference model. Lastly, the spearman correlation for all regions and metrics and for both models is close to 1, indicating that, while the values might be different, the patterns of glacier loss over time are not significantly different.

The most likely reason that the differences in Gt are so small compared to the differences in m.w.e. is the way the GlaMBIE model accounts for glacier area shrinkage. When converting values from m.w.e. to gt, they first remove the effect of glacier area shrinkage using the formula below, relating all glacier areas to that regions reference glacier area. They then use that area to convert the values from m.w.e. to Gt. When they convert the values the other way around, they first use the reference glacier area and then apply the glacier area change to the result. As a result, glacier area shrinkage has little to no effect on the results in gigatonnes. It is unclear to us why they remove the effect of glacier shrinkage before converting units to gigatonnes.

Most likely this is also the reason why changing the RGI did not meaningfully impact the results in m.w.e. but did impact the results in gigatonnes: the gigaton calculation is more affected by the reference glacier area value but not by the glacier area shrinkage calculation, while for the m.w.e. calculation this is the other way around.

converting from m.w.e. to Gt (remove area glacier change):

$$change_{adjusted} = \frac{change_{mwe}}{\left(\frac{Area_{reference}}{area_{adjusted}}\right)}$$

$$change_{Gt} = change_{adjusted} * \rho * Area_{reference}$$

where:

$change$ = the values that are being converted

$area_{adjusted}$ = the glacier area calculated using the formula's above

ρ = density of water

converting from Gt to m w.e. (add area glacier change):

$$change_{adjusted} = \frac{area_{reference}}{area_{adjusted}} * change_{Gt}$$

$$change_{mwe} = \frac{change_{adjusted}}{Area_{reference} * \rho}$$

where:

$change$ = the values that are being converted

$area_{adjusted}$ = the glacier area calculated using the formula's above

ρ = density of water

To summarise: while the different glacier area loss calculations of models 2012 and 2023 have a larger impact on the m w.e. results than on the Gt results, the total relative difference stays under 5% and the absolute difference stays largely within the uncertainty of the reference model for all regions and both models, indicating that changing the glacier area loss calculations did not have a significant impact on the results.

4. Test excluded datasets

The GlAMBE paper uses 233 datasets of which 38 datasets were excluded. Zemp et al. (2025) give the following reasons for data exclusion of gravimetry datasets due to

- implausible trends and high measurement noise
- leakage between regions
- interference of hydrological mass changes

Besides for gravimetry, there are datasets from other methods excluded as well. The methods section only states that exclusions were based on "expert evaluation" assigning "no confidence" at a regional level during a workshop. No further reasons are provided for individual datasets. we decided to test on global and regional results.

```
In [27]: dataset_meta = pd.read_csv(
    'capstone_glambe/input/input_datasets/dataset_inclusion.csv', encoding='latin1'
)

# Long format
id_cols = ['Research team', 'Observation method', 'Sensor / Product']
region_cols = [col for col in dataset_meta.columns if col not in id_cols]

long_data = (
    dataset_meta
    .melt(id_vars=id_cols, value_vars=region_cols, var_name='region', value_name='inclusion')
    .dropna(subset=['inclusion'])
)

long_data['dataset'] = long_data['Research team'] + " (" + long_data['Observation method'] + ")"

# Build heatmap pivot, sorted by observation method
heatmap_data = long_data.pivot_table(
    index='dataset',
    columns='region',
```

```

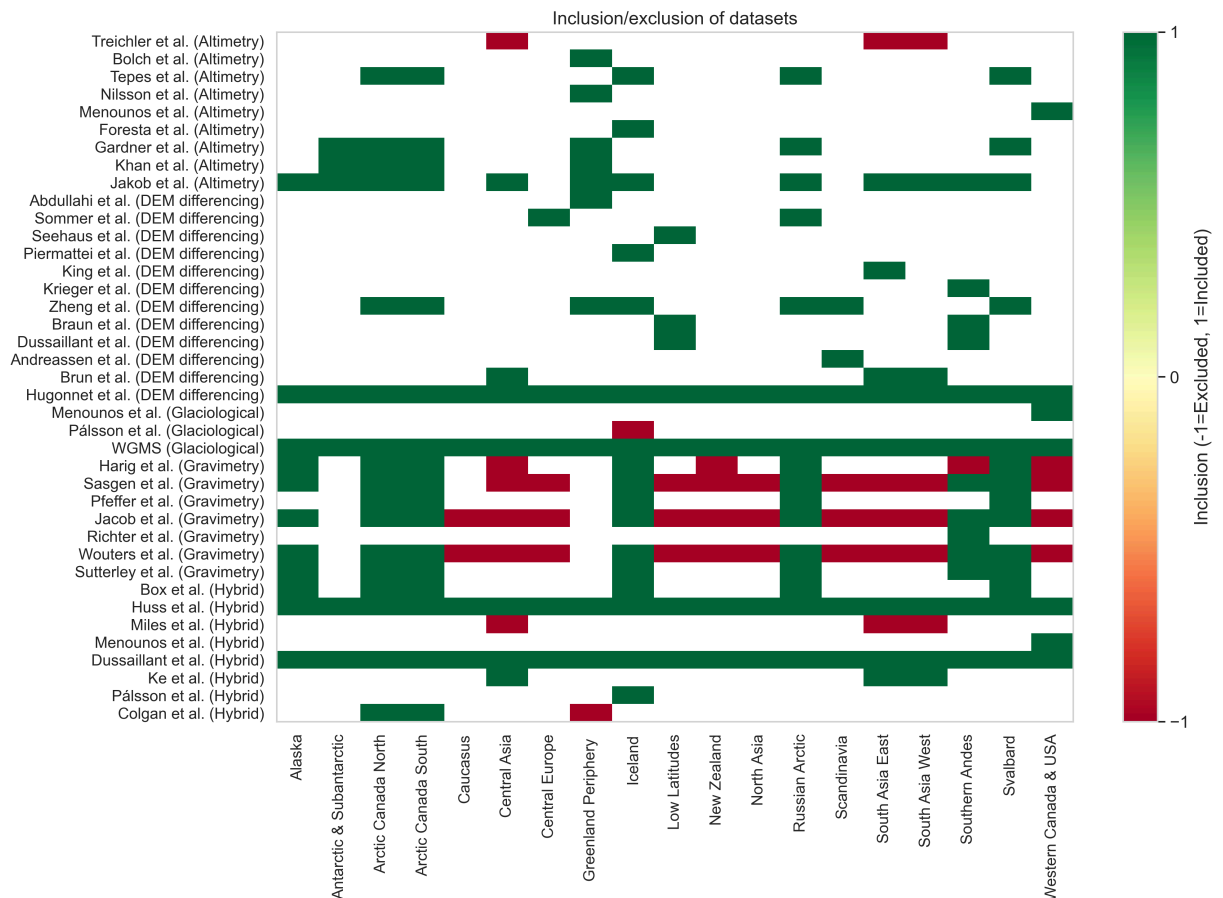
        values='inclusion',
        aggfunc='first'
    )
heatmap_data = heatmap_data.sort_index(key=lambda idx: idx.str.extract(r'\((.+)\)')[0].fillna(''))

# Plot
fig, ax = plt.subplots(figsize=(14, 10), dpi=300)
im = ax.imshow(heatmap_data, cmap='RdYlGn', aspect='auto', vmin=-1, vmax=1)
fig.colorbar(im, ax=ax, ticks=[-1, 0, 1], label='Inclusion (-1=Excluded, 1=Included)')

ax.set_title("Inclusion/exclusion of datasets", fontsize=14)
ax.set_xticks(range(len(heatmap_data.columns)))
ax.set_xticklabels(heatmap_data.columns, rotation=90, fontsize=12)
ax.set_yticks(range(len(heatmap_data.index)))
ax.set_yticklabels(heatmap_data.index, fontsize=12)

plt.tight_layout()
plt.show()

```



Heatmap showing the inclusion (green, +1) or exclusion (red, -1) status of all datasets submitted to GlaMBIE, arranged by research team and observation method (rows) against glacier region (columns). Based on Zemp et al. (2025) Source Data (Extended Data Fig. 1).

The heatmap shows the inclusion and exclusion based on the metadata given by Zemp et al. (2025) [Source Data Extended Data Fig. 1]. Overall, while there is thorough documentation of the collected datasets and their original publication, trying to connect them with the configuration was challenging and possibly inconsistent. We found that some datasets were excluded in the configuration (with which we were able to reproduce the results) and not able to run the model which were included according due to the metadata.

Input datasets

The input datasets are visualised per region (`scripts/plots_datasets.ipynb` , outputs in `capstone_glaMBIE/output/datasets/`), separated by included and excluded status. We can see some datasets with obvious deviations in uncertainty and trend from the rest such as Jacob et al. (2012) which is always excluded. Other datasets though, like Gardner et al. (2013) or Tepes et al. (2021), are excluded despite them being not being

remarkably different from other included datasets (though one could argue about very tight confidence intervals in these two cases). Subsequently, we wanted to test the effect of their inclusion.

It is possible to change the included datasets simply by editing the configuration file. While it would have been ideal to test the sensitivity of the estimates for every individual dataset, the scope of this was outside our ability, partially due to the runtimes of the model with manifold combinations of datasets. Instead, we decided to compare three dataset configurations to the default configuration:

- **Include all** possible datasets that were removed to see the overall effect of excluded datasets.
- **Include all gravimetry datasets** to verify the statement from the paper of unrealistic trends for those gravimetry datasets.
- **Include all except gravimetry datasets** to see the effect of other datasets.

However, we were not able to include all datasets as intended, as the model failed on some inclusions. Running excluded datasets required a small code fix: datasets with a temporal resolution of one year or greater that are not already aligned to the desired annual grid must first be shifted onto it before conversion to annual trends. This was achieved by calling `shift_timeseries_to_annual_grid_proportionally()` prior to `convert_timeseries_to_annual_trends()` in the `processing_helpers.py`:

```
# Fix for running excluded datasets that are not on the desired annual grid
if ds.data.max_temporal_resolution >= 1 and not
ds.timeseries_is_annual_grid(year_type=year_type):
    ds = ds.shift_timeseries_to_annual_grid_proportionally(year_type=year_type)
datasets.append(ds.convert_timeseries_to_annual_trends(year_type=year_type))
```

Even with this fix applied, the following datasets still failed to run:

- Bolch_2013 in 1 region: greenland_periphery
- Gardner2013_icesat in 2 regions: antarctic_and_subantarctic, greenland_periphery
- Khan in 2 regions: arctic_canada_north, arctic_canada_south
- Treichler_snowfall in 3 regions: central_asia, south_asia_east, south_asia_west

This means, from a total of 233 datasets, we were not able to include 8 datasets due to technical issues with the model. Additionally, the dataset UHZ_GlaciolSineWave as not included as it seems to be the calibration dataset for seasonal homogenization (Zemp and Welty, 2023), while WGMS-beta was excluded as to be a secondary WGMS glaciological series to the already included WGMS_mean-ba dataset, though no documentation was found to specifically confirm this.

```
In [28]: reference_run = Path(reference_run_dir)
grav_run = Path(datasets_including_gravimetry)
all_run = Path(datasets_including_all)
most_except_run = Path(datasets_including_except_gravimetry)

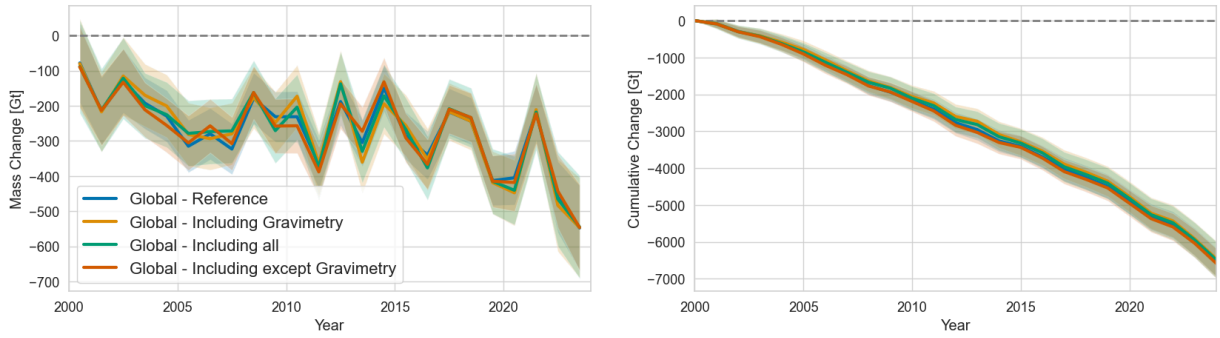
for region_file in list_of_folders:
    ref_df = pd.read_csv(reference_run / region_file).drop(columns=["Unnamed: 0"], errors="ignore")
    grav_df = pd.read_csv(grav_run / region_file).drop(columns=["Unnamed: 0"], errors="ignore")
    most_df = pd.read_csv(all_run / region_file).drop(columns=["Unnamed: 0"], errors="ignore")
    most_except_df = pd.read_csv(
        most_except_run / region_file
    ).drop(columns=["Unnamed: 0"], errors="ignore")

    region_name = region_file.split("_", 1)[1].replace(".csv", "").replace("_", " ").title()

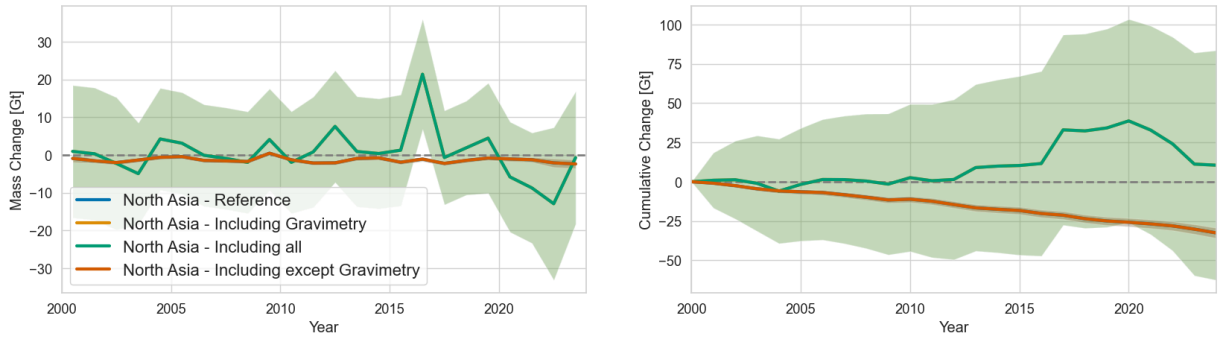
    four_runs_annual_cumulative_plot(
        region_name,
        "Reference", "Including Gravimetry", "Including all", "Including except Gravimetry",
        ref_df, grav_df, most_df, most_except_df
    )

plt.show()
```

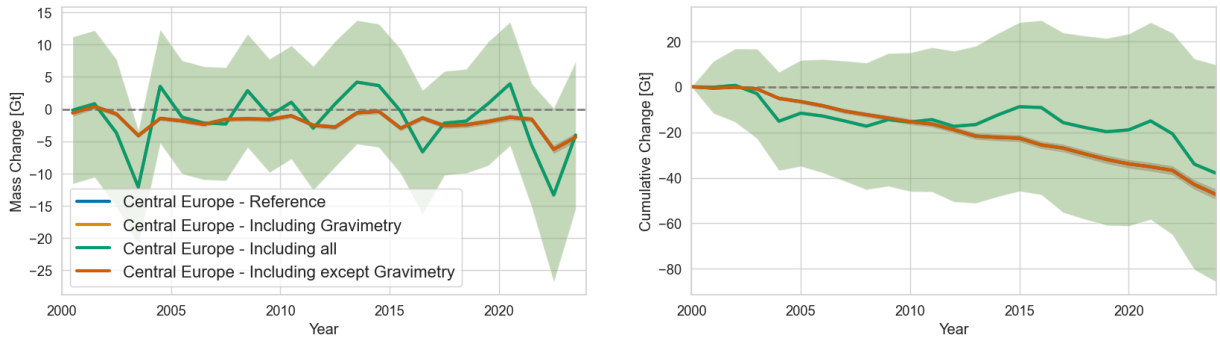
Global: annual and cumulative comparison



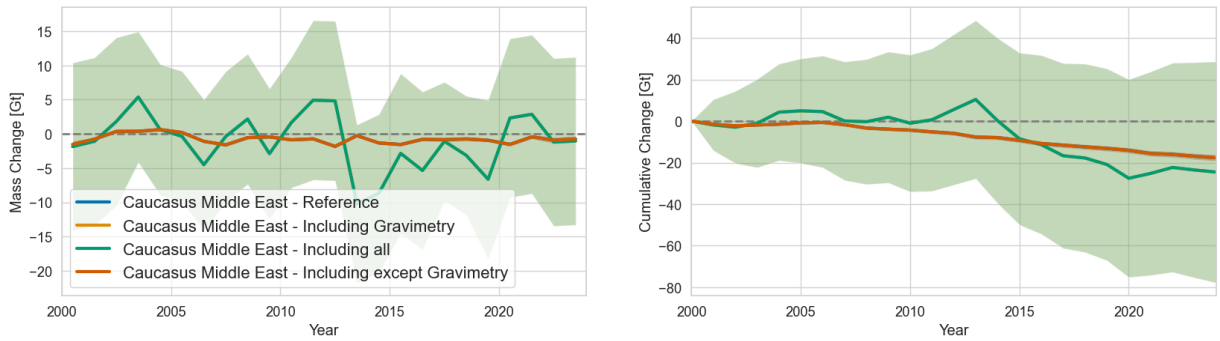
North Asia: annual and cumulative comparison



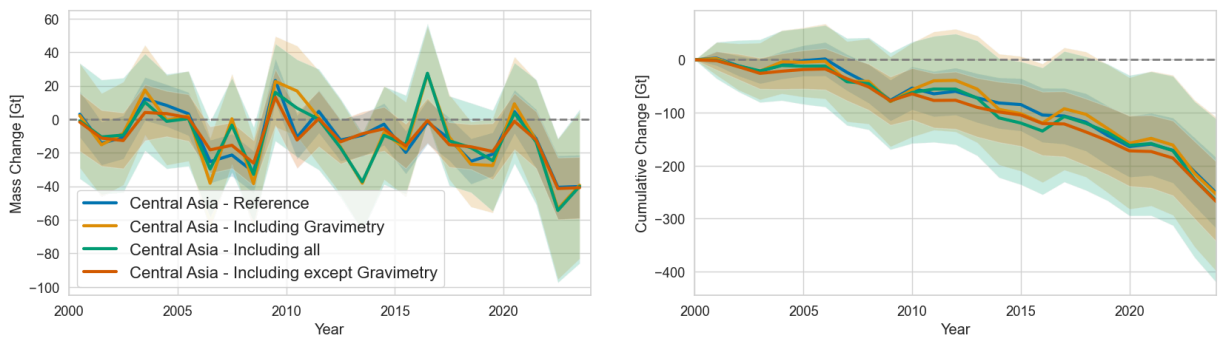
Central Europe: annual and cumulative comparison



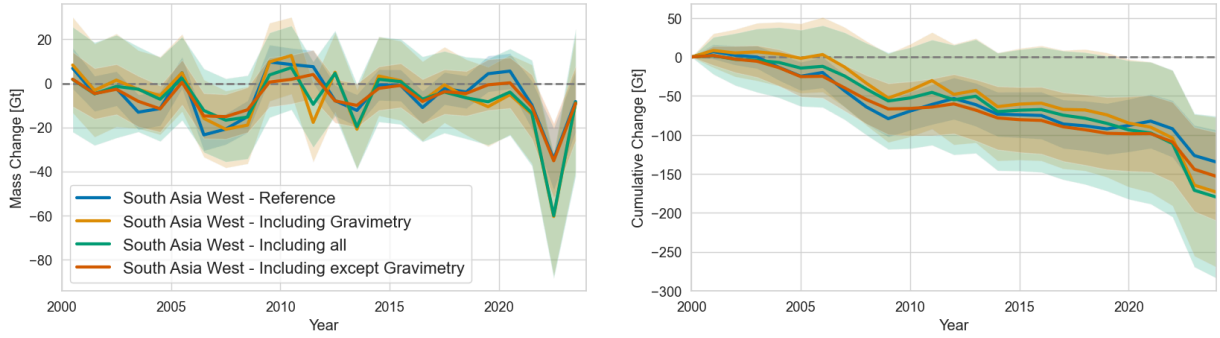
Caucasus Middle East: annual and cumulative comparison



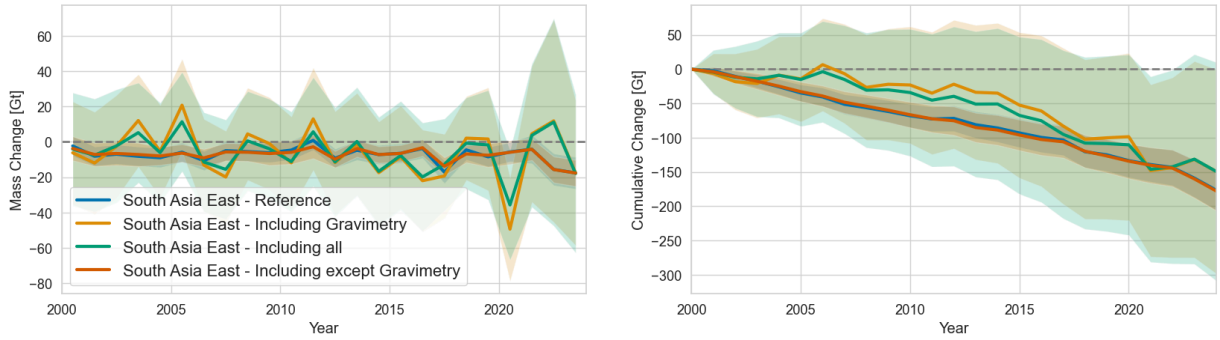
Central Asia: annual and cumulative comparison



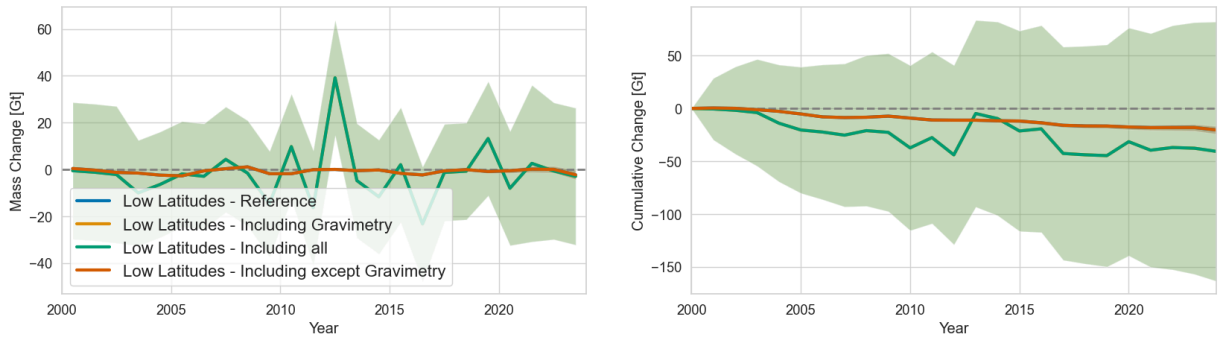
South Asia West: annual and cumulative comparison



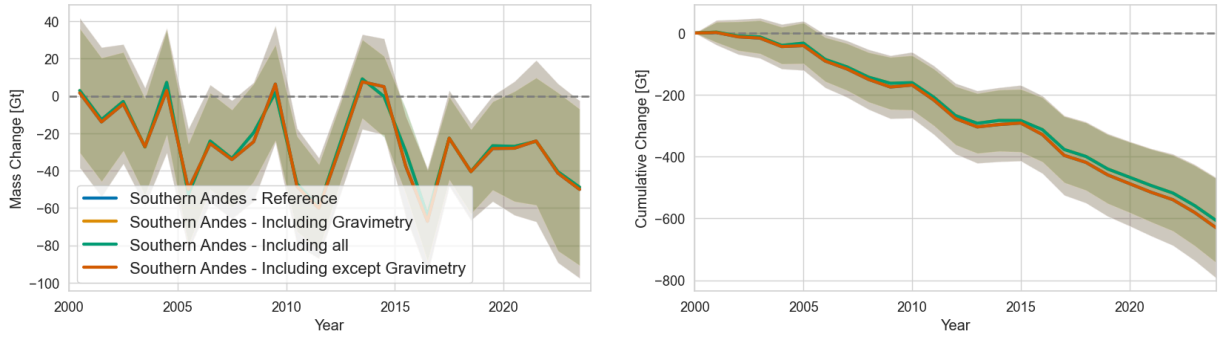
South Asia East: annual and cumulative comparison



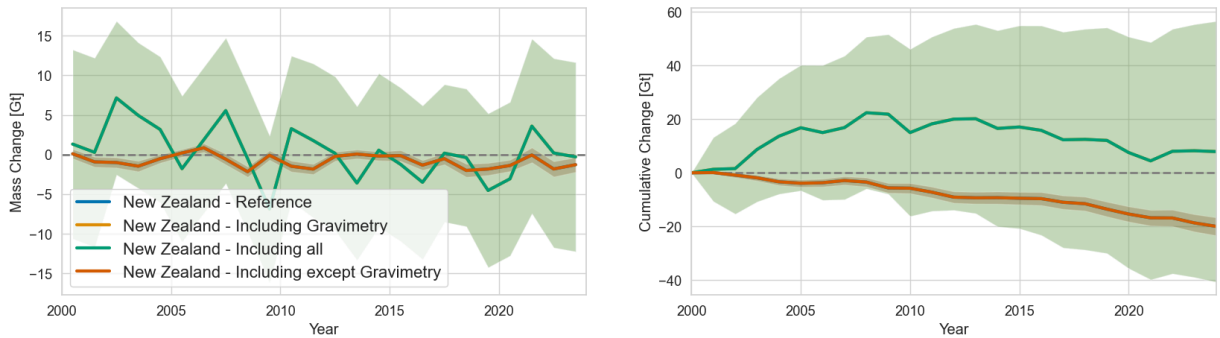
Low Latitudes: annual and cumulative comparison



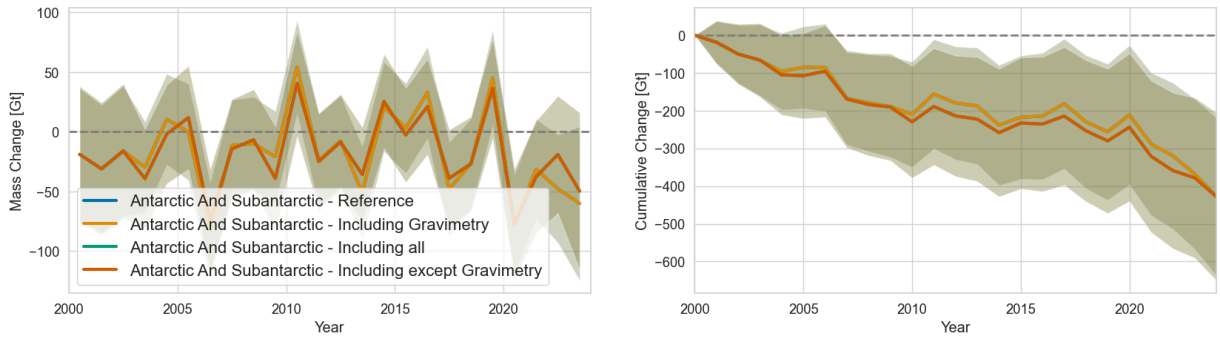
Southern Andes: annual and cumulative comparison



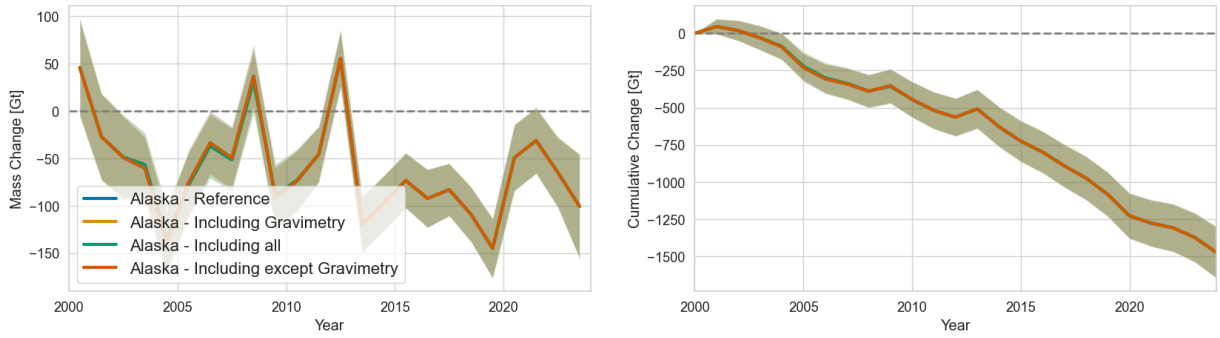
New Zealand: annual and cumulative comparison



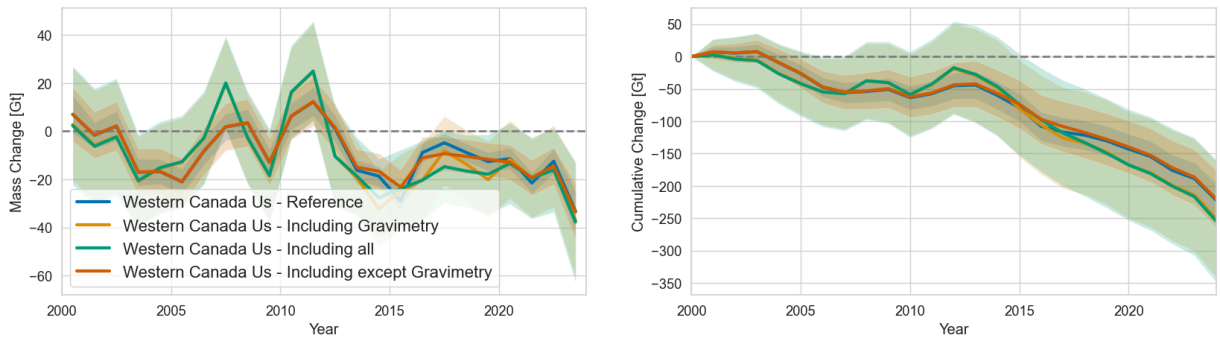
Antarctic And Subantarctic: annual and cumulative comparison



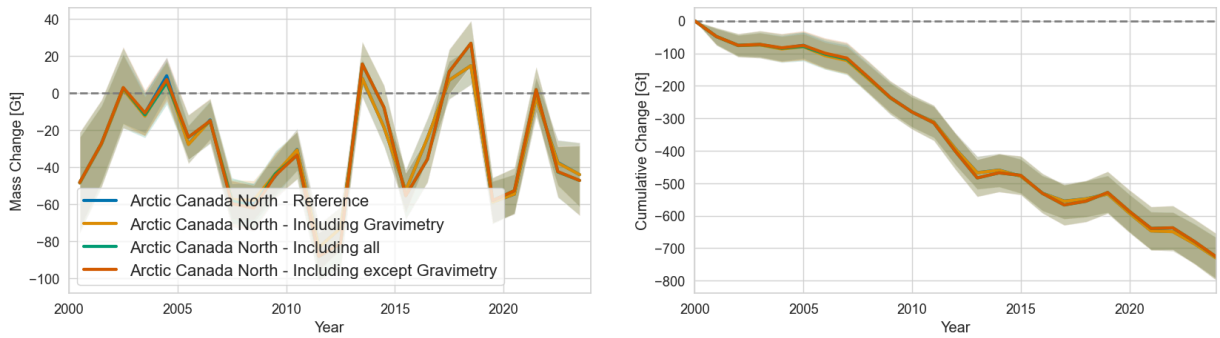
Alaska: annual and cumulative comparison



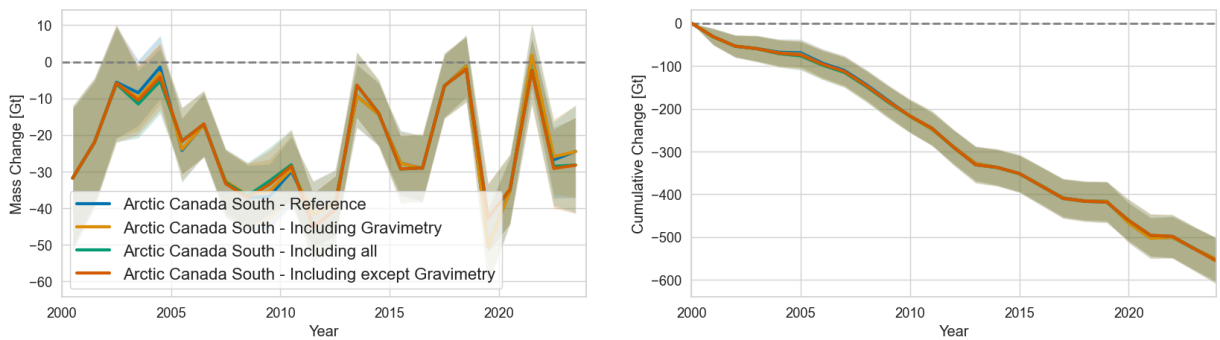
Western Canada Us: annual and cumulative comparison



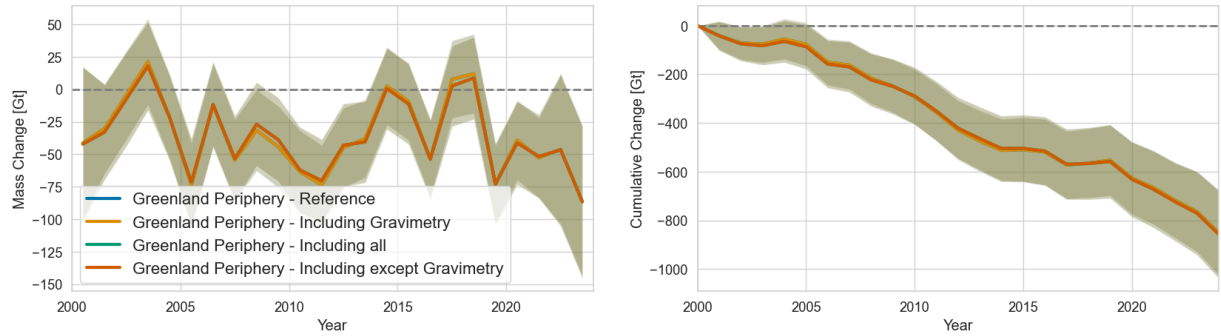
Arctic Canada North: annual and cumulative comparison



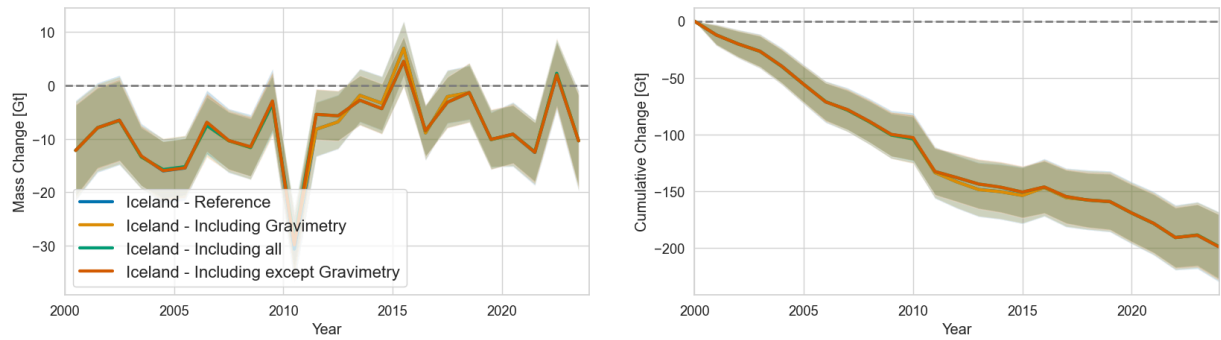
Arctic Canada South: annual and cumulative comparison



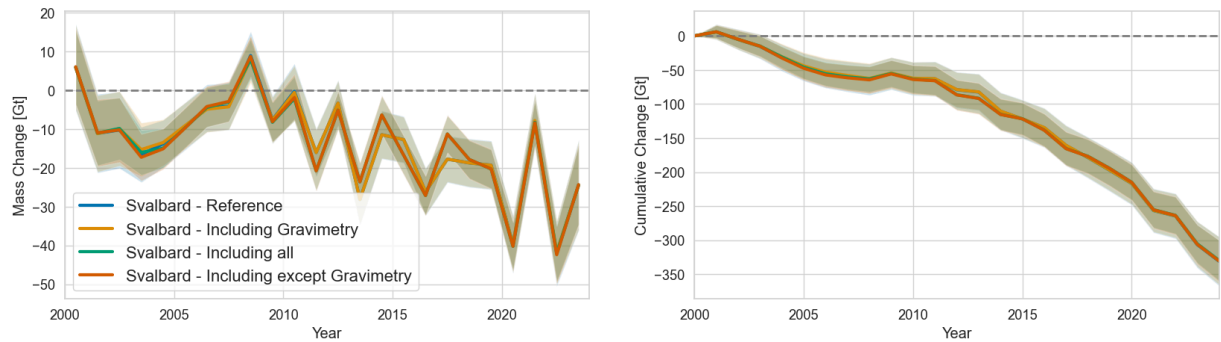
Greenland Periphery: annual and cumulative comparison



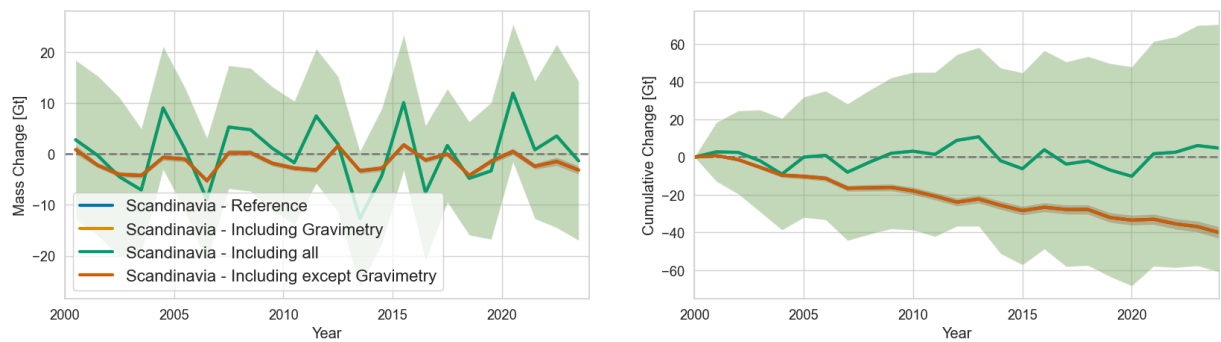
Iceland: annual and cumulative comparison



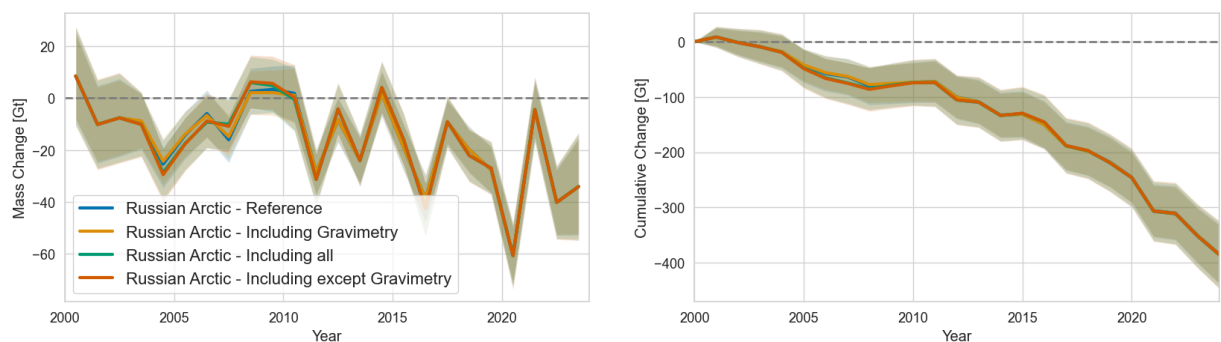
Svalbard: annual and cumulative comparison



Scandinavia: annual and cumulative comparison



Russian Arctic: annual and cumulative comparison



Annual and cumulative glacier mass change (Gt), comparing four dataset configurations relative to the reference run: including gravimetry, Including all available datasets, and Including all datasets except gravimetry.


```

In [29]: def compare_runs(run1_path, run2_path):
    region_names, rel_diffs, abs_diffs, corrs = [], [], [], []

    for region_dir, display_name in regions.items():
        csv1 = run1_path / f'{region_dir}.csv'
        csv2 = run2_path / f'{region_dir}.csv'

        if not csv1.exists() or not csv2.exists():
            continue

        df1, df2 = pd.read_csv(csv1), pd.read_csv(csv2)
        rel_diff, abs_diff, corr = compute_all_metrics_gt(df1, df2)

        region_names.append(display_name)
        rel_diffs.append(rel_diff)
        abs_diffs.append(abs_diff)
        corrs.append(corr)

    # Global comparison
    csv1_g, csv2_g = run1_path / '0_global.csv', run2_path / '0_global.csv'
    if csv1_g.exists() and csv2_g.exists():
        rel_diff, abs_diff, corr = compute_all_metrics_gt(pd.read_csv(csv1_g), pd.read_csv(csv2_g))
        region_names.append('Global')
        rel_diffs.append(rel_diff)
        abs_diffs.append(abs_diff)
        corrs.append(corr)

    return region_names, rel_diffs, abs_diffs, corrs

def create_comparison_barchart(region_names, rel_diff_values, abs_diff_values,
                              corr_values, run1_name, run2_name, output_prefix):
    """Create 3-panel horizontal barchart comparing runs."""
    fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(18, 8), sharey=True)
    y_pos = range(len(region_names))

    # Panel 1: Relative difference
    ax1.barh(y_pos, rel_diff_values)
    ax1.set_yticks(y_pos)
    ax1.set_yticklabels(region_names)
    ax1.invert_yaxis()
    ax1.set_xlabel("Relative Change (%)", fontsize=11)
    ax1.axvline(0, color="black", linewidth=0.8)

    max_abs_rel = max(abs(v) for v in rel_diff_values) or 1
    for i, val in enumerate(rel_diff_values):
        x_pos = val + (0.02 * max_abs_rel if val >= 0 else -0.02 * max_abs_rel)
        ax1.text(x_pos, i, f"{val:+.1f}%", va="center", ha="left" if val >= 0 else "right", fontsize=9)
    ax1.set_xlim(min(rel_diff_values) - 0.35*max_abs_rel, max(rel_diff_values) + 0.35*max_abs_rel)

    # Panel 2: Absolute difference
    ax2.barh(y_pos, abs_diff_values)
    ax2.set_xlabel("Absolute Difference (Gt)", fontsize=11)
    ax2.axvline(0, color="black", linewidth=0.8)

    max_abs_diff = max(abs(v) for v in abs_diff_values) or 1
    for i, val in enumerate(abs_diff_values):
        x_pos = val + (0.02 * max_abs_diff if val >= 0 else -0.02 * max_abs_diff)
        ax2.text(x_pos, i, f"{val:+.2f}", va="center", ha="left" if val >= 0 else "right", fontsize=9)
    ax2.set_xlim(min(abs_diff_values) - 0.30*max_abs_diff, max(abs_diff_values) + 0.30*max_abs_diff)

    # Panel 3: Correlation
    ax3.barh(y_pos, corr_values)
    ax3.set_xlabel("Spearman Correlation", fontsize=11)
    ax3.set_xlim(-0.1, 1.1)

    for i, val in enumerate(corr_values):
        ax3.text(val + 0.02, i, f"{val:.2f}", va="center", fontsize=9)

    fig.suptitle(f"{run1_name} vs {run2_name}", fontsize=13)
    fig.tight_layout()
    plt.show()

```

```

In [30]: # Reference run and regions map (as defined previously)
regions = {

```

```

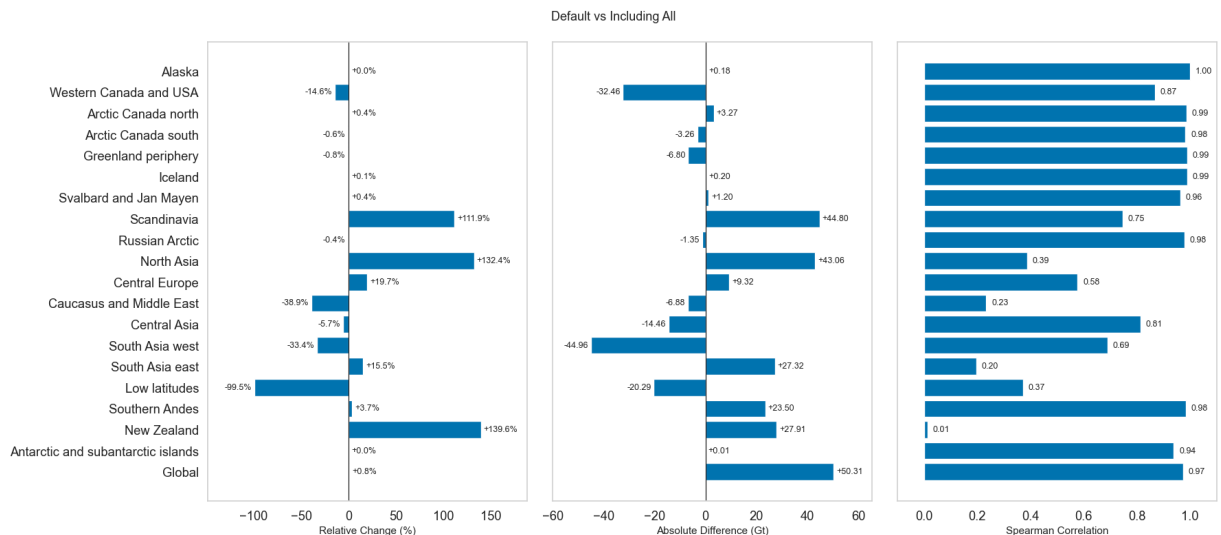
"1_alaska": "Alaska",
"2_western_canada_us": "Western Canada and USA",
"3_arctic_canada_north": "Arctic Canada north",
"4_arctic_canada_south": "Arctic Canada south",
"5_greenland_periphery": "Greenland periphery",
"6_iceland": "Iceland",
"7_svalbard": "Svalbard and Jan Mayen",
"8_scandinavia": "Scandinavia",
"9_russian_arctic": "Russian Arctic",
"10_north_asia": "North Asia",
"11_central_europe": "Central Europe",
"12_caucasus_middle_east": "Caucasus and Middle East",
"13_central_asia": "Central Asia",
"14_south_asia_west": "South Asia west",
"15_south_asia_east": "South Asia east",
"16_low_latitudes": "Low latitudes",
"17_southern_andes": "Southern Andes",
"18_new_zealand": "New Zealand",
"19_antarctic_and_subantarctic": "Antarctic and subantarctic islands",
}

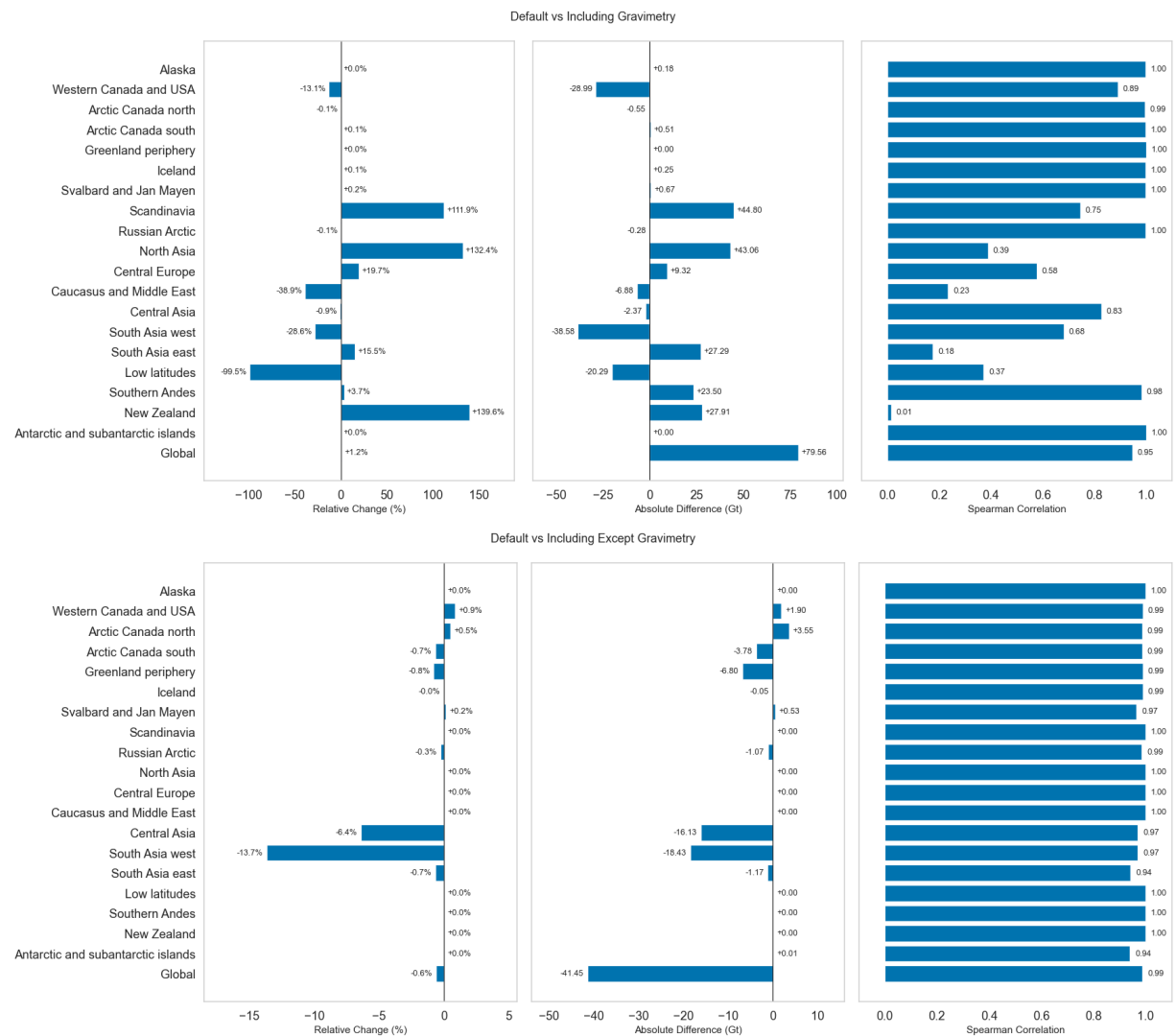
all_results = {}
for run in dataset_run_dirs:
    # Get metrics lists explicitly
    names, rel_diffs, abs_diffs, corrs = compare_runs(reference_run, run)

    # Store in dictionary using simple keys
    all_results[run.parent.name] = {
        'region_names': names,
        'rel_diff_values': rel_diffs,
        'abs_diff_values': abs_diffs,
        'corr_values': corrs
    }

    # Create the plot using the variables explicitly
    label = run.parent.name.replace('datasets_', '').replace('_', ' ').title()
    create_comparison_barchart(
        names,
        rel_diffs,
        abs_diffs,
        corrs,
        'Default',
        label,
        f'sensitivity_{run.name}'
    )

```





Relative difference (%), absolute difference (Gt), and Spearman correlation of cumulative glacier mass change per region, comparing each alternative dataset configuration against the reference run: including gravimetry (top), Including all available datasets (middle), and Including all datasets except gravimetry (bottom).

```
In [31]: # Create summary table using global results from all_results
summary_rows = []
for run_name, results in all_results.items():
    # Use the same labeling logic as the charts
    run_label = run_name.replace('datasets_', '')

    # Explicitly find Global index
    global_idx = results['region_names'].index('Global')

    summary_rows.append({
        "Run": run_label,
        "Relative Difference (%)": results['rel_diff_values'][global_idx],
        "Absolute Difference (Gt)": results['abs_diff_values'][global_idx],
        "Spearman Correlation": results['corr_values'][global_idx]
    })

summary_df = pd.DataFrame(summary_rows)
display(summary_df)
```

	Run	Relative Difference (%)	Absolute Difference (Gt)	Spearman Correlation
0	including_all	0.768998	50.311443	0.973913
1	including_gravimetry	1.216083	79.561878	0.946957
2	including_except_gravimetry	-0.633565	-41.450831	0.987826

We see big differences in the estimates caused by changed dataset selection. When evaluating the inclusion of all available datasets, the global cumulative mass change deviates by 50.31 Gt, representing a 0.77% relative difference compared to the reference run. Regionally, these variations are much more extreme, with absolute changes ranging from -44.96 Gt to 44.80 Gt (-99.48% to 139.64%). The sensitivity to specific data types is particularly large for gravimetry datasets, which drives the highest global shift of 79.56 Gt (a 1.22% relative difference). Conversely, when incorporating all available datasets except gravimetry, the global deviation flips to -41.45 Gt (-0.63% relative difference), and the regional relative discrepancies narrow significantly to a smaller range of -13.69% to 0.86%.

One dataset contributing to these large differences seems to be Jacob et al. (2012) (see other model runs in `glambie_runs_clean/datasets_including_jacob_2012_dmdt` and the dataset plots in `output/datasets`). Their results based on GRACE (Gravity Recovery and Climate Experiment) were later corrected by Gardner et al. (2013), who integrated altimetry and glaciological data to lower mass loss estimates and further refined by Wouters et al. (2019).

The figures showing the cumulative mass changes show unreasonable results when including all gravimetry datasets: there are much larger errors and mass accumulation in contradiction to other methods for many regions, e.g. North Asia and New Zealand. This confirms their exclusion by Zemp et al (2025) and highlights the influence of dataset selection on estimates.

While we could not use every excluded dataset due to model failures, this limitation is itself a finding. It highlights that the integration still faces some technical challenges. The valid exclusion of gravimetry datasets means that the final results are dominated by a substantial extent by DEM-differencing, which also becomes apparent as some regions are only covered by DEM-differencing, something the paper set out to change: DEM-differencing covers 19 regions, while altimetry and gravimetry only 13 and 7 regions respectively (Zemp et al., 2025).

Dataset choice has a large impact on results, and exclusion decisions could be more transparent. The paper is marketed as novel due to the inclusion of other methods. However, if they are mostly excluded or technically restricted to specific regions, is this really the case? It is no surprise that the results are similar to other DEM-differencing-based studies - a point also mentioned in the peer review. Overall though, we found, while there could be more transparency on dataset inclusion/exclusion, we found no indication of problematic decisions- the study appears sound.

Beyond the four primary methods, Zemp et al. (2025) include 58 hybrid estimates from 7 research teams, reclassified into the best-fitting method category for processing. Our dataset sensitivity runs do not isolate hybrid estimates, a dedicated run excluding all hybrid submissions would be a logical next step, particularly for regions where hybrid datasets pose a high proportion of the included datasets.

Overview of results

```
In [33]: rows = []

comparisons = (
    [("Area", "RGI 7", rgi7_dir),
     ("Area", "Model 2012", d_dir_2012),
     ("Area", "Model 2023", d_dir_2023)]
    + [("Methods", r.parent.name.replace("methods_", ""), str(r)) for r in method_run_dirs]
    + [("Datasets", r.parent.name.replace("datasets_", ""), str(r)) for r in dataset_run_dirs]
)

for group, comp_name, comp_dir in comparisons:

    abs_gt = {}
    rel_gt = {}
    corr_gt = {}
    abs_mwe = {}
    rel_mwe = {}
    corr_mwe = {}

    base_calendar = Path(reference_run_dir)
    comp_calendar = Path(comp_dir)

    for region_file in [p for p in list_of_folders]:
```

```

df_base = pd.read_csv(base_calendar / region_file).drop(columns=["Unnamed: 0"], errors="ignore")
df_comp = pd.read_csv(comp_calendar / region_file).drop(columns=["Unnamed: 0"], errors="ignore")

m = pd.merge(df_base, df_comp, on="start_dates", suffixes=("_base", "_comp"), how="outer")

# gt metrics
a = m["combined_gt_base"]
b = m["combined_gt_comp"]
abs_diff = b.sum() - a.sum()
abs_gt[region_file] = abs_diff
rel_gt[region_file] = (abs_diff / abs(a.sum()) * 100) if abs(a.sum()) > 1e-12 else 0.0
corr, _ = spearmanr(a, b, nan_policy="omit")
corr_gt[region_file] = 0.0 if pd.isna(corr) else corr

# compute ranges over regional files (exclude global)
gt_reg_val = [v for k, v in abs_gt.items() if k != "0_global.csv"]
rel_gt_reg_val = [v for k, v in rel_gt.items() if k != "0_global.csv"]
corr_gt_reg_val = [v for k, v in corr_gt.items() if k != "0_global.csv"]
mwe_reg_val = [v for k, v in abs_mwe.items() if k != "0_global.csv"]
rel_mwe_reg_val = [v for k, v in rel_mwe.items() if k != "0_global.csv"]
corr_mwe_reg_val = [v for k, v in corr_mwe.items() if k != "0_global.csv"]

rows.append({
    "Group": group,
    "Run": comp_name,
    "Δ Gt Abs (Global)": f"{abs_gt.get('0_global.csv'):.2f}",
    "Δ Gt Abs (Regional Range)": f"{min(gt_reg_val):.2f} to {max(gt_reg_val):.2f}",
    "Δ Gt Rel % (Global)": f"{rel_gt.get('0_global.csv'):.2f}",
    "Δ Gt Rel % (Regional Range)": f"{min(rel_gt_reg_val):.2f} to {max(rel_gt_reg_val):.2f}",
    "Gt Spearman r (Global)": f"{corr_gt.get('0_global.csv'):.2f}",
    "Gt Spearman r (Regional Range)": f"{min(corr_gt_reg_val):.2f} to {max(corr_gt_reg_val):.2f}"})

combined_metrics_table = pd.DataFrame(rows).set_index(["Group", "Run"])
display(combined_metrics_table)

```

		Δ Gt Abs (Global)	Δ Gt Abs (Regional Range)	Δ Gt Rel % (Global)	Δ Gt Rel % (Regional Range)	Gt Spearman r (Global)	Gt Spearman r (Regional Range)
Group	Run						
Area	RGI 7	-0.42	-15.75 to 22.43	-0.01	-9.67 to 23.75	1.00	1.00 to 1.00
	Model 2012	0.44	-1.32 to 0.53	0.01	-0.15 to 0.22	1.00	1.00 to 1.00
	Model 2023	-1.74	-1.00 to 0.28	-0.03	-0.12 to 0.09	1.00	1.00 to 1.00
Methods	regression-proportional	0.00	0.00 to 0.00	0.00	0.00 to 0.00	1.00	1.00 to 1.00
	regression-seasonal_homogenization	26.12	-12.66 to 26.34	0.40	-4.08 to 6.16	1.00	0.92 to 1.00
	start_vs_end-proportional	31.98	-19.03 to 73.79	0.49	-9.56 to 11.70	1.00	1.00 to 1.00
	start_vs_end-seasonal_homogenization	58.11	-20.05 to 70.71	0.89	-10.07 to 11.21	0.99	0.92 to 1.00
Datasets	including_all	50.31	-44.96 to 44.80	0.77	-99.48 to 139.64	0.97	0.01 to 1.00
	including_gravimetry	79.56	-38.58 to 44.80	1.22	-99.48 to 139.64	0.95	0.01 to 1.00
	including_except_gravimetry	-41.45	-18.43 to 3.55	-0.63	-13.69 to 0.86	0.99	0.94 to 1.00

```

In [34]: rows = []

comparisons = (
    [("Area", "RGI 7", rgi7_dir),
     ("Area", "Model 2012", d_dir_2012),

```

```

    ("Area", "Model 2023", d_dir_2023)]
+ [{"Methods", r.parent.name.replace("methods_", ""), str(r)} for r in method_run_dirs]
+ [{"Datasets", r.parent.name.replace("datasets_", ""), str(r)} for r in dataset_run_dirs]
)

for group, comp_name, comp_dir in comparisons:

    abs_gt = {}
    rel_gt = {}
    corr_gt = {}
    abs_mwe = {}
    rel_mwe = {}
    corr_mwe = {}

    base_calendar = Path(reference_run_dir)
    comp_calendar = Path(comp_dir)

    for region_file in [p for p in list_of_folders]:

        df_base = pd.read_csv(base_calendar / region_file).drop(columns=["Unnamed: 0"], errors="ignore")
        df_comp = pd.read_csv(comp_calendar / region_file).drop(columns=["Unnamed: 0"], errors="ignore")

        m = pd.merge(df_base, df_comp, on="start_dates", suffixes=("_base", "_comp"), how="outer")

        # mwe metrics
        a = m["combined_mwe_base"]
        b = m["combined_mwe_comp"]
        abs_diff = b.sum() - a.sum()
        abs_mwe[region_file] = abs_diff
        rel_mwe[region_file] = (abs_diff / abs(a.sum()) * 100) if abs(a.sum()) > 1e-12 else 0.0
        corr, _ = spearmanr(a, b, nan_policy="omit")
        corr_mwe[region_file] = 0.0 if pd.isna(corr) else corr

    # compute ranges over regional files (exclude global)
    gt_reg_val = [v for k, v in abs_gt.items() if k != "_global.csv"]
    rel_gt_reg_val = [v for k, v in rel_gt.items() if k != "_global.csv"]
    corr_gt_reg_val = [v for k, v in corr_gt.items() if k != "_global.csv"]
    mwe_reg_val = [v for k, v in abs_mwe.items() if k != "_global.csv"]
    rel_mwe_reg_val = [v for k, v in rel_mwe.items() if k != "_global.csv"]
    corr_mwe_val = [v for k, v in corr_mwe.items() if k != "_global.csv"]

    rows.append({
        "Group": group,
        "Run": comp_name,
        "Δ m w.e. Abs (Global)": f"{abs_mwe.get('_global.csv'):.2f}",
        "Δ m w.e. Abs (Regional Range)": f"{min(mwe_reg_val):.2f} to {max(mwe_reg_val):.2f}",
        "Δ m w.e. Rel % (Global)": f"{rel_mwe.get('_global.csv'):.2f}",
        "Δ m w.e. Rel % (Regional Range)": f"{min(rel_mwe_reg_val):.2f} to {max(rel_mwe_reg_val):.2f}",
        "m w.e. Spearman r (Global)": f"{corr_mwe.get('_global.csv'):.2f}",
        "m w.e. Spearman r (Regional Range)": f"{min(corr_mwe_val):.2f} to {max(corr_mwe_val):.2f}"})

combined_metrics_table = pd.DataFrame(rows).set_index(["Group", "Run"])
combined_metrics_table.to_csv("Combined_table.csv")
display(combined_metrics_table)

```

		Δ m w.e. Abs (Global)	Δ m w.e. Abs (Regional Range)	Δ m w.e. Rel % (Global)	Δ m w.e. Rel % (Regional Range)	m w.e. Spearman r (Global)	m w.e. Spearman r (Regional Range)
Group	Run						
Area	RGI 7	0.01	-0.56 to 0.01	0.14	-2.55 to 0.09	1.00	1.00 to 1.00
	Model 2012	-0.10	-1.13 to -0.01	-1.03	-4.42 to -0.12	1.00	0.99 to 1.00
	Model 2023	0.11	0.02 to 0.84	1.15	0.27 to 4.67	1.00	1.00 to 1.00
Methods	regression-proportional	0.00	0.00 to 0.00	0.00	0.00 to 0.00	1.00	1.00 to 1.00
	regression-seasonal_homogenization	0.04	-0.44 to 0.91	0.38	-4.37 to 6.15	0.99	0.92 to 1.00
	start_vs_end-proportional	0.05	-1.81 to 2.57	0.47	-9.64 to 11.65	1.00	1.00 to 1.00
	start_vs_end-seasonal_homogenization	0.08	-1.91 to 2.47	0.85	-10.17 to 11.18	0.99	0.92 to 1.00
Datasets	including_all	0.07	-9.46 to 29.90	0.72	-93.68 to 130.43	0.96	0.08 to 1.00
	including_gravimetry	0.11	-9.46 to 29.90	1.13	-93.68 to 130.43	0.94	0.08 to 1.00
	including_except_gravimetry	-0.06	-0.59 to 0.12	-0.61	-13.92 to 0.75	0.99	0.93 to 1.00

Table summarizing the variability of the performed changes in methods, glacier area and datasets. Presented values are the differences compared to the estimated change by Zemp et al. (2025).

Conclusion

To conclude, although Zemp et al. (2025) provide a convincing global synthesis of glacier mass loss, the regional estimates are a lot more uncertain and should be interpreted with greater caution. This also highlights the limitations of the current state of glacier monitoring, which future research should address with improved observations. Furthermore, more clarity on the glacier area calculations and the reasoning behind excluding shrinkage from the Gt conversion is needed, as well as the reasons for the the dataset exclusion.

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